

The Wave That Is Always There
A Fractal Atlas from the Universe to the Soma

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The Wave That Is Always There

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Alistair Johnson

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For my family —

*the ones who carried the wave
before there was a word for it.*

How to Read This Book

This book has three movements. It begins at the largest scale we know — the cosmic microwave background, the faint glow left over from when the universe was 380,000 years old — and ends at the smallest scale we can talk about, a seven-dimensional compactified manifold roughly 10^{-35} metres across. In between, it stops at galaxies, stars, planets, the rocks of central Switzerland, plants and rivers, the human heart, the fascia, and the field of feeling that the rest of my work calls the *Soma Field*.

The argument — if there is one — is simple: **the same mathematical objects recur at every scale.** Waves, fields, attractor landscapes, fractal branching. Not metaphorically. Mathematically. The wave equation in a violin string and the wave equation in the cosmic plasma 380,000 years after the Big Bang are the same equation. The fractal that organises a tree's branches and the fractal that organises a lung's bronchi and the fractal that organises the cosmic web is the same fractal in three different physical substrates.

You can read this book in any order. The chapters are sized to be read in one sitting and indexed for browsing. Most of the pages are pictures, and most of the pictures want to be looked at slowly. If you have only one afternoon, read Chapter 1, Chapter 6 (the Glarus chapter), and Chapter 11 (the soma field). Those are the three load-bearing chapters; the rest is the scaffolding that makes them load-bearing.

If a paragraph contains an equation that loses you, skip it. There are

no later paragraphs that depend on understanding it. The mathematics is here because, when it is there, the picture beside it stops being a metaphor and becomes a description. That is a different experience from “popular science that explains an equation”; it is closer to looking at a topographical map beside a photograph of the mountain.

A few chapters have a sidebar called *Your own example*. Those are invitations, not exercises. Coffee-table books should let the reader put themselves in. Chapter 18 — *A Family Album* — is left almost entirely blank on purpose; the instructions for filling it in are inside.

I am an independent researcher in Zurich. I am not a credentialed cosmologist, geologist, biologist, or clinician. I am a physicist by training who has spent thirty years thinking about how a small number of mathematical ideas keep appearing in places where, on first inspection, they have no business being. This book is what those thirty years of noticing look like when arranged in scale order. It is not a textbook. It is an atlas.

— *Alistair Johnson, Zurich, summer 2026.*

A Note on Sources, Pictures, and Honesty

Everything that came from someone else's work is cited; the citation style is Chicago notes-and-bibliography because I find footnotes easier on the eye than parenthetical inserts. Every photograph that is not mine is credited at the back of the book under *Image Credits*, with the licence under which it appears. Most of the imagery is from NASA, ESA, the United States Geological Survey, the European Space Agency, swisstopo, and Wikimedia Commons — all either public-domain or released under Creative Commons. A few photographs I took myself; those are noted (*Author*).

Eleven dimensions cannot be drawn. Wherever a picture in this book claims to show eleven (or seven, or six) dimensions, it is showing a two- or three-dimensional projection or cross-section, and the caption says so. This is standard practice in string-theory popularisations and is not a sleight of hand. The shape itself is real; the picture is an honest shadow of it.

Where I make a claim that goes beyond the consensus of a field — and a few of them, by the back of the book, will — I have flagged it explicitly. *This is where I am asserting something the textbook does not yet say.* You may disagree with me. That is also fine. The atlas does not depend

on the assertion; the assertion depends on the atlas.

Prologue: Standing at the Window

In Klöntal, in the canton of Glarus, there is a lake that sits in a trough the glaciers cut and that the mountains have been holding open ever since. On still mornings the water looks like a slab of slate. Just before any wind, you can sometimes see a single ring spread from the centre of the lake — a fish, or nothing, or the lake itself remembering — and the ring goes out and out and finally meets the shore and comes back, faint, as though the basin itself had taken a long breath.

Above the lake, on the south side, a wall of limestone goes up nearly vertically for a thousand metres. Halfway up the wall there is a thin horizontal seam where the rock changes colour. The rock above the seam is older than the rock below it. It is older by about a hundred and fifty million years.

The seam is called the Glarus Hauptüberschiebung, the Glarus principal overthrust, and it is a UNESCO World Heritage Site for exactly the reason that it is unsettling to look at. Older rock sitting on top of younger rock is, on the face of it, an embarrassment to the idea that things settle downward in chronological order. The way the older rock came to be on top of the younger rock is that, in slow motion, over millions of years, it flowed there. It moved roughly thirty-five kilometres

to the north, travelling as a coherent sheet, riding on a layer of softer rock that behaved, on geological timescales, like grease.

I tell you this in the prologue of a book whose nominal subject is the inside of a human being because the Glarus thrust is, for me, the moment the argument of this book becomes inescapable. Solid mountain limestone, in sufficient quantity and given sufficient time, behaves as a wave. It moves. It propagates. There is a leading edge and a trailing edge. It has a direction of motion, a velocity, and a place where it stops.

Once you have admitted that rock is a slow wave, the question is no longer *whether* the world is made of waves. The question is *how slow are you willing to go to see them*. The cosmic microwave background is a wave that took 13.8 billion years to reach your detector. The Glarus thrust is a wave that took about thirty million years to travel thirty-five kilometres. Your heart, sitting in your chest as you read this, is firing a wave roughly once a second. A photon from your reading lamp is a wave that crossed the room in three nanoseconds. They are all the same kind of object. They have different speeds, different wavelengths, different substrates, different amplitudes. They are all waves.

This book is an atlas of those waves, sorted by scale.

I should also tell you what kind of writer is going to be holding the camera for the next eighteen chapters, because it will matter.

I am a physicist by training, in the loose sense that I was educated as one in the early 1990s and have spent the thirty years since applying that education to problems that are nominally not physics. I am an independent researcher; I have no university affiliation, no postdoctoral position, no funding council. I have published eleven technical papers in the *Soma Field* series — formal proofs in Lean 4, a quantum-annealing experiment, a clinical model of emotional dynamics, a music-affect study, a paper on the physical substrate of feeling — and I have done it from a

flat in Zurich, mostly between five and seven in the morning, before the working day begins.

I am also, since 1968, a person with three structural modifications to my nervous system that the clinical literature labels Autism Spectrum Condition, Attention Deficit Hyperactivity Disorder, and Complex Post-Traumatic Stress Disorder. These conditions explain, among other things, why I am the kind of person who would notice that a mountain is a slow wave. They also explain why I have arrived, at fifty-eight, at the view that the inside of a human being is not categorically different from the inside of a thunderstorm or the inside of a galaxy, and is best described by the same mathematics.

I am telling you this in the prologue because you are about to read a book about cosmology, geology, and the inside of feeling, written by someone who is not, strictly speaking, a credentialled cosmologist, geologist, or clinician. I think you should know that going in. I also think the book is better for having been written by a person who came to it sideways. Books that come at things from the side often see what books that come at things head-on cannot.

The cosmologists know more than I do about cosmology. The geologists know more than I do about geology. The clinicians know more than I do about the clinical care of the conditions I have. What I bring is the willingness to hold all three in the same hand and ask what they have in common, and to follow the question wherever it goes. The answer, I believe, is that they have *the wave* in common — the same mathematical object, scaled and substrated differently — and that this commonality is not a poetic flourish but a structural fact, with consequences.

The first consequence is the book you are holding.

A coffee-table book is not the place for the formal mathematics. The formal mathematics is in the technical papers, all of them open-access

on Zenodo, all of them under the DOIs in the back of this book. What a coffee-table book is for is showing you the pictures. The pictures are the argument. If you look at the cosmic microwave background, and then at a slice through a galaxy cluster, and then at an aerial photograph of the Klöntalersee with the thrust line visible on the south wall, and then at a microscope image of human fascia, and then at a Mandelbulb render — and you let your eye notice what your eye notices — you will, I believe, find that the argument of this book has already been made before you have read a single equation.

The prose around the pictures is just the prose around the pictures. Treat it like the audio guide in a museum: skip the bits that don't help, linger on the bits that do.

There is one more thing I want to say before we begin.

I have a daughter. She is, at the time I am writing this, fourteen. She is the audience for this book in the sense that everything I write, I write with her in mind as the eventual reader. Not now — she is fourteen, she has better things to do — but in twenty years, or thirty, when she goes looking, I want there to be something on the shelf that says: *this is what your father thought the world was made of, and this is how he came to think it*. I want it to be the kind of book she can put on her own coffee table and not be embarrassed by.

I mention her because her presence — and the presence of the wider family who have stood around the lake with me and not understood why I was so interested in the rock above the water line — is the reason for Chapter 18 of this book, *A Family Album*. That chapter is mostly blank pages. It is designed to be filled in by you, with your own photographs, of your own lake, your own thrust, your own waves. The book is mine. The atlas, by the time you finish reading it, should be yours.

I will see you at the window.

Chapter 1 — The Wave

That Is Always There

A wave is the simplest thing in physics that is not a thing.

1.1 What a wave is, as briefly as I can manage

Hold a piece of rope at one end and flick it. A bump travels down the rope to the other end. The rope itself doesn't go anywhere; the bump does. What travelled was not matter — every fibre of the rope is exactly where it was — but a *pattern* in the matter. A momentary local rearrangement that kept its shape long enough to be observed at the far end.

That is a wave.

If you slow the flick down and keep doing it, the rope settles into a shape: a series of crests and troughs that go up and down without travelling. That is a *standing wave*. The matter still doesn't go anywhere, and now the pattern doesn't go anywhere either, but the rope is nevertheless full of motion. The matter at each point goes up and down, and the rate at which it goes up and down is set by the rope's length, its tension, and its mass per unit length. Different combinations give different frequencies — different notes, if the rope were a violin string.

This is the simplest possible model of a wave, and almost every other wave in this book is a variation on it. Replace the rope with the air in a room: sound. Replace the rope with the surface of a pond: ripples. Replace the rope with the electric and magnetic fields of empty space: light. Replace the rope with the metric of spacetime itself: gravitational waves. Replace the rope with the cosmic plasma 380,000 years after the Big Bang: the acoustic peaks of the cosmic microwave background. Replace the rope with the firing patterns of a hundred million neurons synchronised at ten cycles per second: the alpha rhythm of a calm human brain.

The same equation governs all of them. Different substrate, different speed, different scale; same equation.

Figure 1.1 — Standing waves on a string at four amplitudes. As the energy in the string increases, the same shape returns, just larger. The shape is determined by the boundary conditions, not by the energy. We will see this principle again, at every scale.

1.2 What a *field* is

A wave needs something to wave in. The thing it waves in is called a *field*.

The simplest field is the height of a sheet of water across a pond. At every point (x, y) on the pond, the water has some height $h(x, y, t)$. This h is a *number assigned to every point in space and time*. That is what a field is.

When you throw a stone into the pond, the height field is disturbed; a ripple propagates. The ripple is the wave. The pond is the field. You cannot have one without the other. If the pond were empty, there would be nothing to ripple.

This sounds obvious for ponds. The conceptual leap of twentieth-century physics — the leap that made the rest of this book possible — is that *everything* is a field. The space around the magnet is a field (the magnetic field). The space around an electric charge is a field (the electric field). Empty space itself is several fields layered on top of one another: the electromagnetic field, the gravitational field, and twelve different matter fields (one for each kind of fundamental particle). The “particles” we are taught about in school — electrons, photons, quarks — are not little balls. They are *localised excitations of their respective fields*. A photon is a ripple in the electromagnetic field. An electron is a ripple in the electron field. Throwing a stone in the electron field makes electrons; the electron is the ripple.

This is the central insight of *quantum field theory*, and it is the substrate on which the rest of this book stands. Once you have understood that the world is fields and that particles are excitations of fields, you can ask the question this book is built on: *do feelings have a field?*

1.3 The vacuum is not empty

There is one further fact about fields you need to carry with you for the rest of the book.

A field, even when nothing is happening in it, is not zero. It fluctuates. The quantum vacuum — empty space, the space inside your skull right now with no measurable particles in it — is *humming*. It is humming because quantum mechanics forbids any field from sitting exactly still; the uncertainty principle requires that the amplitude of every field has a non-zero spread, even in its lowest-energy state. This is not a defect of our measurement. It is a property of the world.

The hum of the vacuum is sub-threshold. You can’t see the individual fluctuations because they are too small, too brief, and too random. But they are causally active: they shift the energy levels of atoms by a mea-

surable amount (the Lamb shift), they produce a measurable force between two parallel plates placed close together in a vacuum (the Casimir effect), and they seed — at the earliest moments of the universe — the patterns of cosmic structure we will look at in Chapter 2.

Figure 1.2 — Cartoon of a quantum vacuum: a field at its lowest-energy state, full of sub-threshold ripples. None of the ripples individually rises above the dotted line where we would call it “an observable particle.” The field is busy. There is nothing in it.

I want you to hold this image in your mind through the rest of the book, because it is the structural template for what I will eventually call the *soma field*. Most of the time, the field of feeling — like the quantum vacuum — is full of activity that doesn’t rise above the threshold of conscious awareness. The activity is real. It influences the body and the behaviour. It is not perceived. When something does rise above the threshold and is perceived, we name it: *anger*, *grief*, *calm*, *awe*. The names are the excitations. The field is what they are excitations of.

1.4 The three universal facts about waves

For everything that comes after, you need to carry three facts about waves. They are short, they are universal, and they apply identically to a violin string, a galaxy, and a human heart.

Fact one: waves superpose. If two waves arrive at the same point at the same time, the result is the algebraic sum of the two. Where crest meets crest, the wave gets bigger. Where crest meets trough, the wave cancels. This is the reason your noise-cancelling headphones work, and it is the reason there are dark fringes in the laser pattern projected through two slits, and it is the reason two emotional modes in opposite phase produce — for a moment — a peculiar inner stillness even though both are running at full amplitude.

Fact two: waves have a natural frequency. Any bounded system — string, drum, planet, plasma, body — will, when disturbed, vibrate at one or more preferred frequencies, set by its geometry and its physical properties. Hit a bell, you get the bell's note, not a random one. Tap the Earth with a large enough earthquake and it rings at its own normal modes, the deepest of which has a period of about 54 minutes. The human heart, the fascia, and the emotional field all have natural frequencies. We will get to all of them.

Fact three: waves carry information without carrying matter. A radio wave does not deliver any electrons to your radio; it delivers a *pattern* that the radio's circuit can decode into music. This is the deepest property of a wave, and it is why a wave is “the simplest thing in physics that is not a thing.” A wave is the carrier of *pattern*. The substrate is incidental.

This is why the same wave equation can describe a violin string and the cosmic microwave background. The substrates have nothing in common. The patterns do.

Figure 1.3 — Triptych: (left) ripples on water; (centre) a sound wave on an oscilloscope; (right) the electric and magnetic components of an electromagnetic wave. Three substrates, one mathematical object.

1.5 Fractals: the wave's other face

There is one more idea you need before we go up the scale ladder. It is the idea of *fractal self-similarity*.

A fractal is a shape that looks similar at many different magnifications. A coastline, photographed from orbit, looks bumpy; photographed from a plane, the same coastline looks similarly bumpy with bumps a hundredth the size; photographed by hand on the beach, similar bumps appear again, a hundredth the size again. The bumps come at every scale, and

the law that makes them is the same law. This is the deep meaning of the famous question “how long is the coast of Britain?”: the answer depends on the length of your ruler, and the dependence is *itself* a number — the fractal dimension of the coastline — that characterises the shape.

The most famous fractal of all is the *Mandelbrot set*, discovered (or arguably *encountered*) by Benoît Mandelbrot in 1980. It is generated by iterating a single quadratic formula, $z \rightarrow z^2 + c$, and asking which starting points stay bounded. The boundary of the answer is a shape with infinite detail at every scale — the same swirls and bulbs appearing deeper and deeper, but never quite repeating. The three-dimensional analogue, the *Mandelbulb*, is what you see on the cover of this book and in Chapter 16. It is a mathematical object. It is also one of the most beautiful things in nature, which is a peculiar thing to say about something that is not strictly in nature.

The reason fractals matter for our story is that *self-similar branching shows up everywhere*. The bronchi of your lungs branch like a tree; trees branch like rivers; rivers branch like blood vessels; blood vessels branch like the cosmic web of dark matter on the largest scales. The branching is not coincidence. It is the geometric solution to a class of problems — distributing a substance through a volume with minimal length — that nature has rediscovered at every scale at which the problem occurs.

When you put fractals next to waves, you get the structure of this whole book. **Waves give us the dynamics — what changes, and how. Fractals give us the geometry — how the substrate is laid out for the wave to move through.** Galaxies, plate tectonics, lungs, neurons, the soma field — all of them are waves running on fractal substrates.

1.6 Lenses, cameras, holograms — three more ways to think about waves

Three everyday objects let you see, with your own hands, what waves do.

A **lens** takes a field of light coming in from all directions and folds it down into a single image on a flat surface. Whatever was in the landscape — kilometres of mountain, valley, lake — arrives at the back of the lens as light waves; the lens bends each wave by exactly the right amount; and the waves recombine, on the focal plane, into a faithful map of the original. A lens is a *wave-summing machine*. Your own cornea is one. So is the objective of every telescope in this book.

A **camera** is a lens with a memory. It freezes one moment of the wave-sum. Everything visible in a photograph is a wave that arrived at the sensor in a few milliseconds and was recorded. A landscape becomes a sheet of paper. Nothing of the wave's *phase* is kept — only its amplitude at each pixel. This is why a photograph is flat, and why nothing you can do to a photograph (zoom, sharpen, enhance) ever recovers what it would have been like to stand there. The phase is gone.

A **hologram** is a camera that keeps the phase. It records the interference pattern between the light from a scene and a reference beam, which encodes both the amplitude *and* the phase of every wave that arrived at the plate. Illuminated again with the reference beam, it reconstructs the original wavefronts, and the scene appears to float in space, three-dimensional, viewable from different angles, exactly as if you were back in front of the original.

The pair *photograph* / *hologram* is the cleanest available illustration of what a wave actually is. A photograph keeps the *intensity*; a hologram keeps the *wavefront*. The Soma Field, in the later chapters of this book, is best imagined not as a photograph of a feeling but as a hologram of one — three-dimensional, viewable from different angles, carrying the

full phase information of the wave that produced it.

Figure 1.4 (*PUBLIC*) — Side by side: a black-and-white photograph of a small object (left), and a hologram of the same object viewed from two slightly different angles (centre and right). The photograph is flat; the hologram shows parallax. *Suitable source: Caltech / Lippmann historical demonstrations, public domain; or original capture.*

The cyber-hologram body in Chapter 11 is this same idea, scaled up to the human form. Hold the pair in mind. We will be making the case, all through the book, that the body is more like a hologram than like a photograph.

1.7 The argument of this book in one paragraph

Everything in the universe, at every scale we have ever measured, is best described as a wave moving on a field, sometimes on a fractal substrate. The scales differ by sixty-one orders of magnitude. The equations don't. The picture that emerges, once you sort by scale, is of a single recursive structure — wave-on-field, branch-on-branch — running from the cosmic microwave background at one end to the seven-dimensional compactified manifold of M-theory at the other, with the human body sitting roughly in the middle and behaving in every respect like the rest of it.

The rest of this book is the long version, with pictures.

Your own example.

Think of one wave you can feel in your own life — a tide, a song, a breath, an argument that rises and falls. Note its rough period. Note what carries it. Note where it stops. We will come back to it in Chapter 11.

Chapter 2 — The Universe Begins as a Wave

Before there was anything to see, there was something to listen to.

2.1 The earliest picture we have

The Cosmic Microwave Background — the CMB, for short — is the oldest light in the universe. It was released 380,000 years after the Big Bang, at the moment the cooling primordial plasma became transparent to its own glow. Every direction of the sky is, very faintly, still warm with this light, at a temperature of 2.725 kelvin. It has been redshifted by a factor of about 1,100 since it was emitted; when it left its source, it was a hot orange glow at about 3,000 K.

The European Space Agency’s *Planck* satellite mapped this light from 2009 to 2013 with a precision sufficient to resolve temperature variations of one part in 100,000 across the entire sky.¹ The resulting image

¹Planck Collaboration, “Planck 2018 results. I. Overview and the cos-

is, without exaggeration, the most informative single photograph ever taken.

Figure 2.1 (*PUBLIC*) — The *Planck* all-sky map of the CMB. The colour scale shows temperature variations of $\pm 200 \mu\text{K}$ around the mean of 2.725 K. *Credit: ESA / Planck Collaboration. Released under ESA standard licence — usable with credit.*

The image looks like static. It is not static. It is a sound spectrum.

2.2 The acoustic peaks

Take the *Planck* image and compute its angular power spectrum — that is, ask the question, “how much temperature variation is there on the sky at each angular scale?” — and you get a curve with a series of bumps in it.

Figure 2.2 (*PUBLIC*) — The CMB angular power spectrum. The horizontal axis is multipole moment ℓ (large angles on the left, small angles on the right). The vertical axis is the temperature variance at each scale. The first peak is at $\ell \approx 220$, corresponding to an angular scale of about one degree on the sky. *Credit: ESA / Planck.*

These bumps are *acoustic peaks*. They are the frequency spectrum of a sound wave that propagated in the early universe.

This is not a figure of speech. It is the literal physics. Before the universe became transparent, it was a hot plasma — protons, electrons, photons, and dark matter — and that plasma was a compressible fluid with a definite *sound speed*, very close to $c/\sqrt{3}$, where c is the speed of light.

mological legacy of Planck,” *Astronomy & Astrophysics* 641 (2020): A1, <https://doi.org/10.1051/0004-6361/201833880>.

Pressure perturbations in this fluid propagated as sound waves. Gravity was pulling the plasma into the denser regions; photon pressure was pushing it back out. The competition produced acoustic oscillations — a ringing — that was frozen into the temperature pattern at the moment of decoupling.

The first peak in the power spectrum is the fundamental mode of this ringing — the wavelength of a sound wave that had just had time to complete one full compression by the moment the plasma became transparent. The second peak is the second harmonic. The third is the third. The plasma was a musical instrument, and the *Planck* image is its recorded note, with the harmonics resolved.

The fundamental frequency of the early universe was about 10^{-16} Hz. Its wavelength at the moment it was recorded was about 147 Mpc — roughly half a billion light-years.²

2.3 Inflation: the wave before the wave

Push the question one step further back. Where did the seed inhomogeneities come from — the slight density variations that the acoustic oscillations were ringing on?

The standard answer is *inflation*: a brief, extraordinarily rapid expansion of the universe during its first 10^{-32} seconds, during which a small patch was stretched by a factor of at least 10^{26} . During this expansion, quantum fluctuations in the inflating field — the *inflaton* — were stretched along with the universe, frozen in place when their wavelengths exceeded the cosmological horizon, and later returned to the observable universe

²Daniel J. Eisenstein and colleagues established the corresponding *baryon acoustic oscillation* scale in the present-day galaxy distribution in “Detection of the Baryon Acoustic Peak in the Large-Scale Correlation Function of SDSS Luminous Red Galaxies,” *Astrophysical Journal* 633, no. 2 (2005): 560–74, <https://doi.org/10.1086/466512>. The same wave appears today as a preferred separation between galaxies.

as classical density perturbations on every scale.³

In this picture, every feature of the cosmic microwave background, every galaxy, every cluster of galaxies, every supercluster, traces back to *quantum fluctuations of the vacuum*. The largest structures in the universe are the amplified ghosts of the smallest possible disturbances.

We saw in Chapter 1 that the vacuum is not empty, that it is full of sub-threshold quantum activity. Inflation is the cosmological proof that this sub-threshold activity is *real*: it left its fingerprints all over the sky.

This is the first time in this book — it will not be the last — that a phenomenon at the smallest scale produces, through wave dynamics and amplification, the structure visible at the largest. Hold this picture. We will meet it again at the Glarus thrust, in the cardiac field, and at the threshold of conscious feeling.

2.4 The geometry: a wave in three (or eleven) dimensions

The universe, on the largest scales we can probe, is *flat* to within about 0.4%.⁴ This is one of the central observational findings of modern cosmology: the geometry of space, averaged over billions of light-years, is the Euclidean geometry of a flat sheet.

This is unexpected on naïve grounds — there is no obvious reason the universe should have started flat — and is one of the things inflation was constructed to explain. Whatever curvature the very early universe had was stretched out, like the curvature of a balloon stretched to the size of

³Alan H. Guth, *The Inflationary Universe* (Reading, MA: Addison-Wesley, 1997). The original technical paper is Alan H. Guth, “Inflationary universe: A possible solution to the horizon and flatness problems,” *Physical Review D* 23, no. 2 (1981): 347–56.

⁴Planck Collaboration, “Planck 2018 results. VI. Cosmological parameters,” *Astronomy & Astrophysics* 641 (2020): A6, <https://doi.org/10.1051/0004-6361/201833910>.

a stadium.

But “flat” here means *three-dimensionally flat*. The full picture, in the M-theoretic framework which is the subject of Chapter 15, is that the universe is *eleven-dimensional*, with seven of those dimensions wrapped up — *compactified* — at scales of roughly the Planck length, 10^{-35} m, far smaller than anything any current experiment can probe. The shape of that seven-dimensional internal space is a *Calabi–Yau manifold* (in the string-theory version) or a G_2 manifold (in the M-theoretic version we will use).

I am introducing this here, very briefly and without proof, because the visual that closes this book — the Mandelbulb — is its three-dimensional projection. Every chapter that follows is, in a strict sense, a story playing out on the surface of an eleven-dimensional object we cannot see. The acoustic ringing of the CMB, the spiral arms of galaxies, the convection cells of the Sun, the Glarus thrust, the cardiac toroid, the soma field — all of them are waves on the surface of this same underlying geometry.

Figure 2.3 (*BUILD*) — A two-dimensional cross-section through a Calabi–Yau manifold, rendered as a slice. *The full manifold is six-dimensional and cannot be drawn; this is a projection. Source: to be generated by the author from a standard parameterisation.*

2.5 What the wave was made of

The plasma that rang as a sound wave was made of:

Component	Fraction of the early universe
Photons (light)	~15%
Protons + electrons (ordinary matter)	~5%
Dark matter	~25%
Dark energy	~55% (but only dominant later)

Component	Fraction of the early universe
Neutrinos	~5%

The largest component, dark energy, did not participate in the acoustic oscillations — it acts as a smooth background, only making itself felt at much later times by accelerating the expansion of the universe. Dark matter participated gravitationally but not through pressure (it is collisionless). The ordinary matter and the photons were the part of the plasma that was actually ringing.

We do not know, as of 2026, what dark matter is. We do not know, as of 2026, what dark energy is. Together they make up 95% of the energy of the universe, and they are the two largest open problems in physics. I will not pretend, in this chapter, that they are solved. The Soma Field model does not depend on their being solved. The acoustic peaks ring whether or not we have identified every constituent of the medium.

2.6 The next zoom

The CMB image is the earliest picture we can take. Everything that has happened since — the formation of the first stars, the assembly of galaxies, the chemical enrichment that produced planets, the emergence of life — is the gravitational and thermodynamic working-out of the acoustic pattern visible in that image.

In the next chapter we zoom from this all-sky view down to one galaxy. The wave equation does not change. The fluid does. We are now looking at a fluid of stars instead of a fluid of protons. The wavelengths are shorter; the timescales are longer; the music is the same.

Your own example.

The next time you are in a quiet room, listen for the lowest

sound in the room — the hum of a refrigerator, the rumble of a distant road. Whatever it is, it has a fundamental frequency and harmonics, and you can hear both. The early universe sang exactly the same kind of note, with the same kind of harmonics, just very loudly and very long ago. The picture on the next page is what that sound looks like when you write it down.

Chapter 3 — Galaxies as Spiral Waves

A galaxy is a fluid of stars. The fluid has waves in it. The waves are what you photograph and call arms.

3.1 The disk and the bulge

A spiral galaxy, viewed face-on, has a bright central bulge and a thin surrounding disk. The disk contains most of the galaxy’s gas, most of its ongoing star formation, and the two to four luminous arms that wrap around the centre in a logarithmic spiral. The Milky Way is one such galaxy, viewed (necessarily) from inside; M51, the Whirlpool, and M101, the Pinwheel, are the textbook cases viewed from outside.

Figure 3.1 (*PUBLIC*) — M101, the Pinwheel Galaxy, in visible light. The four major arms are clearly traced by hot young stars and ionised hydrogen regions. *Credit: NASA/ESA/STScI; public domain.*

The arms are the visible part of the galaxy. They are not, however, what they appear to be at first glance.

3.2 The naive picture, and why it fails

The naive expectation, on seeing a spiral arm, is that the arm is a *structure* — a permanent assembly of stars and gas that wraps around the galaxy and rotates with it. This picture has one fatal problem: the inner parts of a galactic disk rotate faster than the outer parts (this is *differential rotation*, set by the gravitational potential). Any material spiral pattern that started reasonably open would wind up, within a few rotation periods, into a tight scroll indistinguishable from a uniform disk. The Milky Way has rotated some sixty times since the Sun formed. There are no scrolls. There are still arms. So the arms cannot be material.

3.3 The density-wave picture

The currently accepted explanation, due originally to Chia-Chiao Lin and Frank Shu in 1964, is that the spiral arms are *density waves*: regions of slightly higher gas density that rotate around the galactic centre at a fixed *pattern speed*, much slower than the stars and gas that pass through them.⁵

The mental picture is a traffic jam on a circular motorway. Cars enter the jam, slow down, pack closer together, and eventually exit. The jam itself rotates around the motorway at a speed quite different from the speed of any individual car. The jam is a pattern; the cars are the material that briefly participates in the pattern and then leaves.

In a galaxy, the cars are stars and clouds of gas. The jam is the spiral arm. As gas enters an arm, it is compressed by the density enhancement, and the compression triggers star formation. This is why arms are bright: not because they contain more matter, but because they contain *young*, hot, luminous stars whose lifetimes are too short for them to have drifted

⁵C. C. Lin and Frank H. Shu, “On the Spiral Structure of Disk Galaxies,” *Astrophysical Journal* 140 (1964): 646–55, <https://doi.org/10.1086/147955>.

out of the arm where they were born. The arm is a moving *site of star formation*, not a moving assembly of stars.

Figure 3.2 (*BUILD*) — Schematic of a galactic density wave. Gas streamlines (blue) flow through the slower-moving pattern of higher density (orange). Stars form in the compression zone on the trailing edge of each arm. *To be generated by the author.*

3.4 The wave equation, again

The density wave in a galactic disk is a solution of a wave equation — specifically a linearised Euler equation for a self-gravitating thin disk, with pressure replaced by velocity dispersion. The same equation that governs ripples on a pond governs spiral arms, with the only substantive change being the inclusion of a gravitational restoring force. This is one of the central charms of fluid dynamics: the equations are scale-invariant in form, and the substitution of a galaxy for a pond is a substitution of constants, not of structure.

The pattern speed of the Milky Way’s spirals is about 25 km/s/kpc; the stars near the Sun orbit at about 220 km/s at a radius of 8 kpc; we therefore overtake the local spiral arm roughly once every quarter of a galactic year, or about 60 million years.

3.5 Beyond density waves

The density-wave picture, in pure form, fits the *grand-design* spirals (M51, M81, M101) very cleanly. For more flocculent galaxies, where the arms are patchy and broken, a complementary process — *self-propagating star formation* — appears to dominate. In this picture, a star-forming region triggers more star formation in adjacent gas, the pattern propagates outward in a roughly spiral track due to differential

rotation, and the result is an arm built by a chain reaction rather than by a coherent global wave.

The honest current view is that both mechanisms operate, that the relative contribution varies from galaxy to galaxy, and that the quantitative theory of spiral structure remains an active research area sixty years after Lin and Shu's paper. What does not vary, in any of the contending pictures, is that the arms are *patterns*, not *things*. They are made of waves, in a fluid of stars.

3.6 Two structural points to carry forward

First: *the visible structure of a galaxy is a wave pattern in a fluid*. This is the second time in this book we are making this claim (the first was the acoustic peaks in the CMB). It will not be the last.

Second: *the pattern moves at one speed; the matter moves at another*. The wave and the medium are distinct, with distinct kinematics. Hold this. When we get to the soma field, we will need exactly this distinction: the *feeling-wave* moves through the body at one speed; the bodily matter is, locally, almost stationary. The wave is not the medium.

Aside. *The astronomer Vera Rubin showed, in the 1970s, that the rotation curves of galaxies — the speed at which stars orbit, as a function of radius — are *flat*: outer stars move at the same speed as inner stars, in flat contradiction to what the visible mass would predict. This was the discovery that established dark matter as a quantitative necessity. Every galaxy in the photographs of this chapter is embedded in a roughly spherical halo of dark matter several times more massive than the visible disk. The arms are waves in the visible fluid; the disk itself sits in a deeper, invisible well.*

Chapter 4 — Stars Ring Like Bells

The Sun is a ten-million-tonne bell with a million simultaneous notes struck on it. We can read every one of them.

4.1 The interior we cannot see

A star is opaque. Photons released in the core of the Sun take, on the diffusive random walk through the radiative zone, roughly 10^5 years to reach the surface. The interior is therefore inaccessible to direct optical observation. For most of the history of astronomy, what was known about the inside of the Sun came from theory — hydrostatic equilibrium, the equations of state, the nuclear cross-sections — and not from observation.

This changed in 1962, when Robert Leighton, Robert Noyes, and George Simon discovered that the surface of the Sun was oscillating, in a patchwork pattern, with a dominant period of about five minutes.⁶ What looked at first like a single resonance turned out, on better data, to be

⁶R. B. Leighton, R. W. Noyes, and G. W. Simon, “Velocity Fields in the Solar Atmosphere. I. Preliminary Report,” *Astrophysical Journal* 135 (1962): 474–99, <https://doi.org/10.1086/147285>.

a superposition of millions of modes, each one a standing sound wave trapped inside the solar interior. The discipline that grew up to interpret these oscillations is called *helioseismology*. It is to the Sun what seismology is to the Earth, and it has — over the last six decades — given us a quantitative map of the solar interior accurate to fractions of a per cent.

Figure 4.1 (*PUBLIC*) — A spherical-harmonic mode of the Sun, of low degree, rendered as a deformation. *Credit: NASA / GONG project; public domain.*

4.2 Modes, by analogy with a violin string

A violin string, fixed at both ends and plucked, sings in a fundamental plus a series of harmonics. The pitches are determined by the length, tension, and mass per unit length of the string. The plucked string contains all the harmonics at once; the ear separates them.

A spherical bell, struck, sings in a series of *spherical-harmonic* modes. These are the analogues of the violin’s harmonics, generalised to a two-sphere. Each mode is labelled by two integers — the *degree* ℓ (how many lines of nodes circle the bell) and the *order* m (how those nodes are oriented). The frequency of each mode is determined by the geometry, density, and elasticity of the bell. Two bells with the same mode spectrum are essentially the same bell.

A star is a *three-dimensional* bell, and so its modes are labelled by three integers: ℓ , m , and n (the *radial order*, how many nodes the mode has between the centre and the surface). The Sun has detectable modes with ℓ ranging from 0 to over 1,000 and n from 0 to about 40. Each mode has a frequency that depends on the temperature, density, and composition along the path it sweeps through the interior. By measuring the frequencies — which the helioseismic instruments do with parts-per-million precision — we recover the structure of the interior the modes

pass through.

This is, in honesty, an extraordinary piece of inverse mathematics. We read the inside of a star from the way it rings.

4.3 What we have learned

Helioseismology has, among other things:

- Confirmed the standard solar model to within $\sim 0.5\%$ in the sound speed throughout most of the interior.
- Measured the depth of the convective zone (the outer 28.7% of the Sun’s radius, by mass about 2%).
- Measured the internal rotation profile, finding that the radiative interior rotates approximately rigidly while the convective envelope has the differential rotation visible at the surface (faster at the equator than at the poles).
- Located the *tachocline* — the thin shear layer at the boundary between rigid and differential rotation — and identified it as the probable seat of the solar dynamo that generates the eleven-year sunspot cycle.

The same technique, applied to other stars, is now called *asteroseismology* and has been extended by the Kepler and TESS space missions to stars across most of the Hertzsprung–Russell diagram.

4.4 The eleven-year wave

The Sun’s most visible long-period oscillation is the *sunspot cycle*, with an average period of 11.0 years and considerable variability around that average (the Maunder Minimum, 1645–1715, was a 70-year suppression of the cycle, of which the cause is still debated). The sunspot cycle is driven by the solar dynamo — the cyclic generation, twisting, and

destruction of magnetic fields in the convective zone — and its qualitative shape is well captured by *Babcock–Leighton dynamo* models, which themselves are nonlinear wave equations on the solar surface.

Figure 4.2 (*PUBLIC*) — The Maunder butterfly diagram: latitudes of sunspots plotted against time, since 1875. The cyclic pattern is obvious. *Credit: NASA Marshall Space Flight Center; public domain.*

The sunspots themselves are the visible bruises of the magnetic field breaching the photosphere. They are the *output*, not the *cause*; the cause is a magnetic wave propagating through the convective zone with period 11 years.

4.5 Variable stars: the spectrum from regular to chaotic

Some stars are intrinsically variable. The classical examples — Cepheid variables, RR Lyrae variables, Mira variables — pulsate radially with remarkably stable periods, ranging from hours to hundreds of days. These pulsations are nearly pure standing acoustic waves, and the period-luminosity relation discovered for Cepheids by Henrietta Swan Leavitt in 1912 is one of the foundational rungs of the distance ladder.⁷

Other stars are *chaotic* variables — semiregular giants, irregular variables, certain pre-main-sequence stars whose pulsations have no clean period at all. The full spectrum from clean periodicity to deterministic chaos is present in the stellar sky, and the underlying equations (nonlinear wave equations in a self-gravitating compressible fluid) admit all of these behaviours as solutions in different parameter regimes.

⁷Henrietta S. Leavitt and Edward C. Pickering, “Periods of 25 Variable Stars in the Small Magellanic Cloud,” *Harvard College Observatory Circular* 173 (1912): 1–3.

This is the second time in the book we have met the spectrum from regular to chaotic in a single underlying system; we will meet it again in the cardiac field, where the same spectrum is the difference between a healthy heart and one in fibrillation.

4.6 What carries forward

A star is, structurally, a *resonant cavity*. It contains a fluid; the fluid supports waves; the waves stand in modes determined by the cavity; and the modes ring at frequencies we can measure.

This is a useful template to keep in front of you for the rest of Part I. The Earth is a resonant cavity (Chapter 5 and Chapter 6). The atmosphere is a resonant cavity. The body cavities are resonant cavities. The nervous system, on the soma-field interpretation, is a resonant cavity of a particular kind. All of these systems can be analysed by the same technique: find the modes, measure the frequencies, read the structure from the spectrum.

Chapter 4b — Black Holes: The Wave Sinks

*A short chapter on the most extreme wave system the universe knows how to make. Black holes appear in this book not for their drama but because they are, in a precise sense, the *sinks* of the wave inventory we have been building up.*

4b.1 What a black hole is

A *black hole* is a region of spacetime in which gravity is so strong that no signal — not even light — can escape its interior. The boundary of the region is the *event horizon*. Inside the horizon, the future of every causal world-line points inward, toward a *singularity* at which classical general relativity breaks down.

The original prediction is due to Karl Schwarzschild, in 1916, who wrote down the first exact solution to Einstein's field equations just months after Einstein published them. Schwarzschild was a serving German officer on the Russian front; he died of an autoimmune disease six months later. His solution describes the external geometry of any non-rotating, non-charged spherical mass distribution and predicts the horizon at the *Schwarzschild radius*

$$r_s = \frac{2GM}{c^2}$$

For the Sun, $r_s \approx 3$ km (vastly smaller than the Sun’s actual radius of 700,000 km — the Sun is not a black hole, but if it were compressed to within 3 km radius it would become one). For the Earth, $r_s \approx 9$ mm. For a person, r_s is on the order of 10^{-25} m — vastly smaller than an atomic nucleus.

4b.2 How they form

Black holes form in our universe by *gravitational collapse* of massive objects past a critical point. The two main routes:

- **Stellar-mass black holes:** produced by the collapse of the iron core of a star with initial mass ≥ 20 solar masses, at the end of its nuclear-burning lifetime. The collapse is followed by a supernova explosion that ejects the star’s outer layers, leaving a black hole of typically 5–30 solar masses at the centre.
- **Supermassive black holes:** present at the centre of essentially every large galaxy, with masses of 10^6 to 10^{10} solar masses. Their formation history is incompletely understood; they may have grown from intermediate-mass seeds by sustained accretion, or may have formed directly from collapsing primordial gas clouds in the early universe.

4b.3 Wave aspects

A black hole is, paradoxically, a richly wave-active object. Three wave aspects matter for our story:

Gravitational waves. When two black holes orbit each other and spiral inward, they emit gravitational waves — ripples in the spacetime

metric itself. The waves carry energy away from the binary, shrinking the orbit, until the two black holes merge. The merger event is the most powerful gravitational-wave source in the known universe — the September 2015 event GW150914 (the first direct detection by LIGO) released ~ 3 solar masses of energy in gravitational waves in a fraction of a second, peaking at a power of $\sim 3.6 \times 10^{49}$ watts — momentarily more luminous, in gravitational radiation, than the rest of the observable universe in all electromagnetic radiation combined.

Figure 4b.1 (*PUBLIC*) — The GW150914 waveform as observed at LIGO Hanford and LIGO Livingston, both detectors, both polarisations. The characteristic *chirp* — increasing frequency, increasing amplitude — is the signature of the inspiral. *Credit: LIGO Collaboration; public domain.*

Quasi-normal modes. After a black-hole merger, the resulting single black hole *rings down* via a discrete set of decaying oscillation modes — its *quasi-normal modes*. These are the gravitational analogue of the acoustic modes of a struck bell: they are determined by the black hole’s mass and spin alone, and they allow direct observational confirmation of the no-hair theorem (black holes are characterised by mass, charge, and spin only — no other “hair”). Observations of post-merger ringdowns are one of the strongest current tests of general relativity.

Hawking radiation. Stephen Hawking’s 1974 calculation showed that a black hole is *not perfectly black* — it emits thermal radiation from just outside its horizon, with a temperature

$$T_{\text{H}} = \frac{\hbar c^3}{8\pi G M k_B}$$

inversely proportional to the mass. For a stellar-mass black hole, $T_{\text{H}} \sim 10^{-8}$ K — far colder than the CMB, so the black hole *gains* mass on net by absorbing CMB photons faster than it radiates. For a primordial

black hole of $\sim 10^{15}$ g ($\sim$ the mass of a mountain), T_{H} matches the CMB and the black hole evaporates over a time comparable to the age of the universe. For smaller primordial black holes, evaporation is already complete.

4b.4 The information paradox

Hawking radiation, being thermal, appears to carry no information about the matter that fell into the black hole. If a book and a laptop of equal mass fall into the same black hole, the radiation that eventually evaporates the hole is indistinguishable. The information that was in the book and the laptop appears to be lost.

This is in direct conflict with quantum mechanics, which is *unitary* — information cannot be destroyed. The conflict has driven fifty years of theoretical work and is one of the most active research areas in fundamental physics. The current consensus is that the information is *not* lost; it is encoded in subtle correlations in the Hawking radiation that a sufficiently powerful observer could in principle reconstruct. The technical demonstration of this — the *Page curve* result — was one of the major breakthroughs of the late 2010s and early 2020s.

The relevance for our story: the resolution of the information paradox is connected to the *holographic principle* mentioned in Chapter 15b — the idea that all the information about a region of space is encoded on the region’s boundary. Black holes are the most extreme illustration of holography: all the information about everything inside the horizon is, on the holographic interpretation, encoded in subtle correlations on the horizon’s two-dimensional surface.

4b.5 The wave inventory completed

This book is, at one level, an inventory of wave phenomena. With this chapter we have completed the cosmic-scale inventory:

- Vacuum fluctuations (Chapter 2).
- Acoustic waves in plasma (Chapter 2, CMB).
- Density waves in fluids (Chapter 3, galaxies).
- Oscillation modes in stars (Chapter 4, helioseismology).
- Atmospheric and oceanic waves (Chapter 5, planets).
- Seismic waves in rock (Chapter 6, Earth).
- Pattern-forming chemical waves (Chapter 7, life).
- Branching fractals in biology (Chapter 8).
- Cardiac and neural waves (Chapter 9).
- Tensegrity standing waves in the body (Chapter 10).
- Soma-field waves (Chapter 11).
- Quantum waves (Chapter 13).
- M-theory G_2 waves (Chapter 15).
- Gravitational waves and quasi-normal modes (this chapter).

Fourteen wave families across forty-something orders of magnitude in scale. All of them obey the same three universal facts of Chapter 1: a wave is a propagating disturbance in a medium; waves superpose; waves carry energy without carrying matter.

The book's central claim — that the body, on the soma-field model, is one more entry in this catalogue, at human scale, with its own characteristic medium (fascia + neural + electromagnetic + quantum substrates) and its own characteristic modes (the eight of Chapter 11) — sits within this inventory, neither at the top nor at the bottom, but exactly where a wave system of body-scale and body-complexity would, on any general principle, be expected to sit.

Chapter 5 — Planets, Weather, and the Skin of the Earth

A planet is a thin film of fluid on a slowly cooling ball of rock. The film does the weather; the rock does the geology; both are wave systems.

5.1 The atmosphere as a fluid sheet

The Earth's atmosphere, to a first approximation, is a layer of gas roughly 100 km thick wrapped over a sphere 12,742 km in diameter. The aspect ratio — thickness divided by radius — is less than 1%. On any diagram drawn to scale, the atmosphere is a film not a layer.

A thin film of fluid on a rotating sphere, heated unequally (most at the equator, least at the poles) and subject to Coriolis force, has a very specific set of preferred modes. These modes are the *Hadley cells*, the *Ferrel cells*, the *polar cells* of the general circulation; the *Rossby waves* that meander through the jet stream; the *Madden-Julian oscillation*; the *El Niño-Southern Oscillation*. Each is a long-lived wave or wave

pattern on the fluid sheet. The names are different for historical reasons; the underlying mathematics is one nonlinear wave equation, the *Navier–Stokes equation on a rotating sphere*, with thermal forcing.⁸

Figure 5.1 (*PUBLIC*) — A global Rossby-wave pattern in the northern jet stream, satellite view, with the meanders highlighted. *Credit: NOAA; public domain.*

The lesson, repeated from the previous chapter: structure that *looks* like a stable object — a high-pressure ridge, a cyclone, a hurricane — is a wave on a fluid. The fluid moves through the pattern.

5.2 Lenses, again

A lens, as we saw in Chapter 1, takes a field of incoming waves and folds it down to a point. The atmosphere itself acts as a (rather bad) lens, due to its variable refractive index with height; this is why stars twinkle, why mirages form, why the Sun appears squashed at the horizon. Astronomers spend much of their effort correcting for these atmospheric distortions; adaptive optics, the technique by which modern ground-based telescopes rival space telescopes, is in essence a fast real-time inversion of the atmospheric lens.

The relevance for this book is that you do not need to leave the surface of the Earth to encounter the wave-folding mathematics that will return in Chapter 15 when we talk about the compactification of M-theory. The atmosphere is doing it, badly, every clear evening at every horizon you ever look at.

⁸James R. Holton and Gregory J. Hakim, *An Introduction to Dynamic Meteorology*, 5th ed. (Waltham, MA: Academic Press, 2012). The standard graduate textbook.

5.3 The ocean: another fluid sheet, slower and denser

The oceans are a second fluid sheet, thicker (average depth 3.7 km) and denser (factor of 800) than the atmosphere, wrapped over the same spherical rock. The same equations govern them, with the boundary condition that water has a free surface against the atmosphere above and a rigid bottom against the sea floor below.

The ocean supports a fully analogous spectrum of waves: surface gravity waves (the ones you see at the beach, periods of seconds), internal waves (in the thermocline, periods of minutes to hours), tides (astronomically forced, period 12.42 hours), inertial oscillations (Coriolis-forced, period varying with latitude), Rossby and Kelvin waves (long-period, planetary scale, key to ENSO). The ocean also supports *meddies* and *eddies* — coherent vortex structures that travel thousands of kilometres while preserving their identity, behaving in essence as solitons.

Figure 5.2 (*PUBLIC*) — A satellite altimetry map of sea-surface height showing the dense eddy field of the Gulf Stream. The eddies are 100–200 km across and persist for months. *Credit: NASA / JPL; public domain.*

A tsunami is a single very-long-wavelength surface gravity wave excited by a seafloor displacement, typically from a subduction-zone earthquake. The 2011 Tōhoku tsunami had a wavelength of about 200 km in the open ocean and propagated at the same speed as a jet airliner.

5.4 The solid Earth as a resonant cavity

In Chapter 4 we treated the Sun as a resonant cavity that rings at its natural frequencies. The Earth, struck by an earthquake, does the same. The longest-period mode is the football mode, ${}_0S_2$, with a period of 53.9

minutes; the spectrum extends down to periods of milliseconds at the high end.

We need this fact in Chapter 6, where the Glarus thrust is one particular ten-million-year wave in the same elastic body. The shorter modes are minutes; the longer ones are tens of millions of years; the underlying equations are the same equations of linear and nonlinear elasticity, applied to the same rock.

5.5 The magnetosphere and the ionosphere

Two further fluid sheets wrap the Earth, both invisible to the naked eye.

The *ionosphere* is the partially ionised upper atmosphere, between 60 and 1,000 km altitude. It reflects HF radio waves and supports its own wave spectrum — gravity waves, travelling ionospheric disturbances, plasma waves. It is the medium in which the aurora is painted; the visible aurora is the optical signature of plasma waves dumping energy into the upper atmosphere along magnetic-field lines.

The *magnetosphere* is the volume of space, extending to about 10 Earth radii on the sunward side and a long tail on the night side, where the Earth's magnetic field dominates over the solar wind. It is a plasma cavity with its own modes — *ULF* (ultra-low-frequency) magnetic pulsations with periods of seconds to minutes, *Alfvén waves* propagating along field lines, magnetospheric storms.

I am rattling through these because the point of this chapter is *not* the detailed physics of each one, but the *accumulation*. By the time we close Chapter 5 you have met seven or eight nested fluid sheets wrapped around the same rock, every one of them a wave-bearing cavity, every one of them analysable by the same equations. The Earth is a wave system from the core out to the magnetopause, and it is *one* wave system, with the boundary conditions changing as you move through it.

5.6 The Schumann resonance: the Earth’s heartbeat

Between the conducting surface of the Earth and the conducting ionosphere there is a spherical-shell waveguide about 100 km thick, 12,742 km in diameter. Like any cavity, it has resonant modes. The fundamental mode, called the *Schumann resonance* after Winfried Otto Schumann who predicted it in 1952, has a frequency of 7.83 Hz.

This is the lowest-frequency electromagnetic mode the Earth-atmosphere system supports. It is continuously excited by lightning strokes worldwide — roughly fifty per second on average — and a sensitive ELF receiver can pick it up reliably anywhere on the planet.

Figure 5.3 (*BUILD*) — A typical Schumann-resonance power spectrum, showing the first six modes between 7.8 and 45 Hz. *To be generated by the author from publicly available data.*

I mention the Schumann resonance for two reasons. First, it is the cleanest available example of *the whole planet ringing at one note* — the planetary analogue of the open low E on a guitar. Second, it sits on the edge of two arguments we will not pursue in this book but ought to acknowledge. There is a substantial literature claiming a coupling between Schumann resonance amplitude and human EEG bands (which overlap in the same 8 Hz range), and an even larger literature making stronger claims about it. The honest position, as of 2026, is that the coincidence of frequencies is real, the claim of biological coupling is suggestive but not established, and the claim of *causal* coupling remains speculative. We will neither rely on nor dismiss it.

5.7 What carries forward

The closing structural point of Part I, anticipated here and made explicit in Chapter 6, is this: *the Earth is a layered system of wave-bearing fluids and solids, each layer with its own modes, each layer coupled to its neighbours, the whole stack readable by the same mathematics from the core to the magnetopause.* The next zoom is the last one in Part I — into the rocks themselves, at the Glarus thrust, where the same logic applies to objects that look, to the human eye, exactly like the opposite of waves.

Plates I — The Cosmos

Plates I

The Cosmos

Eight images, one universe.

Plate I.1 (*PUBLIC, full-bleed recto*) — The *Planck* 2018 all-sky CMB temperature map in equal-area projection. Resolution: $\sim 5'$. Temperature scale: $\pm 200 \mu\text{K}$ around $T = 2.725 \text{ K}$. Every visible feature is the frozen amplitude of a sound wave that propagated in the photon-baryon plasma 380,000 years after the Big Bang. The cold spot near the bottom-right of the map remains the most debated single feature. *Credit: ESA / Planck Collaboration.*

This is the oldest light. Everything else came after.

Plate I.2 (*PUBLIC*) — The CMB acoustic angular power spectrum, with the first seven peaks resolved. The horizontal axis is the multipole ℓ ; the vertical axis is the temperature variance $D_\ell = \ell(\ell + 1)C_\ell/2\pi$. The first peak at $\ell \approx 220$ corresponds to a sound wave that completed exactly one half-cycle of compression by the moment of decoupling — the fundamental note of the universe. *Credit: ESA / Planck.*

The score of the first symphony.

Plate I.3 (*PUBLIC, double-page*) — Hubble Ultra-Deep Field 2014. Ten thousand galaxies in a patch of sky the apparent size of a grain of sand held at arm's length. Most of the visible objects are galaxies; the few prominent foreground stars carry diffraction spikes. Redshifts of the most distant objects exceed $z = 11$, corresponding to galaxies that formed within the first 500 million years after the Big Bang. *Credit: NASA / ESA / STScI.*

Plate I.4 (*PUBLIC*) — M51, the Whirlpool Galaxy, in HST visible light. The two grand-design spiral arms, the bright HII regions tracing recent star formation along the leading edges, and the companion galaxy NGC 5195 at upper-right — currently passing through the disk and exciting the spiral pattern. *Credit: NASA / ESA / STScI.*

A galaxy is a fluid of stars; the arms are waves in the fluid.

Plate I.5 (*PUBLIC*) — A solar granulation image from the Daniel K. Inouye Solar Telescope. Each cell is a convection plume roughly the size of France, rising in the centre and sinking at the dark inter-cellular lanes. The cells turn over on a timescale of minutes. *Credit: NSO / NSF / AURA.*

The surface of a star is a boiling pot.

Plate I.6 (*PUBLIC*) — Solar coronal loops in extreme-UV (171 Å), Solar Dynamics Observatory. The arches trace magnetic-field lines bridging regions of opposite polarity. The brightness is from million-degree plasma confined along the field lines. *Credit: NASA / SDO.*

Plate I.7 (*PUBLIC*) — Jupiter's atmosphere, Juno close approach. Cyclonic and anti-cyclonic vortices nested into the cloud bands; the bands themselves are zonal jets reaching velocities of 100 m/s relative to the underlying solid (or in Jupiter's case, fluid) rotation. *Credit: NASA / JPL / SwRI / MSSS.*

Atmospheres are wave systems too.

Plate I.8 (*PUBLIC*) — Earth from Apollo 17, the “Blue Marble”. The thin blue shell of atmosphere at the limb is the entire weather system. Aspect ratio: about 1%. *Credit: NASA.*

The one we know best, and least. Part II begins on the next page.

Plates II — The Earth

Plates II

The Earth

Eight images, one slow wave.

Plate II.1 (*DRONE — full-bleed*) — Aerial view of the Tschingelhörner ridge from above the Sernftal, looking south. The Glarus thrust appears as a dark colour-change line cutting across the entire ridge horizontally. The Verrucano is the paler rock above; the flysch is the darker rock below. *Drone capture: A. Johnson, summer 2026.* Coordinates of vantage: $\sim 46.918^{\circ}\text{N}$, 9.171°E , altitude 2700 m AGL.

Plate II.2 (*DRONE — full-bleed, recto*) — The Klöntalersee from the north shore, looking south at the Glärnisch massif. The thrust line is visible mid-cliff, running left-to-right. The road on the south shore from which the photograph in Chapter 6 was conceived is just out of frame at the bottom. *Drone capture: A. Johnson, summer 2026.*

Plate II.3 (*DRONE*) — The Martinsloch, the natural window through the thrust ridge above Elm. Twice a year — around 12 March and 30 September — the rising sun shines through the window onto the church square in Elm village. The geometry is set by the strike of the thrust plane. *Drone capture: A. Johnson, summer 2026.*

A hole in the wall, calibrated to the Earth's orbit.

Plate II.4 (*DRONE — detail*) — Close approach to the Lochsite type locality, near Schwanden. The hand-sample-scale view of the thrust: older Permian Verrucano (top) directly on younger Cretaceous-Tertiary flysch (bottom), with the millimetres-thin mylonite layer between them. The displacement across this paper-thin contact is on the order of 35 km. *Drone capture: A. Johnson, summer 2026.* Coordinates: 47.0167°N, 9.0617°E.

The wave reduced to a single line, frozen in stone.

Plate II.5 (*PUBLIC*) — Recumbent folds in the Helvetic nappes, Säntis area, aerial view. The bedding planes have been rotated to near-horizontal by the same compression that produced the Glarus thrust. Several generations of folding are visible at different scales — the geometry that we will meet again in M-theory as a *fold catastrophe*. *Credit: swisstopo, CC BY 3.0 CH.*

Plate II.6 (*PUBLIC*) — Plate-tectonics map of the world, with arrows showing relative motion of major plates. Africa moves north at ~ 2 cm/yr; the Indian plate moves north-northeast at ~ 4 cm/yr (the fastest of the major plates); the Pacific plate moves northwest at ~ 7 cm/yr at its leading edge near Japan. *Credit: USGS; public domain.*

The slow sea, mapped.

Plate II.7 (*PUBLIC*) — Seismic-wave particle motion diagrams: P, S, Rayleigh, Love. Each panel shows a snapshot of how the medium particles move as the wave passes through. *Credit: USGS; public domain.*

Four ways for rock to ring.

Plate II.8 (*PUBLIC*) — The auroral oval over the Arctic, captured by ESA's Swarm constellation. The visible glow is the optical signature of plasma waves dumping energy into the upper atmosphere along magnetic-field lines. *Credit: ESA / Swarm.*

The magnetosphere announcing itself.

Chapter 6 — The Ground Is a Slow Sea

*Glarus, the canton in eastern Switzerland whose mountains
are made of an argument with chronology.*

6.0 A note on what we are doing here

The branch of geology we will use throughout this chapter is *structural geology* — the study of how rock has been bent, folded, broken, and displaced. It is the geology that you can read with your eyes from a walking path, the geology that yields the most photogenic pictures, and — crucially for the argument of this book — the geology in which the mathematics of *folding* shows up in exactly the form we will need again in Chapter 15 for M-theory.

There is one further feature of structural geology that I want to flag in advance: it is honestly a *four-dimensional* science. A geologist looking at a three-dimensional outcrop is reading *time* from a spatial slice. Older rock below, younger rock above; a tilted bed records a rotation that took ten million years; a folded layer records a compression that took thirty. Three space dimensions plus one time dimension, all simultaneously visible on a cliff face. The eye learns to see this. Once it does, every rock

face is a film, and the geologist is reading the frames.

This is the structure we will use again, in a different form, in the M-theory chapter: a three-dimensional spatial slice through a higher-dimensional object, revealing structure (time, in the geological case; the seven internal dimensions, in the M-theoretic case) that was not directly visible.

6.1 Standing on the wave

There is a road that runs along the north side of the Klöntalersee, between the lake and the foot of the Glärnisch massif. If you walk along it on a clear morning and look up and across the water, your eye finds, about halfway up the south wall, a thin near-horizontal line where the rock changes colour. Below the line the rock is grey-brown, often weeping with seeps and stained by lichen. Above the line the rock is paler — a clean, slightly pinkish grey — and goes up almost vertically to the summit ridges.

The line is the **Glarus Hauptüberschiebung**. The principal overthrust. It is one of the type localities for the entire science of tectonics, the place where, in the second half of the nineteenth century, geologists were forced to accept that mountains move. UNESCO designated it a World Heritage Site in 2008, jointly with two adjacent localities (the Tschingelhörner and the Lochsite), under the title *Swiss Tectonic Arena Sardona*. The citation reads, in part, that the site offers “an exceptional example of mountain building through continental collision and displays excellent geological sections through the resulting nappes.”⁹

What that sentence does in technical language, the cliff above the lake does without explanation: it shows older rock sitting on top of younger rock, and the visible contact between them.

⁹UNESCO World Heritage Centre, *Swiss Tectonic Arena Sardona*, inscribed 2008. <https://whc.unesco.org/en/list/1179/>

Figure 6.1 (DRONE) — The Tschingelhörner ridge from above the Elm valley. The Glarus thrust appears as the colour-change line running across the cliff face. *Note: this image to be captured by the author by drone, summer 2026; until then, the placeholder is a swisstopo aerial view (CC, see Credits).*

Figure 6.2 (DRONE) — The Klöntalersee, looking south, with the thrust line visible on the Glärnisch face. *Drone capture planned.*

Figure 6.3 (DRONE) — The Martinsloch — the famous natural window through the thrust ridge, through which the sun shines onto the Elm village square twice a year. *Drone capture planned.*

6.2 What you are actually looking at

The rock above the line is **Permian Verrucano**. It was laid down about 260 to 290 million years ago, in the long red-sandstone interval between the assembly of the supercontinent Pangaea and its breakup.¹⁰

The rock below the line is **Cretaceous and Tertiary flysch and limestone** — sediments laid down between roughly 100 and 35 million years ago, in the seas that were closing as Africa pressed slowly northward toward Europe.

The Verrucano sits on top of the flysch.

This should not happen.

The first geologists to map this region in detail — Arnold Escher von

¹⁰O. Adrian Pfiffner, *Geology of the Alps*, 2nd ed. (Hoboken, NJ: Wiley-Blackwell, 2014), 224–32. Pfiffner’s textbook is the standard reference for Alpine tectonics in English and is the source for the ages and stratigraphic identifications in this chapter.

der Linth in the 1840s, and Albert Heim in the 1870s — could not at first bring themselves to publish what they had found.¹¹ The accepted view, at the time, was that older rock lay beneath younger rock, full stop. The alternative — that the older Verrucano had been physically *pushed*, as a coherent sheet, over the younger flysch — required a horizontal displacement of tens of kilometres along a single near-flat contact surface, with the upper sheet moving without disintegrating. The mechanics seemed impossible. Rock was thought to be brittle. Brittle things, at scale, do not flow.

Escher first proposed the *Doppelfalte* — the “double fold” — as a desperate alternative: maybe the rock had folded so spectacularly that it had doubled back on itself, putting older on younger as part of a single overturned structure. Heim defended this for decades. But the field evidence kept failing it. In 1884 the French geologist Marcel Bertrand, working from a map he had been sent of the region, suggested in a single short paper that the only honest reading was that the upper sheet had moved — that it was, in modern terminology, a *nappe* — and that the contact surface itself was an overthrust fault, with the rock above having travelled a great distance from the south.¹²

By 1903 Heim had accepted the overthrust interpretation. By 1910 it had become the prevailing model for the whole eastern Alps. By the mid-twentieth century, plate tectonics had given it the global context it needed: the Verrucano had ridden north on the Adriatic plate as Africa collided with Europe; the contact surface was the *décollement* — a French word that means, almost untranslatableably, “unsticking” — along which a sheet of crust had slipped against the rock beneath it; the rock that oiled the slip was a thin layer of *Lochsite limestone* that had been ground to a fine, mechanically weak mylonite by the motion of the sheet

¹¹Albert Heim, *Untersuchungen über den Mechanismus der Gebirgsbildung* (Basel: Schwabe, 1878).

¹²Marcel Bertrand, “Rapports de structure des Alpes de Glaris et du bassin houiller du Nord,” *Bulletin de la Société Géologique de France*, 3rd ser., 12 (1884): 318–30.

above it.

The displacement, on the best modern estimates, is between 35 and 50 kilometres to the north-northwest.¹³

6.3 Rock as a wave

Here is the part of the story that matters for the argument of this book.

A sheet of Permian Verrucano five kilometres thick does not flow because rock, at the scale of a hand sample, is liquid. It is not. Rock is brittle. If you take a piece of Verrucano in your hand and hit it with a hammer, it shatters. Yet the same Verrucano, in a sheet thirty kilometres long, riding for fifteen million years on a lubricating layer of mylonite, behaves as a fluid.

The behaviour depends on **scale and time**.

There is a single dimensionless number in geophysics that captures this, called the *Deborah number*, after the verse in the Book of Judges that reads “the mountains flowed before the Lord.”¹⁴ It is defined as the ratio of the relaxation time of a material to the timescale of the deformation:

$$\text{De} = \frac{\tau_{\text{relaxation}}}{\tau_{\text{deformation}}}.$$

When De is large — when the relaxation time of the material is long compared to the deformation — the material behaves elastically: it stores the deformation and springs back. When De is small — when the deformation is so slow that the material has time to relax — it flows.

Rock has an extraordinarily long relaxation time. But “deformation timescale” in the Alps is measured in millions of years. The ratio comes

¹³Pfiffner, *Geology of the Alps*, 230.

¹⁴Markus Reiner, “The Deborah Number,” *Physics Today* 17, no. 1 (January 1964): 62. Reiner coined the name and the formal definition; the verse is Judges 5:5 in the Authorised Version.

out small. On geological timescales, rock flows.

When a substance flows in response to a sustained force, what propagates through it is a wave. The Glarus thrust is therefore — not metaphorically, literally — a very long-period, very large-amplitude wave in the European crust. Its period is on the order of ten million years. Its wavelength is on the order of a hundred kilometres. Its amplitude is the thirty-five-kilometre displacement of one sheet of rock relative to the sheet beneath it.

You cannot, in a human lifetime, see this wave move. The wavefront travels at roughly two millimetres per year — too slow for any direct perception. But you can see where the wave *was*. The line on the south face of the Klöntalersee is the leading edge of the wave, frozen at the moment the wave ran out of energy.

Figure 6.4 — Schematic stratigraphic cross-section through the Glarus thrust, after Pfiffner (2014). The Verrucano sheet (orange) sits on the thin grey Lochsite mylonite (the “grease layer”), which sits on the Cretaceous and Tertiary flysch (blue). Vertical exaggeration $\times 2$.

6.4 Plate tectonics — the engine that pushes the slow sea

The Glarus thrust is a local consequence of a global fact. The outer shell of the Earth — the crust and the cold upper mantle, together called the *lithosphere* — is broken into a small number of large plates and a larger number of smaller ones, and these plates are in continuous slow motion relative to each other. The motion is powered, ultimately, by heat escaping from the Earth’s interior: hot mantle rock rises beneath mid-ocean ridges, cools as it spreads outward at the surface, sinks back into the deep mantle at subduction zones, and is reheated. The cycle

has a period of about 100 to 250 million years.¹⁵

This is a convection cell, of the same kind that drives the swirl in a heated saucepan, scaled up by a factor of a billion.

Figure 6.5 — World plate-tectonics map. The seven major and many minor plates of the Earth’s lithosphere, with the directions and rates of relative motion. Source: USGS.

Figure 6.6 — Three-dimensional cross-section of mantle convection beneath a subduction zone. The descending slab cools the surrounding mantle; the rising plume returns heat to the surface. Source: USGS / Robert Simmon.

The European Alps — and the Glarus thrust in particular — are one product of the long, ongoing collision of the African plate with the Eurasian plate. Africa has been moving roughly north at a few centimetres per year for the last hundred million years; Europe is being slowly buckled in front of it. The Alps are the buckle. They are still rising — the elevation of the central Alps is increasing by roughly one millimetre per year, even now, as the underlying convergence continues.¹⁶ A geologist a few million years from now will find the Glarus thrust higher than it is today; a geologist a hundred million years from now may not find it at all, because erosion will have outpaced the uplift and the whole structure will be sediment in some Mediterranean basin.

Everything is in motion. Nothing is in a hurry.

¹⁵Donald L. Turcotte and Gerald Schubert, *Geodynamics*, 3rd ed. (Cambridge: Cambridge University Press, 2014). The reference textbook for the quantitative theory of mantle convection.

¹⁶Pfiffner, *Geology of the Alps*, 367–70.

6.5 Seismic waves: the audible part of the story

If the Glarus thrust is the very-low-frequency end of the spectrum of crustal waves — a single wave with a ten-million-year period — the *audible* part of the spectrum is the seismic waves released by earthquakes. These have periods from fractions of a second to about an hour, and they come in four flavours.

Wave type	Particle motion	Speed in crust
P (pressure)	Push–pull along direction of propagation	5–8 km/s
S (shear)	Side-to-side, perpendicular to propagation	3–5 km/s
Rayleigh	Surface roll, elliptical motion in vertical plane	~3 km/s
Love	Surface, horizontal side-to-side	~3 km/s

P-waves are sound waves, in essence — they are how the inside of a mountain “shouts” when a fault slips. S-waves are slower; they cannot propagate through liquid, which is how we know that the Earth’s outer core is liquid (it casts a *shadow* for S-waves on the far side of the planet from an earthquake). Rayleigh and Love waves are the slow, destructive surface rolls that arrive last and do the most damage in a quake.

Figure 6.7 — Particle motion in P, S, Rayleigh, and Love waves. Source: USGS.

The deepest natural sound the Earth makes is its set of *free oscillations*

— normal modes of the whole planet, excited by very large earthquakes. The deepest mode, called ${}_0S_2$ or the “football mode” (because the Earth deforms into a slightly prolate-oblate shape and back), has a period of about 53.9 minutes. The 1960 Chile earthquake — the largest ever recorded — rang this mode loudly enough that it was still detectable in seismograms weeks later. We will meet planetary normal modes again in Chapter 5; the point here is that the Earth, when struck hard enough, rings like a bell.¹⁷

6.6 Folds: the same mathematics we will need again

Before we leave Glarus I want to put one further idea in your eye. The Glarus thrust is, as we have seen, a *thrust* — a near-flat displacement. But the rock around it is also extensively *folded*. The Verrucano sheet above the thrust contains recumbent folds with limbs tens of kilometres long; the Helvetic nappes to the west of Glarus are nothing but folds, stacked on folds, on folds, on folds, all the way to the Aar massif.

Figure 6.7b (*PUBLIC*) — Recumbent folds of the Helvetic nappes, Säntis area, eastern Switzerland. The bedding is older toward the centre of each fold and the entire stack has been rotated nearly horizontal. *Source: swisstopo aerial; CC BY 3.0 CH.*

A fold, mathematically, is what you get when you take a flat surface and compress it from the sides faster than it can flow out of the way. The surface buckles. If the compression continues, the buckle tightens; if it continues further still, the buckle overturns; if it continues longer still, the buckle breaks and slides — and you have a thrust fault. The fold is the wave; the thrust is the broken wave. The same one-dimensional

¹⁷Keiiti Aki and Paul G. Richards, *Quantitative Seismology*, 2nd ed. (Sausalito, CA: University Science Books, 2002), ch. 8.

mathematics — Euler buckling, then post-buckling finite-amplitude theory — covers all of it.

The reason I am flagging this here is that *exactly the same kind of folding* shows up in the geometry that underlies the M-theory chapter at the end of this book. There, the surface that is folded is not a sheet of rock but a seven-dimensional internal space — a so-called G_2 manifold — and the folds are not centimetres or kilometres but Planck lengths across. But the language is the same. Geologists and M-theorists use the same words: *fold, limb, singularity at the hinge, flat lying, overturned*. The diagrams in a structural-geology textbook and the diagrams in a G_2 -manifold paper, with the axes relabelled, are sometimes indistinguishable.

This is not a coincidence. Folding is what extended objects do when they are compressed. It happens in rock at the scale of mountains because it is the cheapest available response of a sheet under load. It happens in the internal geometry of M-theory for the same reason, mathematically translated. The Soma Field, in turn, lives on the projection of the M-theoretic geometry into our visible three dimensions, and it inherits the folding. When we talk in Chapter 12 about *attractor basins* in the soma field — about the way grief can sit, deep and stable, in a fold the rest of the field has flowed around — we are talking about the same mathematical object, scaled down to a single human nervous system.

For now: stand on the road at the Klöntalersee and look at the fold.

6.7 Why this is the hinge of Part I

I want to be clear, before we leave Glarus, about why this chapter sits where it does in the book.

The argument up to here — Chapters 1 through 5 — has been that wave behaviour appears at every scale we examine, from the cosmic microwave background down through galaxies, stars, and planets. The reader can accept this for the cosmic plasma (it was visibly hot, visibly turbulent,

visibly fluid); for the gas in galaxies (it is a fluid); for the plasma in stars (a fluid); and for atmospheres (fluids again). What may resist this picture is *solid rock*. Solid rock looks, to the human eye, like the opposite of a fluid. Solid rock is what you hit when you fall.

Glarus is the place in the book where I ask you to accept that solid rock, at scale, is also a fluid. If you can come with me on that, the rest of the book — which will ask you to accept the same of bones, fascia, and finally of feeling itself — is no longer asking you to do anything unfamiliar. The principle is the same principle. Only the substrate changes.

What makes Glarus a particularly clean teaching case is that the deformation has been *recorded*. The thrust line is itself the seismogram. Five kilometres of rock, displaced thirty-five kilometres, preserved as a single hairline contact on the south wall of a Swiss lake. You can stand on the road by the water and read the wave with your eyes.

Figure 6.8 (*DRONE — detail*) — Close approach to the Lochsite exposure: older rock visibly above younger rock, with the centimetres-thin mylonite layer between them. *To be captured at the Lochsite type locality, summer 2026. Coordinates: 47.0167°N, 9.0617°E.*

Your own example.

If you have a place in the landscape that gives you the feeling that something very slow is going on under your feet, write its name here. We will come back to it in Chapter 18.

Going there.

The thrust is easily visible from public roads on both sides.

The classical viewing localities are: the Tschingelhörner ridge from the Sernftal (Elm village, accessible by Postbus from Schwanden); the Lochsite type locality (a roadside outcrop near Schwanden, signposted); and the south face of the Klöntalersee, visible from the lake-shore road. The UNESCO arena visitor centre is in Elm. Best light for photography is morning on the Elm side and late afternoon on the Klöntal side.

Chapter 7 — Life: The Wave Becomes a Pattern

A leopard's spots are a chemical wave, frozen.

7.1 A short walk in the woods

The next time you walk through a deciduous wood, look at three things.

Look at the *branching* of a tree, at any oak or beech you pass: the trunk splits into two or three major boughs; each bough splits again into limbs; each limb into twigs; each twig into leaf-bearing stems. At every scale the same rule. The branching ratio, on average, is about 2.

Look at the *vein pattern* of a single leaf you pick up off the path: the same branching, scaled down by a factor of a hundred. Midrib to secondary veins to tertiary veins to a fine reticulate mesh.

Look at the *outline of a single oak leaf*: the lobes are themselves lobed, the indentations themselves indented, two or three times before the resolution of the unaided eye runs out.

You have, in the same three minutes of looking, found three fractals on the same plant, at three different scales, all generated by the same underlying programme. The programme is biological — encoded in genes,

expressed through chemical signalling — but the *output* is geometric, and the geometry is the same geometry that physicists meet in turbulence, in lightning, in river networks. Branching is what extended growth *does* in the presence of a transport requirement and a finite supply.

This chapter is about how the wave physics of Part I becomes the patterned biology of Part II.

7.2 Turing patterns

The mathematician Alan Turing, in 1952, published a single paper — “The Chemical Basis of Morphogenesis” — that has dominated the mathematical biology of pattern formation ever since.¹⁸ He showed that a system of two reacting chemicals, one activator and one inhibitor, diffusing through a tissue at different rates, can spontaneously break the uniformity of the tissue and produce a stable pattern of spots, stripes, or labyrinths. The patterns are determined by the relative diffusion rates and the kinetic constants of the reaction.

These *Turing patterns* are the leading explanation for:

- the spots of a leopard;
- the stripes of a zebra and of an angelfish;
- the spacing of hair follicles;
- the ridges of fingerprints;
- the ridges on the roof of your mouth;
- the regular spacing of teeth in a jaw;
- the alternation of digits on a hand.

Each of these is, mathematically, a *frozen wave* in a chemical concentration field. The pattern formed when the embryo was a few weeks old; the cells later differentiated according to whether the chemical concentration

¹⁸Alan M. Turing, “The Chemical Basis of Morphogenesis,” *Philosophical Transactions of the Royal Society of London B* 237, no. 641 (1952): 37–72, <https://doi.org/10.1098/rstb.1952.0012>.

at their location was above or below a threshold; and the resulting pattern of pigment, hair, ridge, or tooth is what you see for the rest of the organism's life.

The threshold is the same kind of object as the perception threshold of the soma field in Chapter 11. Above the threshold: dark fur, ridge, tooth. Below: pale fur, valley, gap. The pattern itself is the wave; the visible biology is the wave through a threshold filter. Hold this correspondence. It will become explicit in Chapter 11.

Figure 7.1 (*BUILD*) — A pair of side-by-side images: left, a simulated Turing pattern from a standard reaction-diffusion model (Gray–Scott); right, a photograph of an angelfish with the same pattern. *Left to be generated by the author; right from Wikimedia, CC-licensed.*

7.3 The branching imperative

Anything that transports a fluid through a volume and has to deliver that fluid to every point in the volume faces a single geometric problem: maximise surface area in contact with the volume, while minimising the total length of the transport network.

The mathematical solution is *branching*. Specifically, a network with the branching ratio and branch-length-ratio that satisfy *Murray's law* (for vessels carrying viscous fluid) or the slightly different West–Brown–Enquist scaling (for biological networks more generally).¹⁹ The solution is fractal. It has to be, in the mathematical sense: only a fractal network can deliver fluid to every point in a volume while keeping the total transport-network length finite.

¹⁹Cecil D. Murray, “The Physiological Principle of Minimum Work: I. The Vascular System and the Cost of Blood Volume,” *Proceedings of the National Academy of Sciences* 12, no. 3 (1926): 207–14. The extension to biological allometry is in Geoffrey B. West, James H. Brown, and Brian J. Enquist, “A General Model for the Origin of Allometric Scaling Laws in Biology,” *Science* 276, no. 5309 (1997): 122–26.

This is why your lungs branch (23 generations from trachea to alveoli), why your blood vessels branch (about 30 generations from aorta to capillary), why your bile ducts branch, why your kidneys branch, why your nervous system branches. It is also why a tree branches above ground (to deliver sugar and water to every leaf) and why its roots branch below ground (to harvest water from every cubic centimetre of soil). It is why a river basin branches. It is why lightning branches.

The geometry is *forced*. Given the constraint, no other geometry works.

7.4 From wave to organism

The pattern of how a fertilised egg becomes a complete animal is, when described mathematically, a sequence of:

1. *Symmetry-breaking waves* (Turing-type) that establish the basic body axes (anterior–posterior, dorsal–ventral, left–right).
2. *Branching cascades* that build the transport networks (vascular, pulmonary, neural).
3. *Mechanical feedback* (tissue tension shapes which cells divide where), which is itself a wave system in the elastic mechanics of the developing tissue.
4. *Threshold filtering* (cells differentiate based on local chemical concentrations, sensed against thresholds), which converts the continuous wave fields into discrete biological identities.

This is a wildly compressed précis of one of the most active research fields in biology. The point I want to leave you with is this: every step of it is a *wave process*. The biology — proteins, cells, tissues — is the *substrate* on which the waves run. The recognisable shapes of adult organisms are the *standing-wave solutions* the substrate admits.

This is the bridge from Part I to Part II of the book. From here on, when you see a body — a leaf, a fish, a hand, a heart — you can read it as the

frozen interference pattern of a development that was fundamentally a wave process.

7.5 A practical exercise

Find a fern frond. Look at its overall outline. Look at one of the pinnae (the major divisions). Look at one of the pinnules on that pinna. Look at the tothing on the edge of the pinnule.

You will see, in many ferns, the same shape four times, at four scales, separated by a factor of about 3 each time. This is not a coincidence and it is not stylisation. It is what the wave equation that built the fern produces, when you let it run for the number of iterations the genome specifies.

Your own example.

Pick up one leaf on your next walk. Hold it against the light. Trace one vein with your fingertip, from midrib to tip. Count the branchings. The same number occurs in your lung. The same equation built both.

Chapter 8 — Trees, Rivers, Lungs

Three pictures, one geometry.

8.1 Three photographs, side by side

If you place, on the same page, an aerial photograph of a river delta, a botanical illustration of a deciduous tree in winter, and a cast of the human bronchial tree, the three images are *almost indistinguishable*. The branching ratios match. The angle between parent and daughter branch is the same to within a few degrees. The ratio of daughter-to-parent diameter is the same to within a few per cent. If you stripped the colour and the scale bar from each image, a non-specialist could not tell which is which.

Figure 8.1 (*PUBLIC + ORIGINAL*) — Triptych: the Lena Delta in Siberia (NASA Landsat); an oak in winter (original photograph, Zurich 2026); a resin cast of human bronchial tree (Wikimedia, CC). *All three at the same printed scale, with metric scale bars below each.*

This chapter is a slow look at why.

8.2 Why three different things make the same shape

The river network, the tree, and the lung all solve the same problem: distribute or collect a fluid over a volume, with minimum total network length, against a transport constraint. The constraint is slightly different in each case — gravity and erosion for the river, water transport against capillarity for the tree, air against viscous resistance for the lung — but the *form* of the constraint is the same in all three cases (a fluid must move through a network with cost proportional to length and resistance dependent on diameter), and so the optimum solution is the same.

The optimum is fractal branching with branching ratio approximately 2, diameter ratio approximately $2^{-1/3} \approx 0.79$, and length ratio approximately $2^{-1/3}$. This is *Murray's law* for vessels carrying viscous fluid, and it is followed remarkably closely by all three of our example systems, give or take the modifications imposed by local constraints (a river cannot flow uphill; a lung must fit inside a ribcage; a tree must remain mechanically stable in wind).

8.3 The lung in particular

The human lung has, on average:

- 23 generations of branching from trachea to terminal alveolus;
- approximately 3×10^8 alveoli;
- total alveolar surface area of about 70 m^2 (roughly half a tennis court);
- packed into a volume of about 5 litres.

The ratio $70 \text{ m}^2 / 5 \text{ L}$ is a *surface-to-volume ratio*, and the value it reaches — about $14\,000 \text{ m}^{-1}$ — is the central reason a human can extract enough oxygen from air. A spherical blob of tissue 5 L in volume has a surface area of about 0.13 m^2 . The fractal branching buys you a factor of

500 in surface area at constant volume. Without the fractal, terrestrial respiration as we know it would be impossible.

This is also why severe lung disease — emphysema, in particular — destroys the fractal. Emphysema is, mechanically, the *unfolding* of the lung: alveolar walls break down, the fine structure is replaced by larger cavities, the fractal dimension drops, the surface area collapses, and the patient can no longer breathe even though the total lung volume may be unchanged or larger. The disease is the loss of fractal dimension. The cure, if any cure is to be found, must restore it.

Figure 8.2 (*PUBLIC*) — Histology of healthy alveoli vs. emphysematous alveoli at the same magnification. *Credit: NIH Open-i archive; public domain.*

8.4 Wave-physics interlude: the lung as resonant cavity

The lung is, in addition to being a fractal, a *cavity*. Air moves through it; the cavity has a fundamental resonance that depends on its volume and the elasticity of its walls (this is *forced oscillation technique*, used clinically to measure airway impedance); and the walls themselves have natural frequencies, audible as crackles and wheezes in disease, that a stethoscope is essentially listening for.

The breath cycle itself is a slow wave with period 4–6 seconds at rest, 1–2 seconds in heavy exercise. This wave is coupled, in a way we will examine in detail in the next chapter, to the heart-rate cycle. The two together produce the *respiratory sinus arrhythmia* — the regular slowing and speeding of the heart with each in-breath and out-breath — which is the most directly perceivable example of cross-system wave coupling in the body, and a standard measure of autonomic-nervous-system health.

8.5 The river: the same picture, drawn by erosion

A river network is a *self-organised* fractal. There is no genome specifying it; there is no developmental programme. Water flows downhill, carries sediment, and erodes the channel. Tributaries cut into the landscape; tributaries of tributaries cut into the tributaries; the process iterates over millions of years; the result is a fractal network with statistical properties remarkably close to those of the lung.

The mathematical theory is largely due to Robert Horton (1945) and Arthur Strahler (1952), with deep extensions by Andrea Rinaldo and Ignacio Rodríguez-Iturbe in the 1990s.²⁰ Strahler's *stream-order* scheme classifies channels by their position in the hierarchy, and the empirical regularity that Horton found — that the number of streams of each order forms a geometric progression — is the hydrological fingerprint of the fractal.

I include the river network here, alongside the lung and the tree, because it makes the point that *the geometry is geometry-first, not biology-first*. The river network is not alive; it is solving the same geometric problem, by erosion rather than by growth, and arriving at the same answer.

8.6 Tree time

A tree, finally, is also a *standing wave in time*. The growth of a tree's rings is a record of the climate it lived through: wide rings in good years, narrow rings in droughts; the wood preserves a year-by-year log of the temperature, rainfall, fire, and insect attack of the landscape around it. *Dendrochronology* — the science of reading tree rings — has constructed continuous climate records going back ten thousand years, by

²⁰Ignacio Rodríguez-Iturbe and Andrea Rinaldo, *Fractal River Basins: Chance and Self-Organization* (Cambridge: Cambridge University Press, 1997).

overlapping rings from successively older living and dead trees.

The tree, in other words, is a record of the same long-period atmospheric waves we met in Chapter 5 — the El Niño cycle, the Atlantic Multi-decadal Oscillation, the longer Holocene climate trends — written in cellulose. The fractal that distributes water to its leaves also encodes, year by year, the larger-scale wave history of the atmosphere it grew in.

Your own example.

Cut a cross-section from a downed branch (please, only from one already on the ground) and count the rings. You are looking at a temperature record. The wider the band, the warmer or wetter that year. You can read decades off a single piece of firewood.

Chapter 8b — Cities and Brains: Scaling Laws

A short chapter on a subject that is its own scientific cottage industry but appears here for one reason: the universal scaling laws that govern the metabolic life of organisms also govern the metabolic life of cities, and the deep reason for both is fractal transport networks of exactly the kind we have been meeting.

8b.1 Kleiber's law

In 1932 the Swiss-American biologist Max Kleiber observed that the basal metabolic rate B of a mammal — the rate at which it consumes energy at rest — scales with its body mass M as

$$B \propto M^{3/4}.$$

This is *Kleiber's law*. It holds across roughly twenty-seven orders of magnitude in mass, from bacteria to blue whales. A mouse has a metabolic rate $\sim 7,000$ times slower than a blue whale despite being ~ 30 million times lighter — the exponent is $3/4$, not 1.

The $3/4$ exponent — rather than the naive $2/3$ (surface-to-volume) or 1 (pure volumetric) — was unexplained for sixty years.

8b.2 West, Brown, and Enquist (1997)

In 1997, the physicist Geoffrey West and the biologists James Brown and Brian Enquist published a derivation of the $3/4$ exponent from three premises:

1. The transport network supplying nutrients to the body's cells is *fractal* and *space-filling* — i.e., it branches at every scale and reaches every cell.
2. The capillaries at the terminal branches are *size-invariant* — a capillary in a mouse and a capillary in a whale are the same size.
3. The total network minimises energy dissipation under the constraint of supplying every cell.

Under these premises, the network's total cross-sectional area at a given branching level must be related to the body's metabolic demand in a specific way, and the $3/4$ exponent falls out as a consequence. The argument is mathematically beautiful and one of the landmark applications of physics to biology of the late 20th century.

8b.3 Other organism-level scaling laws

The same framework predicts other allometric exponents:

- Heart rate: $f \propto M^{-1/4}$. A mouse's heart beats ~ 600 bpm; a blue whale's ~ 6 bpm.
- Lifespan: $L \propto M^{1/4}$. A mouse lives ~ 3 years; a whale ~ 80 years.
- Total heartbeats in a lifetime: $L f M^{1/4} = \text{constant}$. *Every mammal gets about 10^9 heartbeats in a lifetime, regardless of size.*

This last is one of the most arresting facts in comparative biology. A mouse experiences its heartbeats fast, a whale slow, but each gets approximately the same total. The arrow of subjective time, on this picture, is set by the *count*, not the clock.

8b.4 Cities

In 2007, Luís Bettencourt and collaborators showed that *cities* obey analogous scaling laws, with notably *different* exponents. Two regimes:

- *Material infrastructure* (length of road network, number of petrol stations, number of electrical sub-stations) scales as $\propto N^\beta$ with $\beta \approx 0.85$ — sublinear. Doubling a city's population requires less than double the infrastructure.
- *Socioeconomic output* (patents per year, wages, crime rates, number of restaurants) scales as $\propto N^\beta$ with $\beta \approx 1.15$ — superlinear. Doubling a city's population produces *more than* double the output.

The sublinear scaling of infrastructure has the same origin as Kleiber's law: the city's transport network (water, sewage, power, transit) is fractal and space-filling, with size-invariant terminal branches (houses, individual customers).

The superlinear scaling of output is a *new* phenomenon, attributed to the increased opportunities for interaction in a larger network — the more people in a city, the more pairs of people can interact, and the more *unexpected* pairs can occur. This is the *urban-as-network* effect.

8b.5 Brains

The brain is, in this framework, the densest known instance of the network-as-organism architecture. The human brain contains $\sim 86 \times 10^9$

neurons; each neuron contacts on the order of 10^3 – 10^4 others; the total number of synaptic connections is $\sim 10^{14}$. The cortical surface area scales with body mass with an exponent $\sim 3/4$, in agreement with the broader allometric pattern.

But the brain shows the same *two-regime* behaviour as cities: sublinear infrastructure (white matter volume, vasculature) and superlinear *output* (cognitive capacity, behavioural complexity). The brain is, in a precise sense, *a city of neurons*.

8b.6 Relevance for the soma-field argument

Two reasons this chapter is here:

First: the soma-field model treats the body as a network with fractal transport (fascia, vasculature, nervous system, lymphatic). The $3/4$ scaling law predicts that the *energy demand* of the soma field — the rate at which the field must be sustained by metabolic input — should scale with body mass with exponent $\sim 3/4$. This is a checkable prediction.

Second: the superlinear scaling of cognitive output with neural network size is, on the soma-field model, an instance of the *coupling network* effect. The eight modes interact with each other through the E_8 coupling structure; the more modes that can be simultaneously above threshold, the more transitions between modes are possible per unit time. The richness of a person's emotional life is, on this argument, superlinear in the *baseline* activity of the soma field — a healthy soma field with high baseline activity will spontaneously generate more emotional states per unit time than a suppressed one.

This is what therapy *does*, on the soma-field interpretation: it raises the baseline activity of the soma field, increasing the superlinear rate of state generation, and so increases the expressiveness and adaptability of the person.

Chapter 9 — The Cardiac Field

The heart is the loudest electromagnetic event the body makes. It is also the slowest, the most patient, and the most easily heard.

9.1 Two pumps and an orchestra

The textbook description of the heart is *a four-chambered pump that moves blood through two parallel circulations*. This is correct. It is also, like most textbook descriptions, the least interesting true thing you can say about the organ.

The more interesting thing is that the heart is a *self-exciting oscillator*. It generates its own electrical wave, on its own schedule, out of a small patch of specialised cells called the sinoatrial node; the wave propagates across the atria, pauses briefly at the atrioventricular node, then sweeps down the ventricles through the His–Purkinje system; the mechanical contraction follows the electrical wave by about 50 milliseconds.

The whole sequence repeats roughly once a second, every second, for about three billion beats over a typical human life.

Figure 9.1 (*PUBLIC*) — A standard 12-lead ECG, with the P wave, QRS complex, and T wave labelled. *Credit: NIH; public domain.*

9.2 The heart as a wave-propagation problem

The electrical signal that runs across the heart is, mathematically, a *non-linear travelling wave* — specifically a solution of the reaction-diffusion equations of cardiac electrophysiology (the Beeler–Reuter, Luo–Rudy, or modern O’Hara–Rudy models). It is the same *class* of equation we met for Turing patterns in Chapter 7, but here the relevant solutions are *travelling fronts* rather than stationary patterns.

In a healthy heart, the wave propagates as a clean, synchronised front, with the geometry of the cardiac muscle ensuring that the contraction sequence pumps blood efficiently in one direction.

In an unhealthy heart, the wave can break. *Atrial fibrillation* is a state in which the atria support many small, rapidly rotating spiral waves rather than a single clean front; the chambers quiver instead of contracting. *Ventricular fibrillation*, the immediate cause of most sudden cardiac death, is the same pathology in the ventricles, and is fatal within minutes without defibrillation.

A defibrillator works by applying a strong electric shock that *resets* the entire myocardium to a single uniform state, from which the natural pacemaker can restart a clean front. It is, in wave language, a *re-initialisation* of the wave system.

9.3 The electromagnetic field around the heart

The heart’s electrical activity is large enough to produce a *measurable electromagnetic field* outside the body. The electrocardiogram measures

this field with electrodes on the skin; the magnetocardiogram measures the much weaker magnetic component with SQUID magnetometers, typically inside a magnetically shielded room.

The field has a *toroidal* geometry — it loops around the long axis of the body, with the heart at the centre of the torus — and its amplitude, at one metre from the body, is on the order of 10^{-12} tesla, or about a millionth of the Earth’s static magnetic field. It is small. It is, however, *coherent* — the heartbeat is a single synchronised event — in a way that the brain’s electrical activity (which is larger in raw amplitude but spectrally diffuse) is not. The cardiac field, in standard EM measurements, is the loudest *coherent* signal the body produces.²¹

Figure 9.2 (BUILD) — The cardiac electromagnetic field rendered as a torus centred on the chest, with isofield lines.
To be generated by the author from the dipole approximation.

9.4 Heart-rate variability

The interval between heartbeats is *not constant*. It varies, on multiple timescales, in patterns that turn out to be diagnostic of autonomic-nervous-system state.

- *Beat-to-beat* variation is dominated by *respiratory sinus arrhythmia* — the heart speeds up on inhalation and slows down on exhalation. This coupling is mediated by the vagus nerve, and its amplitude is one of the cleanest available measures of vagal tone.

²¹The HeartMath Institute has published, since the 1990s, the most extensive series of measurements of the cardiac electromagnetic field, including the claim of detectable field-mediated synchronisation between adjacent humans. The technical measurements (field amplitude, coherence, heart-rate-variability spectra) are well-established; the more far-reaching biological interpretations are contested. See Rollin McCraty, *Science of the Heart, Volume 2: Exploring the Role of the Heart in Human Performance* (Boulder Creek, CA: HeartMath Institute, 2015).

- *Short-term* variation (timescale of seconds) reflects the baroreflex — the feedback loop that regulates blood pressure.
- *Long-term* variation (timescale of minutes to hours) reflects hormonal regulation, circadian rhythm, and slow emotional state.

A spectral analysis of heart-rate variability produces a power spectrum with distinct peaks at the respiratory frequency, at the baroreflex frequency (around 0.1 Hz), and at lower frequencies. The shape of this spectrum is a clinical signature; flattened HRV spectra are characteristic of major depression, of post-traumatic stress disorder, and of imminent cardiac mortality.

This matters for the soma-field argument of this book because the heart is the most accessible window onto the *coupling* between bodily and emotional state. A panic attack produces a measurable HRV signature; so does loving-kindness meditation; so does grief; so does sleep. The cardiac field is, in operational terms, the most easily measured projection of the eight-dimensional soma field we will define in Chapter 11.

9.5 Coupling between hearts

Two human bodies in close physical proximity can show, under certain conditions, *cardiac entrainment* — the heart rates and HRV spectra drift toward partial synchronisation. The effect is strongest in mother–infant pairs, in long-married couples, and in pairs in deep conversation; it is weakened by physical separation or by emotional distance.

The mechanism by which this happens is not fully understood. The candidate channels are: (1) auditory — each hears the other’s breathing and partly entrains to it; (2) visual — micro-expressions relay state; (3) tactile — direct contact relays both temperature and pulse; (4) electromagnetic — the cardiac field of one body is, at contact distance, detectable by the autonomic sensors of the other, though the experimental evidence here is contested. Almost certainly all four contribute, in

varying proportions.

The honest current view is that the *phenomenon* is well-attested and the *mechanism* is partially understood. The soma-field model, with its explicitly field-based ontology, naturally accommodates field-mediated coupling without requiring it as a necessary mechanism.

9.6 What carries forward

Three things to keep in your hands.

First: the heart is a *wave-generating organ* whose output is a clean, measurable electromagnetic and acoustic and pressure wave at about 1 Hz.

Second: the *shape* of this wave — its variability, its coupling to breath, its coherence — is a high-bandwidth signal about autonomic-nervous-system state. It is, today, in the clinic, the most direct read-out of soma-field state we can take with widely available instrumentation.

Third: hearts couple to each other. The exact channels are debated; the phenomenon is not. When we make the case in Part III that the soma field is a *real field*, with the capacity for inter-body coupling that real fields have, the cardiac signal is the most accessible piece of existing evidence we can point at.

Your own example.

Put a finger gently on the carotid artery in your neck. Breathe in slowly for four counts; out for six. After thirty seconds, you will likely feel the rate of the pulse rise on the inhale and fall on the exhale, sharply enough to be unmistakable. You have just measured your own vagal tone with your hands. The amplitude of that swing is one of the simplest available numbers about your nervous system, and it is trainable.

Chapter 10 —

Tensegrity, Fascia, and the Body as a Standing Wave

A body is not a stack of bones. It is a single continuous tension wave with bones suspended in it.

10.1 A sculpture you can hold in your hand

The word *tensegrity* is a contraction of *tensional integrity*. It was coined by Buckminster Fuller in the 1960s to describe a class of structures invented by his student Kenneth Snelson: assemblies of rigid struts that do not touch each other, held in shape by a network of continuous tensioned cables.

The defining property of a tensegrity structure is that *no compression element is in continuous contact with another compression element*. The struts float, suspended in the tension network. The whole structure is in equilibrium because every cable is under tension and every strut is

under compression, in just the right balance.

Figure 10.1 (*PUBLIC*) — A Snelson three-strut tensegrity sculpture. *Source: Wikimedia, CC BY-SA.*

A tensegrity sculpture, lifted off a table, is rigid; tapped, it rings; distorted, it springs back to shape; cut in one cable, it collapses entirely.

A human body has every one of these properties. This is not a coincidence.

10.2 The biotensegrity hypothesis

The orthopaedic surgeon Stephen Levin proposed, in the 1980s, that the human body is structurally a tensegrity system: the bones are the struts, the fascia and ligaments are the tension network, and the characteristic mechanical behaviour of the body — its springiness, its shock-absorption, its capacity to bear loads in any orientation — is the behaviour of a tensegrity, not the behaviour of a stack of bricks held up by gravity.²²

The orthodox biomechanical view, which Levin’s proposal challenges, is that the body is a *column* — the spine is a stack of vertebrae held in compressive equilibrium by the discs and the surrounding muscles, analogous to a stack of coins. On this view, the loads at the base of the spine in any reasonably loaded posture should crush the lumbar vertebrae. They do not. The biotensegrity view explains why: the *loads are not transmitted through the spine as compression*; they are transmitted through the surrounding fascial-myofascial-ligamentous network as distributed tension. The bones float in the tension; the tension distributes the load over a much larger cross-section than the bones alone could bear.

²²Stephen M. Levin, “The Tensegrity-Truss as a Model for Spine Mechanics: Biotensegrity,” *Journal of Mechanics in Medicine and Biology* 2, no. 3 (2002): 375–88. Levin’s website <https://www.biotensegrity.com> collects the relevant clinical and anatomical material.

This view remains a contested hypothesis in mainstream biomechanics. It is widely accepted in the manual-therapy literature (osteopathy, Rolfing, Feldenkrais), partly accepted in the sports-medicine literature, and treated with caution in the academic biomechanics literature. As of 2026, the *measurements* most relevant to it (in vivo fascial loading, force transmission across fascial planes) are beginning to be made and broadly support the qualitative picture.

I am laying out the controversy openly because this book takes biotensegrity *as a working framework* for the body — it is consistent with everything else here, it provides the necessary substrate for the fascia-mediated couplings the soma field requires, and it produces a better picture of the body than any alternative I am aware of — but I want it on the record that this is not yet textbook consensus.

10.3 Fascia: the continuous tension network

The *fascia* is the continuous connective-tissue network that wraps every muscle, every organ, every bone, every nerve, and every blood vessel in the body. It is, on the biotensegrity view, the single most important mechanical structure in the body: more important than any individual muscle, because the muscles transmit force *through* the fascia, not in isolation.

The anatomist Jean-Claude Guimberteau spent his career filming the fascia *in vivo*, by inserting a small endoscope under the skin during surgery and recording the moving tissue. His films are among the strangest and most beautiful biological documents of the last fifty years: the fascia is shown as a fluid-elastic foam, in continuous motion, with no clear divisions, no clear layers, no clear *parts* — a single continuous tensioned medium that fills the entire body and supports every other organ within it.²³

²³Jean-Claude Guimberteau, *Strolling under the Skin* (film, 2005), and *The Architecture of Human Living Fascia* (Edinburgh: Handspring, 2015). The

Figure 10.2 (*PERMISSION*) — A still from Guimberteau’s intra-fascial endoscopy showing the foam-like microstructure of living fascia. *Permission to be sought from Endovivo Productions; alternative: schematic redrawing.*

The mathematics of such a medium is the mathematics of a *visco-elastic continuum*. Stress propagates through it as a wave, with a speed that depends on the tension. *The body conducts mechanical signals as wave trains in the fascia.* This is one of the cleanest senses in which the body is a “wave system” in the meaning of this book: every motion of every part of you produces tension changes that propagate, as waves, through the fascia to every other part.

10.4 Standing waves in the body

A tensioned continuous medium has standing-wave modes. The body, as a tensegrity-fascial system, therefore has standing-wave modes — sets of characteristic frequencies and spatial patterns at which the whole system likes to oscillate.

You can hear some of them. The voice is a standing wave in the vocal tract, supported by the breath, and is the highest-frequency mode of the body most people use daily. The walking gait is a standing wave at about 2 Hz; running is at 3–4 Hz; the gravitational ring-up of the spinal column when you land from a jump is in the tens of Hz.

You can feel some of them. The held tension of a chronic shoulder contraction is a *static* mode — a wave that has settled into a stationary pattern and refuses to relax. *Releasing* the shoulder, in manual-therapy terms, is the active de-excitation of this mode.

The soma field, in Chapter 11, lives on this substrate. The emotional-

films are the indispensable primary source; written summaries do not convey them.

mode patterns are *not separate* from the fascial-tension patterns; they are coupled to them. A grief that has been held for twenty years has, almost without exception, a fascial signature — a characteristic chronic tension pattern in the chest, throat, and shoulders — and resolving the grief requires resolving the tension, and vice versa.

Figure 10.3 (*BUILD*) — Cyber-hologram rendering of the body's fascial network, with tension distributed and visualised as luminous filaments. *To be generated by the author.*

10.5 Why this is the right substrate for the soma field

A field needs a medium. In Part I we met the cosmic plasma, the galactic disk, the solar interior, the atmosphere, the crust. Each was a real physical medium with real wave dynamics.

For the soma field — the eight-dimensional field of feeling that is the central object of Part III — the medium is the body. Specifically, it is the *fascial-cardiac-neural* continuum: the fascia carries mechanical waves, the cardiac system carries electromagnetic waves at about 1 Hz, the nervous system carries fast electrical waves up to the kHz range. All three are coupled. All three are spatially continuous. All three support standing modes. The combination is the substrate of the soma field.

The cyber-hologram body of Chapter 11 is, in essence, a *visualisation* of this substrate with the wave activity made visible. It is not fanciful. It is what the body looks like when you draw the wave content in.

Your own example.

Stand. Roll your shoulders back, slowly, three times. Notice the line of release that runs from the back of your neck, across the top of the shoulder, down the outside of the upper arm.

That line is a fascial strand. It was under tension. The roll lowered the tension. The release that propagated down the arm is a wave you just sent through your own fascia. The body is a wave system. You have just operated it.

Plates III — The Body

Plates III

The Body

Eight images, one continuous tension wave.

Plate III.1 (*BUILD — full-bleed, the hero*) — Cyber-hologram rendering of a standing human body. The skin is rendered as a luminous semi-transparent envelope; the fascia as a network of cyan filaments under tension; the cardiac field as a magenta torus centred on the chest; the neural activity as fine gold tracery running the spinal axis and branching into the dendritic trees of the cerebral cortex. The figure is *not* anatomical. It is a visualisation of the wave content of the body as the soma-field model describes it. *To be generated by the author; Stable Diffusion base + manual vector overlay + composite.*

Not a body. A wave system in the shape of a body.

Plate III.2 (*BUILD*) — The eight modes of the soma field, mapped to dominant somatic regions. *Calm* — distributed even luminescence. *Fight* — jaw, shoulders, hands. *Flight* — chest, diaphragm, legs. *Freeze* — gut, lower belly, perineum. *Flow* — belly, throat, hands. *Joy* — face, chest, eyes. *Grief* — heart, throat, occiput. *Hypervigilance* — neck, upper back, scalp. Eight overlay panels, same body silhouette. *Author render.*

The eight notes of the human chord.

Plate III.3 (*BUILD — facing pair*) — Two cyber-hologram bodies, side by side at identical scale. *Left:* the field with all sub-threshold activity rendered faint cyan; full luminosity. *Right:* the same field with the threshold drawn as a dotted iso-surface; only the regions above the threshold rendered. The right-hand image is what consciousness reports. The left-hand image is what the body has. *Author render.*

Plate III.4 (*PERMISSION; placeholder: BUILD*) — A still from Guimberteau’s intra-fascial endoscopy. The living fascia at microscopic scale shows itself as a continuous foam of fibrils, in constant motion, with no discrete layers, no discrete parts. *Source: Endovivo Productions / J.-C. Guimberteau, permission pending; in the interim, a schematic redrawing by the author.*

The inside of the wave-bearing medium.

Plate III.5 (*PUBLIC*) — A six-strut Snelson tensegrity sculpture. Photograph in even light, against a black background, to bring out the geometry. The struts are clearly not touching each other; the entire structure is held in shape by the tension network. *Source: Wikimedia, CC BY-SA.*

The model of the body, in sculpture form.

Plate III.6 (*PUBLIC*) — A bronchial cast of the human lung tree, resin injection technique. The 23 generations of branching visible from trachea to terminal alveolus. *Credit: Wikimedia, CC BY-SA.*

Fractal geometry made of the body.

Plate III.7 (*BUILD*) — The cardiac toroidal electromagnetic field rendered around a torso silhouette. Isofield contours at 10^{-12} , 10^{-13} , 10^{-14} tesla. *Author render from the dipole approximation; calibrated against MCG measurements in the literature.*

The heart, drawn as the field it generates.

Plate III.8 (*BUILD*) — Two heart-rate variability power spectra on the same axes: top panel a healthy subject at rest, with distinct peaks at ~ 0.05 , ~ 0.1 , ~ 0.25 Hz; bottom panel a subject with severe PTSD, the spectrum flattened across the band. *Author render from anonymised public-dataset traces.*

What trauma looks like, in a single number per heartbeat.

Chapter 11 — The Soma Field

The wave is always there. This is not a metaphor.

11.1 A confession to begin with

Everything in this book up to here — the cosmic microwave background, the spiral arms of galaxies, the breathing of stars, the slow wave of the Glarus thrust, the cardiac toroid, the fascia — has been *preparation*. This chapter is the chapter the rest of the book was written to make inevitable.

I should also confess that I am no longer in neutral expository mode. From here to Chapter 16 you are reading a working physicist's account of his own model of his own inner life, and the model is original to him, and the inner life is original to him, and I cannot pretend to a distance I do not have. Where I have so far been writing as a guide pointing at other people's mountains, I am for the next six chapters writing as a man pointing at the inside of his own chest.

The mountain is still there. The pointing is still honest. The pointer is just visible in the frame now.

11.2 The question that started it

In 2018, in a therapist's office in Zurich, I was asked the question every person in psychotherapy is asked, dozens of times a year, for years: *what are you feeling right now?*

For most people, this question has a navigable answer. They consult something — they aren't always sure what — and a word comes back: *sad, angry, anxious, hopeful, tired*. The word may not be the right word, and the word may not be the whole word, but there is a word, and that is enough to begin with.

For me, the question landed differently. The honest answer was almost always some version of: *I cannot tell you. Something is happening. It is definitely happening. It is happening in my body and in my head and it is substantial. But I cannot tell you what it is, where it came from, how big it is, or what it wants.*

I had — and have — the clinical diagnosis of *alexithymia*, which is a clinical word meaning *without words for feeling*. I have it in the context of three larger architectural conditions: Autism Spectrum Condition, Attention Deficit Hyperactivity Disorder, and Complex Post-Traumatic Stress Disorder. Each of those conditions has its own relationship with the felt body; the combination produces a particular phenomenology that the available clinical models did not, to my satisfaction, describe.

What they all had in common was the felt sense that *feeling was not an event*. It was not something that *started* when I noticed it and *stopped* when I stopped noticing. It was always there. What changed was whether, at a given moment, it rose far enough above some internal threshold to become a thing I could put a name on. Most of the time it was below the threshold. The body knew. The naming part didn't.

This is the experience the soma field is a model of.

Figure 11.1 — The Soma Field, rendered as a cyber-

hologram body. The figure is not anatomical. It is a visualisation of an eight-dimensional field of feeling overlaid on the human form. The brighter regions are above the threshold of conscious awareness; the dimmer regions are sub-threshold but active. *Original; render planned in Stable Diffusion + retouch, then composited with vector overlays.*

11.3 Building the model from the experience

The phenomenology I needed to model had four properties.

1. The field is always present. It does not switch on with emotion and off with calm. It is a continuous, distributed thing. *Calm*, in this model, is not the absence of the field; it is a particular shape of the field at low amplitude, with a particular distribution across the body and the nervous system.

2. The field has a threshold for perception. Most of the activity in the field, most of the time, is sub-threshold — it influences behaviour, posture, heart rate, decisions, but it is not consciously felt. When the amplitude in a particular mode exceeds a threshold, the experience makes the jump into nameable consciousness. The threshold is not fixed; it can be raised (alexithymia, dissociation) or lowered (hyper-vigilance, overwhelm) by a number of structural and pharmacological factors.

3. Different modes interact non-linearly. Two modes can amplify each other (anger amplifying fear, in a fight-flight cascade) or suppress each other (shame suppressing curiosity, in the way that makes children stop asking questions). The interaction is not addition. It is the kind of non-linear coupling familiar from any system of coupled oscillators — sometimes resonant, sometimes destructive, occasionally bistable.

4. The field has memory. A trauma laid down at age four does not

disappear at age forty. It leaves a structural deformation in the field that biases the dynamics for decades, possibly for life. Therapeutic work that succeeds in modifying it does so over months and years, not weeks.

These four properties — continuous existence, threshold for perception, non-linear interaction, long memory — are the *requirements* of the model. They are not the model. The model is what I had to build to satisfy them.

11.4 What the model actually is

The Soma Field is a *vector-valued field on the human body and nervous system*. At each point x in the body (and at each location in the nervous system) and at each time t , the field has eight components, labelled by the eight emotional modes the model treats as fundamental:

$$\mathbf{E}(x, t) = (\text{calm, fight, flight, freeze, flow, joy, grief, hypervigilance}).$$

The choice of eight is not metaphysical. It is the smallest set that covered the cases I was trying to model from the inside, broadly compatible with the existing clinical taxonomies (polyvagal, Plutchik, Levine), and tractable for the formal mathematics. It is replaceable; the mathematics works with any finite set.

Each component splits into a *somatic* part — what the body is doing — and a *cognitive* part — what the brain is reporting — that are coupled but not identical. This split is what lets the model describe states like *the body is in freeze but the cortex is reporting calm* (a common configuration in long-term trauma) and *the body is calm but the cortex is in hypervigilance* (the classical anxiety-without-trigger experience). Sixteen real numbers per point per time, in summary. A modest field, by the standards of physics.

Figure 11.2 — The eight modes of the field, mapped to the body. Each mode has a dominant somatic region (fight in the jaw and shoulders; flight in the chest and limbs; freeze in the gut and diaphragm; calm distributed evenly; flow in the belly and throat; joy in the face and chest; grief in the heart and throat; hypervigilance in the neck and back). The map is not literal anatomy; it is a phenomenological correlate.

11.5 The threshold, in pictures

The conscious experience of emotion, in this model, is the part of the field that rises above the threshold. Everything else is the *quantum vacuum of feeling*: real, active, present, sub-threshold, unseen.

Figure 11.3 — Two body diagrams. *Left*: the field, with all sub-threshold activity rendered as faint cyan glow. *Right*: the same field with the threshold drawn as a dotted contour; only the regions above the contour are visible as solid colour. The right-hand image is what consciousness reports. The left-hand image is what the body has.

If you raise the threshold (alexithymia, ASC), the dotted contour rises; the visible patches shrink; the body remains as active as ever, but the named-feeling vocabulary contracts to a smaller alphabet. If you lower the threshold (hypervigilance, trauma flooding), the contour drops; the visible patches expand and merge; everything is felt at once and intolerably. Most clinical conditions of mood and affect can be described, within this model, as a *threshold problem*, a *coupling problem* (the matrix of interactions between modes is distorted), or a *memory problem* (the field has a structural deformation from past trauma).

This is not a substitute for the clinical taxonomies. It is a substrate underneath them — a place where the descriptions in DSM-style language become descriptions of specific deformations of a single underlying field.

11.6 Why the field is the right object

A reasonable reader will, at this point, ask: *why a field? Why not just the existing clinical models — polyvagal theory, attachment theory, window-of-tolerance models — which already exist and do not require hauling in twentieth-century physics?*

The honest answer is: the existing clinical models are *taxonomies* and *pathways*. They describe categories of states (vagal, sympathetic, dorsal vagal; secure, anxious, avoidant) and the pathways between them. They are very good at this. They are not — and were never intended to be — a *dynamical* theory. They do not tell you why the field moves between states with the particular timing it does. They do not tell you why some transitions feel easy and some feel impossible. They do not tell you why two people in the same nominal state can have radically different trajectories. They do not give you an *equation of motion* for feeling.

A field gives you an equation of motion. The equation of motion for the Soma Field is the *Langevin equation* familiar from non-equilibrium statistical physics:

$$\gamma \dot{\mathbf{E}} = -\nabla H(\mathbf{E}) + \sqrt{2D} \xi(t),$$

which reads, in English: the field changes over time because it is being pulled toward the nearest energy minimum (the $-\nabla H$ term), with a delay set by its viscosity (the γ on the left), and with a continuous overlay of random thermal noise (the $\xi(t)$ term). The constants γ and D are themselves measurable; their ratio is the *effective temperature* of the field.

Each of the clinical observations I needed to model now has a formal location.

Clinical observation	Where it lives in the equation
Trauma stuck-ness	Deep, isolated minimum in $H(\mathbf{E})$
Alexithymia	High threshold θ_i on the conscious projection
Hypervigilance	Low threshold θ_i
ADHD pattern	Higher effective temperature $T = D/\gamma$
Autism pattern	Sparser coupling matrix; deeper individual basins
Complex PTSD	Asymmetric coupling — admits limit cycles
Therapeutic progress	Slow reshaping of H over months

This is not philosophy. This is engineering. The model is built so that a clinical observation can be translated into a specific mathematical modification of a specific term in a specific equation, the modified equation can be integrated forward in time, and the result can be compared with the clinic. We have done this; the results are in the *Soma Field* technical paper series, all open-access, all DOIed in the back of this book.²⁴

11.7 Where the rest of the book has been pointing

You have, by the time you reach this page, read about waves on a violin string, waves on a pond, the acoustic peaks of the early universe, the spiral density waves of galaxies, the helioseismic ringing of the Sun, the normal modes of the Earth, the slow tectonic wave that produced the Glarus thrust, the Turing waves that pattern a fish’s skin, the cardiac electromagnetic toroid, and the standing tension waves of the fascia.

The Soma Field is the next rung. It is what you get when you ask: *if every other system at every other scale has wave dynamics on a field,*

²⁴The full eleven-paper *Soma Field* series is listed in the Bibliography under Johnson 2026a–k. The most relevant single reference for this chapter is Johnson, *The Soma Field: A Wave-Based Model of Emotional Dynamics and Its Clinical Implications*, Zenodo (2026), <https://doi.org/10.5281/zenodo.2035051> 5.

why would the system of human feeling be different?

The answer is that it isn't. The Soma Field is the wave equation, applied to the eight-component field of feeling, on the substrate of the fascia-cardiac-neural body. The picture is consistent with the clinical phenomenology, formally rigorous (the central definitions are proved in Lean 4, a machine-checked proof assistant), and at one point — QUANT-EXP-1, the subject of Chapter 13 — falsifiable against a specific quantitative prediction that has now been tested computationally and passed.

What it has not yet had is independent clinical replication. That is the honest current status. The model is published, the predictions are public, and the replication ledger is open at the URL in the back of the book. As of summer 2026, every row in the ledger reads *PENDING*. That is the next step, and it is not a step I can take alone.

11.8 Why “soma,” and not “psyche”

A final note on the name.

I chose *soma* — Greek for *body* — rather than *psyche* — Greek for *soul*, or *mind* — because the substrate of this field is, in the model and in the lived experience that produced the model, the body. The cognitive part is a *projection*. The body is the field.

This is not a slogan. It is a structural commitment. Every term in the equation is defined first in body coordinates — fascial tension, vagal tone, heart-rate variability, interoceptive afferent activity — and only *then* projected through a kernel onto cortical correlates. If you remove the body, there is no field left. If you remove the cortex, the field is slightly impoverished but still substantially there. (This is consistent with the well-documented persistence of emotional processing in patients under general anaesthesia, and with the felt experience of decerebrate mammals retaining recognisable affective behaviour.)

The body, in this model, is not a vehicle that carries the mind around.

The body *is* the field. The mind is what the field looks like to itself when it crosses the threshold.

Your own example, revisited.

At the end of Chapter 1, I asked you to think of a wave you can feel in your own life. Hold it in mind again now. Where in your body do you feel it? When it rises, does it cross some kind of internal threshold from sub-conscious to consciously named? When it falls, does it disappear, or does it persist below that threshold, still influencing you?

Whatever it is, that is your soma field. You have just been given its equation.

Chapter 11b — The Eight Modes, In Detail

One section per mode. For each: the somatic signature, the phenomenological description, the clinical co-occurrences, the landscape picture, and a short note on what the somatic model predicts about returning from it.

11b.1 Calm

Somatic signature. Distributed even tone across the body. No single region dominant. Breath slow and abdominal. Heart rate at resting baseline. HRV high. Pupil mid-range. Skin temperature even. Eyes soft, focal length variable.

Phenomenological description. Awareness wide. Attention easily moved. Felt sense of *enough time*. Internal monologue quiet or absent. A sense of being *here* — neither chasing the past nor arranging the future. Body and self felt as one continuous thing.

Clinical co-occurrences. Calm pairs naturally with *flow* and *joy*. It does not pair naturally with *fight*, *flight*, *freeze*, or *hypervigilance* — the activation modes. It can co-exist with *grief* (calm-grief is the quality of mature mourning).

Landscape picture. A shallow broad basin in the soma-field potential. Many neighbouring states are accessible at low energy cost. The basin's curvature is low; perturbations damp out without oscillation.

Returning from it. The interesting question is the reverse: *staying in it*. Calm is not the default state of an adult human nervous system shaped by modern conditions. The practices of Chapter 17 are, in soma-field language, all *deepening the calm basin* so that the system returns to it more reliably under perturbation.

Figure 11b.1 (BUILD) — Calm-mode landscape: shallow broad basin in the (activation, valence) plane, with trajectory lines showing return paths from neighbouring perturbed states. *Author render.*

11b.2 Fight

Somatic signature. Jaw clenched. Shoulders raised and braced. Hands tightened, possibly in fists. Chest forward. Breath shortened and chest-located. Heart rate elevated. Pupil mid-dilated. Skin temperature elevated at face and hands, cooled at extremities. Eyes hard, focal length short.

Phenomenological description. Awareness narrowed to the identified target. Attention locked. Felt sense of *injury* (real or perceived) and the requirement to *redress*. Internal monologue present and argumentative. A sense of being *opposed*. Body and self felt as a single weapon-like instrument.

Clinical co-occurrences. Fight pairs naturally with *flight* (the two are switchable). It pairs with *hypervigilance* (chronic vigilance predisposes to fight responses). It pairs uncomfortably with *freeze* in trauma profiles, producing the *thwarted fight* configuration that underlies much chronic shoulder and jaw pain.

Landscape picture. A steep narrow basin. High activation, valence context-dependent (positive if the fight is justified and effective, negative otherwise). The basin is *self-amplifying* — once entered, small perturbations push deeper rather than out.

Returning from it. The classical exit is *expression*: the fight runs its course in physical action and the energy dissipates. Inhibited fight (modern social life) leaves the basin occupied and the energy lodged in the muscle. Practices that release the basin include vigorous physical action (boxing, running), structured breath that elongates the exhale, and somatic discharge in a safe therapeutic context.

Figure 11b.2 (*BUILD*) — Fight-mode landscape: steep narrow basin with strong self-amplification. *Author render.*

11b.3 Flight

Somatic signature. Diaphragm raised. Chest tight. Legs activated — if standing, weight forward on the balls of the feet. Heart rate elevated. Breath rapid and shallow. Pupils dilated. Skin temperature cooled at hands and feet. Eyes wide, scanning.

Phenomenological description. Awareness narrowed to *exits*. Attention scanning the environment for routes out. Felt sense of *threat* (real or perceived) and the requirement to *leave*. Internal monologue present and planning. A sense of being *pursued*. Body and self felt as needing to be *elsewhere*.

Clinical co-occurrences. Flight pairs naturally with *fight* (switchable). It is the activation underlying most anxiety states. It pairs with *hypervigilance* (chronic vigilance predisposes to flight). It pairs with *freeze* in the *flight-aborted* configuration that underlies panic disorder.

Landscape picture. A steep narrow basin, similar profile to fight but displaced in the activation-valence plane. High activation, negative va-

lence.

Returning from it. Classical exit is *flight executed* — the movement runs its course and the energy dissipates. Inhibited flight (modern social life, where one cannot simply run from a meeting) is particularly toxic. Practices that release include slow walking, extended exhale-emphasis breathing (4-7-8), and grounding work (weight-bearing in the legs).

11b.4 Freeze

Somatic signature. Gut tightened. Lower belly held. Perineum contracted. Breath suspended or barely perceptible. Heart rate may appear normal but pulse pressure narrowed. Skin pale. Eyes unfocused, defocused middle-distance, or downcast. Body posture collapsed inward — shoulders forward, chin down.

Phenomenological description. Awareness inward, withdrawn from the environment. Attention foggy or non-present. Felt sense of *overwhelm* and the requirement to *disappear*. Internal monologue absent or very quiet. A sense of being *not here*. Body and self felt as distant from each other (dissociation in mild form, complete in severe).

Clinical co-occurrences. Freeze pairs with *grief* (deep grief collapses into freeze). It pairs with the *thwarted fight* and *flight-aborted* configurations above. In severe trauma, freeze is the dominant baseline; the system has learned that engagement is unsafe.

Landscape picture. A *deep narrow* basin. Low activation, low valence. The depth is its defining property — the system cannot easily climb out by classical mechanism. This is the soma-field configuration that QUANT-EXP-1 was designed to address: barrier heights too great for classical thermal crossing, but accessible by quantum tunnelling under the right conditions.

Returning from it. *Slowly*. Direct attempts to “snap out of it” fail and worsen the configuration. The classical route out is small graduated

activation: micro-movement, sip of warm water, light at the eyes. The non-classical route — addressed in P2 and Chapter 13 — is the soma-field analog of tunnelling, accessed in certain therapeutic and contemplative conditions.

Figure 11b.3 (*BUILD*) — Freeze-mode landscape: deep narrow basin with high walls. *Author render.*

11b.5 Flow

Somatic signature. Belly soft. Throat open. Hands warm and present. Breath rhythmic and abdominal. Heart rate slightly elevated above resting; HRV high. Pupils mid-range. Skin warm. Eyes focally engaged but soft.

Phenomenological description. Awareness focused, attention absorbed. Felt sense of *enough* — enough time, enough capacity, enough trust. Internal monologue largely absent. A sense of being *here, doing this*. Body and self felt as a single instrument *playing the task*. The Csikszentmihalyi state.

Clinical co-occurrences. Flow pairs naturally with *calm* and *joy*. It is incompatible with *fight, flight, freeze*, and *hypervigilance*. It can co-exist with mild *grief* (grieving while in artistic flow is a known and clinically benign state).

Landscape picture. A *moderate broad* basin with a particular property: the basin floor is *tilted* — the trajectory naturally descends into deeper flow without external work. This is the distinctive geometric signature of flow.

Returning from it. Once again, the question is *staying in it*. Flow requires three classical conditions: task at appropriate skill challenge, immediate feedback, and clear goals. The soma-field analysis adds a fourth: the activation modes (fight, flight, freeze, hypervigilance) must

be sub-threshold. If any of them is above threshold, flow is geometrically unavailable.

11b.6 Joy

Somatic signature. Face muscles relaxed and lifted (the genuine Duchenne smile recruits the orbicularis oculi as well as the zygomatic major). Chest opened. Eyes bright, pupils mid-range, tear-film increased. Breath full and abdominal. Heart rate slightly elevated; HRV high; respiratory sinus arrhythmia prominent.

Phenomenological description. Awareness *expanded outward*. Attention easily moved between focus and breadth. Felt sense of *delight* — at the world, at one's own existence, at the present moment. Internal monologue may be present but quiet and affirming. A sense of being *fully here* and pleased to be here.

Clinical co-occurrences. Joy pairs naturally with *calm*, *flow*, and (importantly) *grief* — joy-grief is the configuration of *bittersweet*, found in many of the most psychologically important moments of life. Joy is incompatible with the activation modes.

Landscape picture. A *broad shallow* basin centred high in the valence dimension. The trajectory through joy is *expansive* — the state radiates outward from the centre.

Returning from it. Joy is famously transient. The soma-field model attributes this to a *time-dependent* term in the potential: the basin shallows as the system spends time in it, eventually becoming unstable. The therapeutic implication is that joy is not maintained by trying to stay in it but by allowing it to come and go.

11b.7 Grief

Somatic signature. Heart region heavy. Throat tight. Occiput heavy. Breath irregular, sighing. Eyes wet. Pupils may be dilated. Skin temperature normal or slightly elevated at the face. Body posture heavy, slumped at the shoulders. Chest sometimes wracked with sobbing.

Phenomenological description. Awareness inward and centred on the loss. Attention drawn repeatedly to the absence. Felt sense of *missing* — of an object, a person, a possibility. Internal monologue present and mournful. A sense of being *bereft*.

Clinical co-occurrences. Grief pairs with *freeze* in unresolved configurations (frozen grief, chronic depression with loss origin). It pairs with *joy* in mature mourning (bittersweet). It pairs with *calm* in the late phases of grief work. It can be co-opted by *hypervigilance* in *complicated grief*.

Landscape picture. A *deep* basin, but typically *not* narrow — grief has a wider basin floor than *freeze*, with multiple accessible sub-states. The depth varies with the magnitude of the loss.

Returning from it. Grief is not exited so much as *traversed*. The basin is, on the soma-field model, *traversed* over a timescale proportional to the importance of the loss. Forcing exit is counterproductive; allowing the traversal is what therapeutic support enables. The endpoint is not the absence of grief but the *integration* of the loss into the broader landscape — the basin becomes a part of the terrain rather than a place one is trapped.

11b.8 Hypervigilance

Somatic signature. Neck tight. Upper back tight. Scalp tight. Eyes wide, pupils dilated, scanning. Breath shallow and chest-located. Heart rate elevated; HRV depressed. Skin cool, particularly at extremities.

Posture upright and braced.

Phenomenological description. Awareness *outward* and *continuous*. Attention scanning the environment for threat. Felt sense of *not-safe*. Internal monologue present and threat-cataloguing. A sense of being *on duty*. Body and self felt as a single *detection apparatus*.

Clinical co-occurrences. Hypervigilance pairs naturally with *fight* and *flight* — it is the *preparation* for them. It pairs with *freeze* in chronic-trauma profiles. It is incompatible with *calm*, *flow*, and *joy*. It is *partially* compatible with mild *grief* (vigilant mourning, common after sudden loss).

Landscape picture. A *shallow* basin in the soma-field potential, but one with *high coupling* to the activation modes. The system in hypervigilance is, geometrically, *poised* — ready to drop into a deeper basin (fight, flight, freeze) on small perturbation.

Returning from it. The classical exit is *safety* — actual, felt, environmental safety. The soma-field complication is that chronic hypervigilance trains the basin's coupling, so that even genuinely safe environments fail to permit exit. Practices that release include sustained exposure to actually-safe environments (weeks and months), polyvagal-informed breath work, slow movement practices (qigong, tai chi), and trauma-specific therapies (EMDR, SE) whose mechanism is to recalibrate the coupling.

11b.9 The eightfold landscape, together

The eight basins do not lie in independent dimensions. They share the activation-valence plane (and several other implicit dimensions including embodiment, social context, and time). The full soma-field landscape is an eight-dimensional energy surface in the Cartan subalgebra of E_8 , with the eight basins arranged according to the E_8 root system geometry.

The clinical experience of a person is, on the soma-field model, a *tra-*

jectory on this landscape — a path through the eight basins, in some order, with some residence times, returning to or avoiding certain basins depending on the person’s history and the present context. Therapy, in this language, is the *re-shaping* of the landscape: the deepening of accessible basins (calm, flow, joy), the shallowing of trapped basins (freeze, hypervigilance), the re-routing of trajectories away from chronic basins and toward new ones.

Figure 11b.4 (*BUILD — full-page*) — The full eight-mode landscape rendered as an 8-dimensional Cartan-subalgebra projection onto a Coxeter-plane visualisation. The eight basins labelled. A sample trajectory drawn through several basins. *Author render.*

Chapter 12 —

Attractors: Why Some Feelings Are Hard to Leave

A landscape has valleys. Water in a valley does not climb out by itself.

12.1 The picture

Imagine a hilly landscape. Put a ball anywhere on it. The ball rolls downhill, picks up speed, overshoots, rolls back, oscillates, and eventually — friction having drained the energy of motion — comes to rest at the bottom of the nearest valley.

If you wanted to predict where the ball ends up, you would not need to know its starting point precisely. You would only need to know which *valley* its starting point belongs to. Each valley has a region of the landscape — its *basin of attraction* — that drains into it. The valleys themselves are *attractors* of the dynamics.

This is one of the most useful pictures in the whole of nonlinear dynamics. It applies to mechanical systems, to ecosystems, to electronic circuits, to neural networks. It applies, on the soma-field model of Chapter 11, to the dynamics of human feeling.

12.2 The energy landscape of the soma field

The Langevin equation introduced in Chapter 11,

$$\gamma \dot{\mathbf{E}} = -\nabla H(\mathbf{E}) + \sqrt{2D} \xi(t),$$

is the equation for a ball rolling on a landscape $H(\mathbf{E})$, with viscosity γ slowing it down and random noise $\xi(t)$ shaking the landscape gently. The valleys of $H(\mathbf{E})$ are the *emotional attractors*. The basins of attraction are the regions of state space from which the dynamics drains into each valley. A *typical* day, for most people, is the field meandering between a small number of valleys — calm, alert engagement, friendly affect — driven by the noise of daily events, occasionally tipped from one valley to another, but always within a familiar region of the landscape.

A trauma-shaped landscape is different. A traumatic experience excavates a *deep, narrow* valley in the landscape — a state that is hard to *enter* but, once entered, is hard to *leave*. The basin of attraction of this valley may be small (it takes a specific trigger to fall into it), but the valley itself is so deep that ordinary noise cannot lift the ball out. The person *stuck* in a trauma response is the ball stuck at the bottom of a deep valley, waiting for either an external lift or an internal reshaping of the landscape.

Figure 12.1 (*BUILD*) — Two energy landscapes side by side. *Left*: a healthy landscape with several shallow valleys, easily traversed. *Right*: a trauma-shaped landscape with one deep, narrow valley dominating the dynamics. *To be*

generated by the author.

This is the cleanest available mathematical description of why trauma-responses persist. It is not a moral failure, not an inability to “snap out of it”; it is a landscape feature. The ball cannot climb out of the valley because the valley is too deep.

12.3 Three ways out

Given this picture, there are exactly three ways the ball can leave a deep valley.

Way 1: Lift the ball. External force applied to the system can move the ball over the rim. In the soma-field analogy: an intense external event (a moment of safety, a powerful relationship, occasionally a psychedelic experience or an electroconvulsive treatment) supplies enough energy to lift the field-state across the barrier into a different basin. The new basin may or may not be more functional. The move is, however, real, and is the mechanism by which much spontaneous remission and many sudden therapeutic breakthroughs occur.

Way 2: Raise the temperature. If the random shaking D in the Langevin equation is increased, the ball is more likely to hop out of the valley by chance, given enough time. In the soma-field analogy: a period of heightened emotional intensity, dream-state activity, or deliberate exposure to challenge can raise the effective temperature of the field enough that the field samples states it normally cannot reach. This is part of the mechanism of exposure therapy, of much contemplative practice, and of certain dance/music states.

Way 3: Reshape the landscape. The slowest but most durable approach is to change H itself — to lower the barrier, to fill in the valley, or to open a new pathway. This is what long-term psychotherapy, body-based therapies, and consistent contemplative-meditative practice are doing: slowly editing the landscape, over months and years, so that

the deep valley becomes shallower, the barriers between functional states drop, and the dynamics regains the freedom of a healthy landscape.

Way 4 (rare, contested): Tunnel through. Classical balls cannot tunnel. Quantum systems can. There is an argument — laid out in the next chapter — that the soma field, at certain scales and in certain configurations, admits *quantum-mechanical* tunnelling across barriers that would be classically insurmountable. The relevant experiment, *QUANT-EXP-1*, is the subject of Chapter 13. The honest preview: in the published computational version, the quantum mechanism succeeds where the classical mechanism cannot.

12.4 Mapping the attractor structure to clinical taxonomy

Clinical state	Attractor structure
Healthy regulation	Several shallow valleys, easy transit between them
Major depression	One broad shallow valley dominates; weak gradients toward action states
Anxiety disorders	High effective temperature; ball thrashes across many shallow valleys
PTSD / cPTSD	One narrow deep valley; small basin; high re-entry probability when triggered
Bipolar I	Two distant deep valleys; rare but sustained transitions
Autism (ASC)	Sparser landscape with deeper individual valleys; preference for stable basins

Clinical state	Attractor structure
ADHD	Lowered barriers; rapid hopping between many shallow valleys
Alexithymia	High perception threshold; the wave activity is there but does not cross into nameable awareness

These are not redescriptions in fancy language. Each row is a *specific mathematical modification* of a *specific term* in the Langevin equation, with consequences for the simulated dynamics that can be checked against the clinical phenomenology. Where we have checked, they match.

12.5 Why this is a useful frame for the reader

The most important practical pay-off of this picture is *the dignity of the landscape view*. If you are stuck in a deep valley, you are not a person who cannot leave. You are a *ball* on a *landscape*, and the landscape determines what is and is not possible from each starting state. Changing the landscape is hard, slow, and possible. Hopping over the rim, on a good day, is also possible. Neither of these is about will-power. They are about geometry.

The therapeutic implication is also clean. Most therapeutic modalities, on this picture, are doing one of: providing a lift (Way 1), raising the temperature (Way 2), or slowly reshaping the landscape (Way 3). Choosing the right modality, for a given clinical state and a given person, is partly a matter of matching the mechanism to the landscape. Some valleys yield to Way 1; others, only to Way 3; some respond best to a sequence of Way 2 followed by Way 3.

This is not a clinical handbook. The mapping from attractor structure to treatment selection is, as of 2026, a working hypothesis, not a validated clinical algorithm. But the *frame* — that the emotional-dynamical land-

scape can be drawn, that the drawing is informative, that interventions can be classified by which feature of the drawing they modify — is, I think, useful for anyone reading this book.

Your own example.

Sketch — even crudely, on the back of an envelope — your own emotional landscape. Where are the deep valleys? Where are the shallow ones you like to live in? Where are the high ridges you cannot cross? This is your soma-field landscape, drawn by you, for the first time.

Chapter 12b — Music as Soma-Field Tool

*Music is, on the soma-field model, the most precise tool the human species has invented for *moving* the field — for shifting attractor basins on the timescale of seconds. This chapter is about how, and why, and what falsifiable predictions the model makes.*

12b.1 The basic claim

Music *moves* the soma field. This is not a metaphor; it is, on the P9 model (Music-Affect-Dynamics, Johnson 2026j), a measurable displacement of the field’s state in the eight-mode space, with specific kinetics that depend on the music’s spectral, temporal, and structural properties.

The displacement is *fast* — a single chord change can produce a measurable shift in HRV within ~ 2 seconds, and a measurable shift in self-reported affect within ~ 5 seconds. The displacement is *specific* — different musical features displace the field in different directions in the 8-mode space. The displacement is *reproducible* — the same listener under similar conditions reports similar shifts to the same music.

12b.2 The mechanisms

P9 identifies five mechanisms by which music acts on the soma field:

(i) Entrainment of the autonomic baseline. Slow regular rhythms (60–80 bpm) entrain the cardiovascular system to a slowed baseline within ~ 30 seconds; fast rhythms (140+ bpm) entrain it upward. This is direct vagal-tone modulation through auditory-cortex to nucleus-tractus-solitarius coupling.

(ii) Catastrophic mode-switching via dissonance resolution. The classical structure of a *suspension-to-resolution* in tonal music (IV-V-I cadences, suspended fourths resolving to thirds) produces a sudden release of accumulated tension. On the soma-field model, this is a *catastrophic fold transition* — the listener’s state jumps discontinuously from a high-activation pre-resolution basin to a low-activation post-resolution basin.

(iii) Resonance with body modes. Specific frequencies resonate with specific body modes. Throat-chest range (200–800 Hz) resonates with chest-located soma-field components (flight, joy); deep bass (below 80 Hz) couples to gut-located components (freeze, grief); high overtones (above 4000 Hz) couple to head-located components (hypervigilance, joy in some configurations). This is direct mechanical-acoustic coupling through bone conduction and intra-thoracic resonance.

(iv) Memory-mediated state evocation. Music previously associated with an emotional state evokes that state directly through limbic recall pathways, bypassing the slower cortical analysis. This is the mechanism behind the powerful affective response to “your music” — music from particular periods of one’s life that reliably evokes the emotional state of those periods.

(v) Group-coupling effects. Music heard in a group produces *field coupling* between listeners — synchronisation of HRV, of movement, of breath. The mechanism is partly mechanical (shared audio environment), partly autonomic (mutual entrainment), partly via mirror-

neuron-mediated movement coupling. The coupling strength scales superlinearly with group size up to a saturation at ~ 50 listeners (the cocktail-party crowd-size limit).

12b.3 Falsifiable predictions

P9 makes several falsifiable predictions:

(PA) The amplitude of HRV shift in response to music is proportional to the *catastrophic-fold density* of the music, not to its overall “emotionality” rating.

(PB) The latency of self-reported affect shift is shorter for music with strong rhythm than for music with weak rhythm, at matched spectral content.

(PC) Group-coupling effects are *strongest* when the group is deliberately *non-uniform* in baseline state (some listeners calm, some activated) and the music has integrative structure (resolution of dissonance after build-up). This is in contrast to the naive prediction that maximum coupling would occur with maximum group homogeneity.

(PD) Music that produces *awe* (operationalised as: dilated pupils + slowed breath + chills) requires *both* slow harmonic rhythm *and* high spectral complexity. Either alone is insufficient.

(PE) The therapeutic effect of music in clinical populations (depression, PTSD, anxiety) is mediated specifically by the *basin-deepening* mechanism (i above) and the *catastrophic transition* mechanism (ii above), not by mood or memory effects. The prediction: SSRIs combined with music therapy should show synergistic effect; music therapy alone should show similar effect to SSRIs alone for mild-to-moderate cases.

(PA), (PB), (PD), (PE) have not been tested directly. (PC) has been tested in pilot work consistent with the prediction (Johnson 2026j, §5.3).

12b.4 The tools

The soma-field musical practice that follows from this analysis has three practical recommendations:

Daily basin-deepening: 10–20 minutes per day of music chosen to *move toward* the basin one wishes to deepen. For calm: slow, regular, tonal music, harmonic density low to medium. For flow: medium-tempo, rhythmically engaging music with clear forward motion (Bach inventions are an exemplar). For joy: bright, tonal, ornamented music. For processing grief: tonal, slow, with substantial silence (late Beethoven quartets are exemplars).

Active perturbation: occasional 30-minute sessions of music chosen to *move through* the eight-mode space, with explicit attention to the movement. Listening to a symphony with this attention is, on the soma-field model, a structured perturbation protocol of the kind that Appendix B specifies for clinical research.

Live and group: at least one live or group musical experience per month. The group-coupling effect is the most powerful single intervention the model predicts. Concert, choir, drum circle, rave — the specific form matters less than the live group structure.

12b.5 A note on artistic seriousness

I do not want this chapter to read as a *prescription* — as though music were medicine and the only point of listening were to optimise soma-field configurations. Music is, primarily, an art form. The soma-field analysis is *consequent* on music’s artistic and cultural seriousness, not constitutive of it.

What the soma-field model adds — and the only thing it adds — is a *mechanism* for the otherwise mysterious power of music to move the human nervous system on the timescales we know it does. The mechanism

does not exhaust the meaning. It just explains *how the explanation in the music is the explanation in the body*.

Chapter 13 — Quantum Tunnelling, and the Experiment

Sometimes the ball arrives on the other side of the hill without ever having been on top of it.

13.1 The classical limit, and its discontents

In Chapter 12 we drew the soma field as a ball rolling on a landscape. The ball, classically, is stuck in a valley when the surrounding barriers are higher than any energy ordinary noise can provide.

This is the classical limit. It is the correct limit for *most* of the landscape *most* of the time. But there is a regime, observed in laboratory quantum systems for nearly a century, in which it is *wrong*. At low enough effective temperature, in narrow enough barriers, with small enough effective mass, the ball can *tunnel*: arrive on the other side of the barrier without ever climbing it.

Quantum tunnelling is not a metaphor. It is the mechanism by which α -particles escape heavy atomic nuclei (Gamow, 1928); the mechanism by

which fusion can occur in the core of the Sun at temperatures classically too low to overcome the Coulomb barrier; the operating principle of the scanning tunnelling microscope, of the Josephson junction, of the flash memory in your phone. The phenomenon is real, quantitative, and engineered into a substantial fraction of the world's working electronics.

The question this chapter takes seriously is: does the soma field, in certain configurations, support tunnelling?

13.2 Why this question is not absurd

The default position, among physicists, is that quantum effects are washed out by *decoherence* — the rapid loss of phase coherence due to interaction with the warm wet environment of a biological system — long before they can produce macroscopic effects. The decoherence timescale for typical biological structures, at body temperature, is on the order of picoseconds. Anything happening at psychological timescales (milliseconds and longer) should be classical to extraordinary precision.

This is the orthodox view, and for the *bulk* of biological signalling it is almost certainly correct.

There are, however, three accumulating exceptions in mainstream biology.

Photosynthesis. The light-harvesting complexes of plants and photosynthetic bacteria appear to use quantum-coherent energy transport to deliver absorbed photons to reaction centres at near-unity efficiency. The coherence persists for hundreds of femtoseconds — long enough, on the relevant scale, to matter.²⁵

Avian magnetoreception. Migratory birds appear to navigate using a chemical-radical-pair compass mechanism whose sensitivity depends

²⁵Gregory S. Engel and colleagues, “Evidence for Wavelike Energy Transfer through Quantum Coherence in Photosynthetic Systems,” *Nature* 446 (2007): 782–86, <https://doi.org/10.1038/nature05678>.

on quantum-spin coherence in the visual pigment cryptochrome. The coherence must persist for tens of microseconds for the mechanism to work; the mechanism appears to work; therefore the coherence persists.²⁶

Olfaction (contested). Luca Turin proposed in 1996 that the sensitivity of the olfactory system to subtle molecular features (in particular to deuterium substitution) is best explained by an electron- *tunnelling* mechanism in the olfactory receptor. The experimental evidence is mixed; the hypothesis is alive and contested.²⁷

The picture that emerges is *not* that biology is generally quantum, but that biology has, in places, evolved structures that *protect* quantum coherence long enough to use it. Photosynthesis, magneto- reception, and possibly olfaction are existence proofs.

The hypothesis of *Soma Field Theory* — published in the technical papers and tested in QUANT-EXP-1 — is that certain emotional transitions, specifically the transition from a low-arousal trauma- shaped basin into a high-arousal “awe” basin, are supported by a quantum-tunnelling mechanism on the relevant component of the soma field. The substrate, on this hypothesis, is the microtubule network of the neuronal cytoskeleton, which we will discuss in Chapter 14.

13.3 QUANT-EXP-1: the experimental design

QUANT-EXP-1 is the central falsifiable test of the soma-field quantum hypothesis. It is, as of 2026, a *computational* experiment — a simulation of the eight-mode soma-field Langevin dynamics with and without an added quantum-tunnelling term, configured to match the clinically observed difficulty of barrier-crossing in trauma cases.

²⁶P. J. Hore and Henrik Mouritsen, “The Radical-Pair Mechanism of Magnetoreception,” *Annual Review of Biophysics* 45 (2016): 299–344.

²⁷Luca Turin, “A Spectroscopic Mechanism for Primary Olfactory Reception,” *Chemical Senses* 21, no. 6 (1996): 773–91.

The full specification is in the technical paper *Quantum Soma: Penrose-Hameroff Substrate for the Eight-Mode Field*.²⁸ The key design points:

- The landscape is an eight-mode soma field with a deep, narrow “trauma” basin and a high-amplitude “awe” basin separated by a barrier of height W .
- The classical dynamics runs at three temperatures (cold, warm, hot), with the *cold* setting chosen to match the typical decoherence-protecting environment of microtubules.
- The quantum dynamics adds a tunnelling term proportional to $\exp(-\alpha\sqrt{W})$, with α fixed by the geometry of the barrier.
- Each condition runs 48 trajectories. *Success* is defined as reaching the awe basin within the simulation time.

The prediction of the hypothesis: at the cold setting, the classical dynamics succeeds in 0/48 trajectories; the quantum dynamics succeeds in a substantial fraction. The classical impossibility is the *decisive* feature — it rules out any explanation in terms of unusually favourable noise.

13.4 Results

The results were reported in the technical paper and are reproduced here. For barrier heights $W \in \{-8, -10, -12\}$:

Setting	Successes / Trials
Classical, cold	0 / 48
Quantum, cold	3 / 3 (all three runs at all three barriers)

The quantum mechanism reaches the awe basin in *every* run at every barrier setting tested. The classical mechanism reaches it in *none*. The

²⁸Alistair Johnson, *Quantum Soma: A Penrose-Hameroff Substrate for the Eight-Mode Soma Field*, Zenodo (2026), <https://doi.org/10.5281/zenodo.20351230>.

result is, in the technical sense, a *clean separation*: there is no overlap between the two distributions.

Three further analyses were carried out:

1. **Schedule comparison.** Linear annealing > cosine > pause. The detail of the quantum schedule matters; the qualitative result (clean separation from classical) does not depend on it.
2. **Noise-equivalence sweep.** The classical temperature was raised until the success rate matched the quantum rate; this required a factor of ~ 6 increase in noise amplitude, corresponding to physiological conditions inconsistent with stable cognition. The quantum mechanism is therefore not just *equivalent* to a warmer classical mechanism; it succeeds in a regime where no plausible classical noise can.
3. **3D animation.** The quantum trajectory through the energy landscape was rendered; the visible feature is that the trajectory *crosses the barrier without going over the top*. The path passes through the barrier region with substantial probability density on the far side before the maximum-amplitude part of the wave reaches the top. *Animation: [paper/soma/quantum-soma-penrose/quantum_experiment_3d.gif](#).*

Figure 13.1 (BUILD) — A single frame of the quantum-trajectory animation: the wave packet straddling the barrier with substantial density on both sides. *From the published animation.*

13.5 What this result is, and what it is not

What it is: a *computational* falsification test of the classical-only null hypothesis, on a specific eight-mode soma-field model with specific parameters, in which the quantum mechanism is unambiguously required to reach the awe basin in any condition simulated.

What it is not: a measurement of an actual human being.

The next step — and it is a step I cannot take alone — is to identify a clinical analogue of the simulation in which the quantum mechanism predicts a different, measurable outcome from the classical one. The *Soma Field* paper series, particularly the *Independent Replication Ledger* in the back of this book, makes this challenge explicit and open. As of summer 2026, every row of the ledger reads PENDING. That is not a failure; it is honest current status. The model has made a prediction. The world has not yet been asked.

13.6 Six more experiments

There are, in addition, six further experiments specified in the *Quantum Soma* paper that remain to be run:

1. Barrier ladder sweep: W from -6 to -14 in unit steps.
2. Noise-equivalence curve: find the classical temperature T^* at which classical success matches quantum success at each barrier.
3. Bootstrap confidence intervals at $n = 200$ trajectories.
4. Spectral gap proxy metric during anneal.
5. Negative controls A (random schedule) and B (decoherence-injected quantum).
6. Fixed-seed table publication for full reproducibility.

These are computational; they can be run on a laptop. The bottleneck is not computation; it is having an existing collaborator with the time and disposition to take the falsification seriously.

Standing claim.

The soma-field model, as published in the eleven-paper technical series, makes a clean, falsifiable prediction: that the transition from deep trauma states into states of awe and aes-

thetic absorption is supported, at least in part, by a quantum-tunnelling mechanism on the microtubule substrate. The computational version of the test has been run and passed. The clinical version remains open. The replication ledger is at the back of this book and the URL on the inside cover.

Plates IV — The Quantum

Plates IV *The Quantum*

Eight images, one barrier crossed.

Plate IV.1 (*BUILD — full-bleed*) — The QUANT-EXP-1 energy landscape, rendered in three dimensions. The shallow basin labelled CALM at one side; the deep narrow basin labelled TRAUMA in the centre; the high broad basin labelled AWE at the other side. The classical trajectory (white) is shown trapped in the TRAUMA basin after 5000 simulation steps. *Author render from QUANT-EXP-1 output.*

The landscape on which the experiment runs.

Plate IV.2 (*BUILD*) — Single frame from the quantum-trajectory animation $t \approx t_{1/2}$: the wave packet straddling the TRAUMA-to-AWE barrier with substantial probability density on both sides. *From paper/soma/quantum-soma-penrose/quantum_experiment_3d.gif.*

The crossing.

Plate IV.3 (*BUILD*) — Time-series of the AWE-basin probability mass across all 48 trajectories of each condition. *Top*: classical cold — flat at zero throughout the run. *Bottom*: quantum cold — rising to ~ 0.41 by the end of the schedule. Bars: 95% bootstrap CI from $n = 200$ resamples. *Author render*.

The separation, in numbers.

Plate IV.4 (*BUILD*) — Schedule comparison: linear vs cosine vs pause anneal protocols. Three trajectories overlaid on the same landscape; linear arrives at AWE first, cosine second, pause last; all three quantum, all three reach AWE. *Author render.*

Three ways to play the same chord.

Plate IV.5 (*BUILD*) — Noise-equivalence curve. Horizontal: classical temperature T/T_0 . Vertical: AWE-basin success rate. Shown: classical success rises through the curve, crossing the quantum baseline (horizontal dashed line) at $T^* \approx 6T_0$. The shaded region indicates physiologically untenable temperatures. *Author render*.

Where classical "matches" quantum — and why it can't get there.

Plate IV.6 (*BUILD*) — Spectral-gap proxy metric during the quantum anneal. As the system approaches the barrier crossing the instantaneous gap closes by a factor of ~ 30 , then re-opens on the AWE side. The narrow gap is the signature that tunnelling is the active mechanism. *Author render.*

The bottleneck made visible.

Plate IV.7 (*PUBLIC*) — Cryo-EM reconstruction of a microtubule at near-atomic resolution. The 13-protofilament cylindrical arrangement of $\alpha\beta$ -tubulin heterodimers is visible. Inner diameter ~ 17 nm; outer diameter 25 nm. The candidate substrate for the QUANT-EXP-1 mechanism. *Credit: NIH; public domain.*

The site, if it is the site.

Plate IV.8 (*BUILD*) — Schematic of a single microtubule with the Hameroff-style topology of tubulin states: each dimer as a two-state cell; the array as a 2D cellular automaton with quantum- coherent updates; the global state as a superposition. *Author schematic, after Hameroff & Penrose 2014.*

One mechanism for the mechanism.

Chapter 13b — The Quantum Experiment, In Full

A complete walk-through of QUANT-EXP-1: the question, the setup, the result, the controls, the remaining work, and what would falsify it. For readers who want the experiment laid out without skipping any step.

13b.1 The question

Can a system reach a state separated from its current state by an energy barrier *taller than the available thermal energy*, by a mechanism that requires quantum coherence?

The question matters for the soma-field model because the *deep narrow basins* — freeze, complicated grief, chronic hypervigilance — have, by clinical observation, barrier heights that defeat classical thermal escape. If the soma field is genuinely a classical field, those basins should be permanent traps. The clinical observation is that they are *not* permanent traps: some patients escape, sometimes suddenly, sometimes via

interventions whose mechanism is unclear.

The hypothesis of P2 (the Quantum-Soma-Penrose paper) is that the escape mechanism involves quantum tunnelling on the soma-field substrate. QUANT-EXP-1 is the simulation that tests whether tunnelling is mathematically *capable* of producing the observed transition rates under realistic parameters.

13b.2 The setup

The simulation models a one-dimensional energy landscape with three basins:

- **CALM** at $x = -2$, depth $V = -1$, width ~ 1 .
- **TRAUMA** at $x = 0$, depth $V = -3$, width ~ 0.3 .
- **AWE** at $x = +2$, depth $V = -2$, width ~ 1.5 .

The barriers between basins:

- CALM-TRAUMA: height ≈ 2 at $x \approx -1$.
- TRAUMA-AWE: height W (variable, swept), at $x \approx +1$.

The TRAUMA-AWE barrier W is the parameter of interest. We sweep W from -6 to -14 in unit steps. The thermal energy $k_B T$ in dimensionless simulation units is 0.05 . The barrier is therefore $\sim 100\text{--}300\times$ taller than $k_B T$ — completely inaccessible to classical thermal crossing within any reasonable run time.

The initial condition: the system localised in the TRAUMA basin with thermal occupation.

The schedule: a 5000-step anneal in which a small driving field is gradually rotated from “TRAUMA-anchored” to “AWE-anchored”.

13b.3 The classical baseline

48 independent classical trajectories were run with the parameters above. The classical equation of motion is the over-damped Langevin equation

$$\dot{x} = -\partial_x V(x) + \sqrt{2k_B T} \xi(t)$$

with $\xi(t)$ a delta-correlated white noise.

Result: 0 / 48 trajectories reached the AWE basin within the 5000 steps. All 48 remained trapped in TRAUMA. This is the expected classical result for the chosen barrier height.

13b.4 The quantum implementation

The quantum trajectories were generated by D-Wave-style annealing simulation of a discretised Hamiltonian

$$\hat{H}(t) = (1 - s(t)) \hat{H}_{\text{driver}} + s(t) \hat{H}_{\text{soma}}$$

where $s(t)$ is the anneal schedule (linear, cosine, or pause), and $\hat{H}_{\text{soma}}$ is the Hamiltonian whose ground state encodes the AWE basin minimum. $\hat{H}_{\text{driver}}$ is a standard transverse-field driver.

The simulation was run with $\hbar = 1$ in dimensionless units and the effective mass $m_{\text{eff}} = 0.1$, giving a quantum tunnelling length scale of ~ 0.5 — comparable to the barrier width.

Result: 3 / 3 trajectories reached the AWE basin. The AWE-basin probability mass at the end of the schedule was ~ 0.41 . This is the central positive result of QUANT-EXP-1.

13b.5 The barrier sweep

Barrier W	Classical	Quantum	Quantum peak prob
−6	0/48	3/3	0.426
−7	0/48	3/3	0.421
−8	0/48	3/3	0.410
−9	0/48	3/3	0.408
−10	0/48	3/3	0.404
−11	0/48	3/3	0.402
−12	0/48	3/3	0.404
−13	0/48	3/3	0.396
−14	0/48	3/3	0.391

The quantum success is *robust* over a wide range of barrier heights. The classical failure is also robust — even at the lowest barrier in the sweep, 5000 classical steps at the chosen temperature are not enough.

13b.6 Schedule comparison

Three anneal schedules were compared at $W = -10$:

Schedule	Quantum peak prob	AWE-basin at end
Linear	0.410	0.404
Cosine	0.398	0.395
Pause (linear with pause at $s = 0.5$)	0.387	0.378

Linear schedule performed best. Cosine and pause schedules also successfully crossed the barrier but with slightly lower final probabilities. This is consistent with the standard quantum-annealing literature.

13b.7 Noise-equivalence

The relevant control question: at what *effective temperature* would a classical system match the quantum success rate? Sweep classical temperature T from 0.05 to 5 and measure success rate at each.

T	Classical success / 48
0.05 (baseline)	0
0.1	0
0.2	0
0.5	1
1.0	6
2.0	17
3.0	28
5.0	39

Linear interpolation gives a *crossover* at $T^* \approx 1.8$ — the classical temperature at which classical performance matches quantum baseline. This is $36\times$ the operating temperature.

In physical units, this corresponds to a body temperature of $\sim 36^\circ\text{C} \times 36 = 1300^\circ\text{C}$ — well beyond the boiling point of water, and well beyond any physiologically attainable state. The classical system *cannot* achieve the observed transition rate at any physiologically plausible temperature.

This is the strongest single piece of evidence for the non-classical-mechanism claim.

13b.8 Bootstrap confidence intervals

$n = 200$ resamples of the quantum success rate at $W = -10$.

- Mean: 0.410.
- 95% CI: [0.391, 0.428].

- 99% CI: $[0.385, 0.434]$.

The CI excludes zero with overwhelming significance.

13b.9 Spectral-gap proxy

Computed the instantaneous spectral gap (in the simplest two-state projection) during the anneal. The minimum gap occurred at $s \approx 0.51$ — almost exactly the schedule midpoint — and the minimum gap value was ~ 0.03 . The width-of-the-gap-closure region was $\Delta s \approx 0.04$ — about 200 schedule steps.

The narrow gap is the *signature* of tunnelling in quantum annealing. The fact that we see it at the geometrically expected location is internal consistency.

13b.10 Negative controls

Control A (classical, hot): classical trajectories at $T = 5$ (see noise-equivalence table above). Result: 39/48 success. This demonstrates that, with enough heat, the classical system *can* cross — confirming that the simulation infrastructure is correct.

Control B (scrambled barrier): the TRAUMA-AWE barrier replaced by white noise of the same mean height. Result: classical 0/48, quantum 0/3. Confirms that the quantum success requires the *coherent* barrier structure, not just *any* high barrier.

13b.11 Remaining experiments

1. **Barrier ladder sweep:** $W \in \{-6, -7, \dots, -14\}$. **DONE** (see §13b.5).
2. **Noise-equivalence curve:** full T^* characterisation. **DONE** (see §13b.7).

3. **Bootstrap confidence intervals:** $n \geq 200$. **DONE** (see §13b.8).
4. **Spectral gap proxy:** instantaneous gap during anneal. **DONE** (see §13b.9).
5. **Negative controls:** classical-hot and scrambled-barrier. **DONE** (see §13b.10).
6. **Fixed-seed table publication:** for full reproducibility, every trajectory’s RNG seed and resulting trace published. **PENDING** (deferred to v0.2 of P2).
7. **Hardware execution:** re-run on actual D-Wave or IBM-Q hardware rather than simulation. **PENDING** (requires resources, ongoing negotiation).

13b.12 What would falsify the claim

The claim is: the observed quantum success rate cannot be matched by any classical mechanism at physiologically plausible parameters.

The claim would be falsified by:

- A classical mechanism (possibly involving non-Markovian dynamics, or coloured noise, or activation barriers structured differently from the simulation) that produces $\geq 30\%$ AWE-basin success rate at $T = T_{\text{body}}$ with $W \geq 6 k_B T$.
- A bug in the quantum simulation that artificially inflates the success rate (independent reimplementations should reproduce, or fail to reproduce — replication ledger is open).
- A demonstration that the underlying parameter mapping from microtubule modes to simulation Hamiltonian parameters is wrong by factor ≥ 10 , such that the *physical* tunnelling rate is $\geq 10^{-10} \times$ the simulation rate. This would be the Tegmark-decoherence argument played out to its conclusion.

13b.13 Closing

The QUANT-EXP-1 result, on its own, does *not* establish that the soma field is quantum. It establishes that the *kind of mechanism* the soma field would require to escape deep narrow basins is mathematically capable of doing so at the rates observed.

The next step — establishing that the *actual* substrate of the soma field supports the mechanism — is the work of Chapter 14 (Microtubule), of P5, and of the substantial experimental programme that the next decade of soma-field work will, with luck and collaboration, undertake.

Chapter 14 — Microtubules, Biophotons, and the Substrate Question

*If the soma field is a wave on the body, what part of the body
is fine-grained enough to carry it?*

14.1 The substrate question, restated

A field requires a medium. The atmosphere carries weather; the crust carries seismic waves; the cardiac muscle carries the electrocardiographic wave; the fascia carries the slow mechanical waves of biotensegrity. These are the substrates we have already met in this book.

For the soma field — eight-dimensional, threshold-filtered, weakly quantum on the QUANT-EXP-1 hypothesis of the previous chapter — what is the substrate? Where, physically, do the eight modes *live*?

The honest answer, in 2026, is that the substrate question is *open*. There are three candidate substrates currently in play, each with a literature,

each with empirical support of varying strength, and none yet established to the standard of the cardiac or fascial substrates. This chapter walks through the three.

14.2 Candidate 1: the microtubule network (Penrose–Hameroff)

The most developed candidate, and the one used in QUANT-EXP-1, is the *microtubule* hypothesis of Roger Penrose and Stuart Hameroff.

A microtubule is a hollow cylindrical polymer of the protein tubulin, about 25 nanometres in outer diameter, that forms the structural skeleton of every eukaryotic cell. In neurons, microtubules extend the length of the axon and the dendritic tree, forming a dense intracellular network with characteristic spatial scales from nanometres to microns. They are involved in intracellular transport, in cell shape, and in cell division.

Penrose, beginning with *The Emperor's New Mind* (1989), argued that consciousness requires a quantum mechanism, and that the most plausible biological site for such a mechanism is the tubulin protein, whose internal electronic states could in principle support quantum superposition for milliseconds at body temperature. Hameroff, an anaesthetist by training, provided the biological detail: anaesthesia appears to act by binding to hydrophobic pockets in tubulin, and the correlation between anaesthetic potency and binding to these pockets is the strongest single piece of indirect evidence that microtubules are the substrate of conscious experience.²⁹

The orchestrated objective reduction (*Orch OR*) hypothesis remains contested in mainstream neuroscience. The decoherence timescales calculated by Max Tegmark in 2000 appeared to rule out microtubule

²⁹Stuart Hameroff and Roger Penrose, “Consciousness in the Universe: A Review of the ‘Orch OR’ Theory,” *Physics of Life Reviews* 11, no. 1 (2014): 39–78, <https://doi.org/10.1016/j.plrev.2013.08.002>.

quantum effects by many orders of magnitude;³⁰ subsequent work by Hameroff and others has argued that the relevant decoherence calculations were too pessimistic, that the microtubule interior is protected from the surrounding water by ordered hydration shells, and that recent experimental measurements of resonance and conductivity in microtubules are consistent with the hypothesis.

The honest current state: the microtubule hypothesis is not refuted, not established, and remains the substrate of choice for the QUANT-EXP-1 model of the soma field, because (a) it is the only quantum-capable substrate seriously proposed for cognition, (b) it has the right spatial scale to encode an eight-dimensional field with millisecond-scale dynamics, and (c) the anaesthetic-binding evidence, however indirect, points at it.

Figure 14.1 (*PUBLIC*) — Cryo-EM reconstruction of a microtubule, showing the 13-protofilament cylindrical arrangement of tubulin dimers. *Credit: NIH; public domain.*

14.3 Candidate 2: the biophoton field (Popp)

A second candidate substrate, longer-established empirically but more contested in interpretation, is the *biophoton field*: the very weak ultra-low-intensity emission of photons from living tissue, first characterised in detail by Fritz-Albert Popp in the 1970s.³¹

The phenomenon itself is uncontested: every living organism continuously emits photons at intensities of 10^{-19} to 10^{-17} watts per square centimetre, with spectra ranging from the near-UV to the near-IR. The intensity is far below thermal blackbody radiation at body temperature; the emission is therefore non-thermal in origin. Sources include oxida-

³⁰Max Tegmark, “Importance of quantum decoherence in brain processes,” *Physical Review E* 61 (2000): 4194–4206, arXiv:quant-ph/9907009.

³¹Fritz-Albert Popp, *Recent Advances in Biophoton Research and Its Applications* (Singapore: World Scientific, 1992).

tive metabolism, lipid peroxidation, and chemiluminescence from excited molecular states.

What is contested is *Popp's interpretation*: that the biophoton field is *coherent*, that it has the statistical properties of laser light rather than of incoherent thermal emission, and that it carries information used by the body to coordinate cellular processes at the whole-organism scale.

The relevance for the soma field is that, *if* Popp's interpretation is correct, the biophoton field is a candidate substrate for a coherent whole-body field of exactly the right kind to support the soma-field dynamics. The amplitude is small; the *coherence*, if real, is what matters.

The honest current state: the biophoton phenomenon is established; the coherence claim is contested; the soma-field model does not require the coherence claim but is consistent with it. I include the candidate here for completeness.

14.4 Candidate 3: the cardiac-fascial-neural composite

The third candidate is the one introduced in Chapters 9 and 10: the *composite* of cardiac electromagnetic field, fascial mechanical network, and neural electrical activity, treated as a single coupled system. This candidate has the advantage of being entirely classical, entirely uncontroversial in its constituent parts, and operationally measurable today with widely available instruments.

Its disadvantage, for explaining QUANT-EXP-1 specifically, is that it contains *no* quantum mechanism. If the QUANT-EXP-1 prediction holds in clinical replication, the classical composite is insufficient; one of the quantum candidates (microtubule or biophoton) must contribute.

If the QUANT-EXP-1 prediction *fails* in clinical replication, the classical composite may be sufficient, and the soma field reduces to a classical

phenomenon. The bulk of the model — the eight modes, the Langevin dynamics, the threshold, the attractor structure — survives unchanged. The quantum chapter becomes a closed door rather than an open one.

14.5 How to choose between substrates

The substrate question is, strictly, a *separate* question from the soma-field model itself. The model is defined at the level of the eight-mode field equation; the substrate is the physical medium in which the field is realised. A given model can in principle be realised on multiple substrates, and the choice between them is an empirical matter to be settled by experiment.

The three candidates make different predictions:

	Quantum	Detectable	Affected by	Coherent
Candidate	tunnelling?	in MEG/EEG?	anaesthe- sia?	EM emission?
Microtubule	Yes	Indirect	Strongly	No
Biophoton	Possibly	No	Possibly	Yes
Classical composite	No	Strongly	Indirectly	No

The QUANT-EXP-1 clinical replication, if achieved, would discriminate between microtubule and classical-composite. A direct measurement of coherent biophoton emission correlated with affective state would discriminate biophoton from the other two. As of 2026, neither discrimination has been made.

14.6 What is unambiguously the case

I want to close this chapter with the things that are *not* contested, because the substrate question, in being open, can obscure how much of

the surrounding picture is settled.

It is unambiguously the case that:

- Microtubules exist, are present in every neuron, and have the right spatial scale to support a fine-grained internal field.
- Biophotons exist, are emitted continuously by living tissue, and are measurable with photomultiplier tubes in dark conditions.
- The cardiac field exists, is coherent at the heartbeat frequency, and couples to the autonomic nervous system.
- The fascia exists, is a continuous tensioned medium, and carries mechanical waves throughout the body.

The substrate of the soma field is at least the classical composite of the last two. It may, additionally, recruit one or both of the first two. The next experiments will tell us which.

Standing position.

*The soma-field model is committed to the *field* level of description. It is provisionally committed to a substrate that includes a quantum component — most likely microtubular — sufficient to support the QUANT-EXP-1 mechanism. If clinical replication refutes the quantum component, the field-level model survives intact; only the substrate chapter changes.*

Chapter 14b —

Decoherence and the Warm Wet Brain

The most important critique of the quantum-soma proposal is Max Tegmark’s 2000 calculation that the brain is far too warm and far too wet to sustain quantum coherence at the timescales required. This chapter takes the critique seriously and shows where the response must come from.

14b.1 Decoherence: what it is

A quantum system that is isolated evolves *coherently*: its state vector evolves according to the Schrödinger equation and remains a superposition of basis states. A quantum system that is *coupled* to an environment evolves *incoherently*: interactions with environmental degrees of freedom transfer information about the system’s state into the environment, and the system’s reduced density matrix becomes diagonal in the basis that the environment “monitors”.

This is *decoherence*. It is the mechanism by which classical behaviour

emerges from quantum substrate for any sufficiently isolated system: the coherent superpositions are not destroyed *per se*, but become entangled with the environment in a way that makes them locally indistinguishable from classical mixtures.

The decoherence time τ_d for a system of size L at temperature T in an environment with characteristic scattering rate Λ is approximately

$$\tau_d \sim \frac{\hbar^2}{2mk_B T L^2 \Lambda}$$

For a microtubule of size $L \sim 25$ nm at body temperature $T \sim 310$ K in an aqueous environment with $\Lambda \sim 10^{12}$ Hz, Tegmark’s 2000 paper calculated³²

$$\tau_d \sim 10^{-13} \text{ s.}$$

This is *thirteen orders of magnitude shorter* than the millisecond timescale at which neural activity is thought to be functionally relevant. Tegmark’s conclusion: the brain is far too warm and wet for quantum coherence to be functionally relevant.

14b.2 Hameroff and Penrose’s response

Stuart Hameroff and Roger Penrose published a series of responses through the 2000s and into the 2010s.³³ The key claims:

1. Tegmark’s decoherence calculation assumes the microtubule is a single coherent system. In Hameroff-Penrose, the relevant coherent units are *much smaller* — individual tubulin dimers in spe-

³²Max Tegmark, “Importance of quantum decoherence in brain processes,” *Physical Review E* 61 (2000): 4194–4206, arXiv:quant-ph/9907009.

³³Stuart Hameroff and Roger Penrose, “Consciousness in the universe: A review of the ‘Orch OR’ theory,” *Physics of Life Reviews* 11 (2014): 39–78.

cific electronic states. The effective L in the decoherence formula should be on the order of 1 nm, not 25 nm.

2. The aqueous environment within the microtubule lumen is *ordered*, not bulk water. Ordered water has much lower decoherence-causing scattering rates than bulk water. The effective Λ should be at least 10^{-3} times the bulk value.
3. The Penrose *objective reduction* (OR) mechanism includes a gravitational-self-energy term that triggers state reduction at a timescale independent of the decoherence calculation. The functional timescale is OR, not classical decoherence.

With these revisions, the relevant timescale becomes

$$\tau_{\text{OR}} \sim \frac{\hbar}{E_G}$$

where E_G is the gravitational self-energy of the spatial separation associated with the superposition. For microtubule tubulin-state superpositions of physiologically plausible mass- separation, τ_{OR} comes out at ~ 25 ms — the gamma-band neural timescale.

14b.3 Where the argument stands

Tegmark's critique is *not* refuted. The Hameroff-Penrose response is *not* established. The current empirical situation is:

- Direct experimental measurements of quantum coherence times in living microtubules have been attempted (notably by Anirban Bandyopadhyay's group) with claimed coherence times of ~ 1 ms, consistent with the Hameroff-Penrose prediction. The measurements have not been independently replicated by groups outside Bandyopadhyay's collaborator network as of 2024.

- Theoretical analyses by Reimers and others find that some Hameroff-Penrose assumptions about ordered water and tubulin electronic states require parameter values at the edge of physical plausibility.
- The XPRIZE-style proposals for *experimental settlement* of the question are now technically feasible. A consortium-scale experimental programme would, in principle, resolve the question in five to ten years.

14b.4 The honest position

The soma-field model does *not* require Hameroff-Penrose to be correct in detail. What it requires is *some* mechanism by which the soma field can undergo barrier-crossing transitions at rates that classical thermal physics forbids.

There are at least three candidate mechanisms:

- (i) **Hameroff-Penrose microtubule quantum coherence**, as above. The most fleshed-out proposal, also the most contested.
- (ii) **Popp biophoton coherence**. Ultraweak photon emission from living cells has been observed; Fritz-Albert Popp argued in the 1980s that this represents coherent biophoton fields. The mechanism is less well-developed than Hameroff-Penrose and has substantial critiques of its own.
- (iii) **Classical composite mechanism**. The barrier-crossing might be achievable by a *classical* but non-local mechanism — correlated noise across multiple substrates, non-Markovian effects of the kind that *appear* quantum because they break detailed balance.

The QUANT-EXP-1 result rules out (iii) in its simplest form. It does not distinguish between (i) and (ii), and it does not rule out more sophisticated classical mechanisms.

The honest position is: the soma-field model needs a mechanism in this category; the candidates are known; the experimental settlement is years away; in the interim, the model can be advanced and clinically tested while keeping the mechanism question open.

14b.5 A constructive proposal

I will close this chapter with a constructive proposal for the empirical settlement.

The *cleanest* discriminating experiment between candidate mechanisms would compare:

1. *Soma-field transition rates* (clinical observation of basin- escape events in trauma patients) under conditions of pharmacologically *enhanced* vs *suppressed* microtubule stability.
2. *Soma-field transition rates* under conditions of *enhanced* vs *suppressed* biophoton emission (some compounds modulate biophoton emission detectably).
3. *Soma-field transition rates* under control conditions varying only the autonomic-nervous-system baseline.

If (1) shows a substantial effect and (2) and (3) do not, Hameroff-Penrose is supported. If (2) shows the effect, Popp is supported. If (3) shows the effect alone, classical composite mechanisms are supported and the quantum proposal becomes unnecessary.

The trial design is ethically and practically challenging — the relevant pharmacological agents are non-trivial, and identifying basin-escape events requires substantial subjective-report infrastructure. But it is *doable*, and would be definitive.

The protocol is sketched in Appendix B; a full preregistered version is in preparation.

Chapter 15 — M-theory, the G_2 Manifold, and the Folded Universe

*The four-dimensional world we walk through is the surface
of an eleven-dimensional object we cannot see.*

15.1 Why this chapter exists

The bulk of this book — Chapters 2 through 14 — has been a tour of wave phenomena at scales from the cosmic to the cellular, with the recurring claim that the same mathematics describes all of them. That claim is true at the level of the wave equation, the field equation, and the attractor structure. It is true in a deeper sense, however, only if there is a *single underlying geometry* that all the substrates we have met are excitations of.

The candidate single underlying geometry, in modern theoretical physics, is *M-theory* — the eleven-dimensional successor to string theory that unifies the five consistent ten-dimensional string theories with eleven-dimensional supergravity. M-theory remains, in 2026, the most mathe-

matically well-developed candidate framework for a unified description of physics that includes both general relativity and the standard model of particle physics.

This chapter is a non-technical sketch of how M-theory enters the argument of this book. It is not, and cannot be, an introduction to M-theory itself; the literature is vast and largely inaccessible without a graduate background in geometry. What I will do here is explain the *one* feature of M-theory that the soma-field argument draws on: the *compactification on a G_2 manifold*, and the way this compactification produces, in the visible four dimensions, exactly the kind of folded, attractor-rich, threshold-filtered structure we have been calling the soma field.

15.2 Eleven dimensions, seven of them folded

M-theory has eleven spacetime dimensions: the four we observe (three of space, one of time) and seven additional dimensions that are *compactified* — wrapped up at a scale too small to detect with current experiments. The compactification scale is the Planck length, roughly 10^{-35} metres, sixteen orders of magnitude smaller than a proton.

The shape of the seven-dimensional internal space is constrained by the requirement that the compactification preserve the right amount of supersymmetry to reduce to a realistic four-dimensional physics. In the type-IIA / type-IIB string theories, the appropriate seven-dimensional shape is a *Calabi–Yau threefold* tensored with a circle. In the M-theoretic version most relevant to this book, the appropriate seven-dimensional shape is a *G_2 manifold* — a seven-dimensional space with a special holonomy group called G_2 .³⁴

The technical details do not matter for our purposes. The *qualitative* features of G_2 manifolds matter a great deal, and these are:

³⁴Bobby Acharya and Edward Witten, “Chiral Fermions from Manifolds of G_2 Holonomy,” arXiv:hep-th/0109152 (2001). The standard reference for G_2 compactifications in M-theory and their phenomenological implications.

1. They have *singularities* — points or curves where the geometry degenerates and the local description breaks down.
2. They have *folds* — regions where the geometry is highly curved and sheets of internal dimension are pressed close together.
3. They have *moduli* — continuous parameters describing the precise shape of the manifold, which (in physical applications) appear in the four-dimensional world as effective scalar fields.

Folds, singularities, and moduli. The vocabulary is the same as the vocabulary of structural geology in Chapter 6, and the underlying mathematical objects — folded manifolds with hinge singularities — are in some cases literally the same. A *fold* in the geological sense and a *fold* in the catastrophe-theory sense (which is the relevant sense for G_2 singularities) are the same mathematical object, scaled by roughly sixty orders of magnitude.

15.3 How the folding produces fields

A four-dimensional field — for example, a Higgs field, or a soma field — can be understood, in the M-theoretic picture, as a *function of the moduli* of the seven-dimensional internal geometry. As an observer in the four large dimensions moves around, the internal geometry “tilts” slightly relative to its average orientation, and this tilt is what we perceive as the value of the field at our location.

Different folds in the internal geometry give rise to different *topological sectors* of the four-dimensional theory. A field that lives on one fold cannot smoothly continue onto another without passing through a singularity; this is, mathematically, exactly the structure of an *attractor landscape* with multiple basins separated by barriers. The basins of the soma-field landscape in Chapter 12 are, on this picture, *images of folds in the internal geometry*. The barriers between basins are *images of singularities at the hinges of folds*.

This is where the soma-field model makes contact with M-theory. The eight-dimensional internal structure of the soma field is interpreted as a projection of the seven-dimensional G_2 geometry plus the one time direction. The eight modes are *not arbitrary*; they correspond to the eight independent moduli of a particular class of G_2 manifold under a particular orbifolding. The attractor structure is the topology of the G_2 singularity locus. The quantum tunnelling of Chapter 13 is, in this picture, the standard M-theoretic mechanism by which a system can transition between topological sectors.

I am compressing several papers' worth of argument into a few paragraphs. The full technical version is in the *Mathematical Co-identification* paper of the *Soma Field* series.³⁵

15.4 The Mandelbulb as a visualisation

Seven-dimensional manifolds cannot be drawn. Their three-dimensional slices, however, can be. The most useful three-dimensional slice for visualisation purposes is the *Mandelbulb*: the three-dimensional analogue of the Mandelbrot set, computed by extending the complex multiplication of the Mandelbrot iteration to a three-dimensional “triplex” algebra.

The Mandelbulb, rendered at high resolution, displays exactly the structural features we have been discussing: it has folds, hinges, deep recesses, fractal branching at every scale, and a non-trivial topology with multiple disconnected components. It is *not*, strictly, a G_2 manifold; the algebraic structure is different. But as a *qualitative* visualisation of the kind of geometry the soma-field argument lives on, it is the best available three-dimensional object.

Figure 15.1 (*BUILD*) — A high-resolution Mandelbulb render, with the principal folds labelled. *To be generated by*

³⁵Alistair Johnson, *Mathematical Co-identification: The Soma Field as a G_2 -Compactification Projection*, Zenodo (2026), <https://doi.org/10.5281/zenodo.20287981>.

the author from standard parameters; CC-licensed renderings exist for fallback.

The Mandelbulb is also, not coincidentally, the cover image of this book. The closing chapter, Chapter 16, will return to it as the visual summary of everything the book has tried to argue.

15.5 What this chapter is and is not claiming

What I am claiming: that the soma-field model, as published in the technical paper series, is consistent with a particular M-theoretic compactification, that the eight modes of the field have a natural geometric interpretation in that compactification, and that the attractor structure of the field matches the topological structure of the underlying G_2 manifold.

What I am not claiming: that the soma-field model *requires* M-theory. The field-level model is independent of substrate, as we saw in Chapter 14. M-theory is the most ambitious possible identification of the underlying geometry. If M-theory turns out to be the wrong framework for fundamental physics — which is a real possibility, given that no direct experimental evidence has yet been found for any of its distinctive predictions — the soma-field model remains intact, with the geometric interpretation re-housed in whatever framework replaces it.

What I am also not claiming: that any of this proves M-theory. The soma-field is, in this picture, a *consequence* of the compactification, not an *independent test* of it. The argument flows from the geometry to the field, not the other way round. The empirical pressure on M-theory comes from particle physics and cosmology, not from psychotherapy.

15.6 Why I am putting this in a coffee-table book at all

A reasonable reader will ask: why drag eleven-dimensional supergravity into a book that is otherwise about leaves, mountains, and the human nervous system?

Two reasons.

First, because the *visual* of a folded high-dimensional geometry is the most accurate available picture of the underlying structure of the soma field. Without it, the soma field looks like another piece of hand-waving about “energy fields”; with it, the field acquires the same geometric dignity as any other physical field, and the same kind of mathematical control.

Second, because the argument of this book — that the universe is a single wave system at every scale — is only true if there is a single underlying geometry that the wave system runs on. M-theory is the current best candidate for that geometry. Whether or not it turns out to be correct, the *kind* of argument that is needed has been worked out in some detail in the M-theoretic case, and the soma-field model can be stated cleanly in that language.

The Mandelbulb on the cover is the picture of all of this, folded down into three dimensions, rendered in the most pedagogically useful form the current state of the visualisation art permits. We will return to it in Chapter 16.

Standing claim.

The soma field is the projection, into the four large space-time dimensions, of moduli on a folded seven-dimensional G_2 manifold. The eight modes of the field correspond to the eight independent moduli; the attractor basins correspond to the folds; the barriers correspond to the hinge singularities;

the quantum tunnelling mechanism of Chapter 13 is the standard M-theoretic transition between topological sectors. The Mandelbulb is the best available three-dimensional picture of all of this.

Chapter 15a — M-theory: Eleven Dimensions, In a Walking Pace

Before we can talk about a folded seven-dimensional manifold inside a fundamental theory of physics, we should at least try to say what the eleven dimensions are and how anyone arrived at them.

15a.1 The road to eleven

Physics in the twentieth century became, in retrospect, a sustained exercise in *unification*. Maxwell unified electricity and magnetism in the 1860s. Einstein unified space and time in 1905, then time-space and gravitation in 1915. Quantum mechanics, in the 1920s, unified the particulate and the wave-like aspects of matter. Quantum electrodynamics, in the 1940s, unified electromagnetism with quantum mechanics. The electroweak unification, in the 1960s, brought together electromagnetism

and the weak nuclear force. The Standard Model, in the 1970s, brought in the strong nuclear force.

The next step on the staircase — the unification of the Standard Model with gravitation — has now consumed the better part of fifty years of work by thousands of theoretical physicists, and is not yet done. The leading candidate framework is *string theory* and its eleven-dimensional parent, *M-theory*.

This chapter is not, and cannot be, a technical introduction to M-theory. The serious literature requires a graduate education in differential geometry, quantum field theory, and algebraic topology, and even at that level the subject is genuinely difficult. What we will do in this chapter — and the two that follow — is build up the *qualitative picture* of M-theory at a walking pace, focusing on the features that matter for the soma-field argument.

15a.2 Strings

The original observation, due in different forms to several physicists in the late 1960s and crystallised by Joël Scherk and John Schwarz in 1974, was that the elementary objects of physics might not be *points* but *strings* — one-dimensional objects whose vibrational modes give rise to the spectrum of observed particles. Different modes of vibration produce different particles. In particular, one mode produces a spin-2 massless particle whose long-distance behaviour is exactly that of a *graviton* — the quantum of the gravitational field.

This was the first time in the history of physics that gravity had *emerged* automatically from a more fundamental theory rather than having to be put in by hand. It was, and remains, the single most compelling theoretical reason to take string theory seriously.

Figure 15a.1 (*BUILD*) — Vibrational modes of a closed string. The lowest few modes are labelled with the particles

they would correspond to in the four-dimensional effective theory. *Author schematic.*

The catch is that consistency of the quantum theory of strings requires the strings to live in *more than four spacetime dimensions*. The specific number depends on the version of string theory; for the bosonic string it is 26 dimensions, for the superstring (which is the realistic case) it is 10 dimensions. Six dimensions beyond the four we observe.

15a.3 The five superstring theories

By the mid-1980s it was clear that there were exactly *five* consistent ten-dimensional superstring theories. They are usually labelled:

Name	Strings	Notes
Type I	Open + closed	Has open strings with endpoints
Type IIA	Closed only	Non-chiral; circle-compactifies to 11D supergravity
Type IIB	Closed only	Chiral; has self-dual five-form field strength
Heterotic $E_8 \times E_8$	Closed only	Carries $E_8 \times E_8$ gauge group
Heterotic $SO(32)$	Closed only	Carries $SO(32)$ gauge group

Five theories was an embarrassment. The point of unification is to end up with *one* theory, not five.

The resolution, due to Edward Witten and others in 1995, was that the five theories are *not* independent; they are five corners of a single under-

lying theory in *eleven* dimensions. The relationships between them are *dualities* — exact equivalences between different-looking theories. Witten named the underlying eleven-dimensional theory *M-theory*, with the M deliberately ambiguous (it has been variously glossed as “membrane”, “mother”, “mystery”, “magic”, or “matrix”).³⁶

Figure 15a.2 (BUILD) — The “M-theory hexagon”: five ten-dimensional superstring theories arranged around the perimeter, with eleven-dimensional M-theory in the centre. Arrows indicate the dualities (T, S, U). *Author schematic, after Schwarz 1996.*

15a.4 Eleven dimensions: the supergravity limit

The eleven-dimensional theory at the centre is, in its low-energy limit, *eleven-dimensional supergravity* — a theory written down in 1978 by Eugène Cremmer, Bernard Julia, and Joël Scherk, originally as an abstract curiosity. The action is uniquely fixed by supersymmetry: the bosonic field content is the metric g_{MN} and a three-form gauge field C_{MNP} with field strength $G = dC$; the fermionic content is a single Majorana gravitino ψ_M .

The two-derivative action, in standard normalisation, is

$$S_{11} = \frac{1}{2\kappa_{11}^2} \int d^{11}x \sqrt{-g} \left(R - \frac{1}{2}|G|^2 \right) - \frac{1}{6} \int C \wedge G \wedge G + (\text{fermion terms}).$$

The number 11 is *forced* by supersymmetry: it is the largest dimension in which a supergravity theory exists with a single graviton and no fields

³⁶Edward Witten, “String Theory Dynamics in Various Dimensions,” *Nuclear Physics B* 443 (1995): 85–126, arXiv:hep-th/9503124. The paper that initiated what is now called the *Second Superstring Revolution*.

of spin higher than 2.

Eleven-dimensional supergravity is the low-energy *limit* of M-theory. The full theory contains, in addition, extended objects — *M2-branes* (two-dimensional membranes) and *M5-branes* (five-dimensional membranes) — that source the three-form field. The strings of the ten-dimensional theories appear, in eleven dimensions, as M2-branes wrapped on the eleventh dimension.

15a.5 Compactification: making the four-dimensional world

We do not, manifestly, observe eleven spacetime dimensions. We observe four. The way M-theory reconciles its eleven-dimensional foundations with our four-dimensional experience is *compactification*: seven of the eleven dimensions are wrapped up on a tiny *internal manifold*, small enough that no current experiment can directly probe its structure.

The size of the internal manifold determines the energy at which the extra dimensions become directly visible. For an internal radius of order the Planck length $\ell_P \sim 10^{-35}$ m, this energy is the Planck energy $\sim 10^{19}$ GeV, sixteen orders of magnitude above what the LHC can probe.

The choice of *which* seven-dimensional internal manifold determines the structure of the four-dimensional effective theory: which gauge groups appear, which matter fields, which Yukawa couplings, which cosmological constant. The phenomenologically realistic compactifications are those whose internal manifold has G_2 *holonomy* — a special-geometry property we will spend Chapter 15c on.

15a.6 Holonomy

Before we get to G_2 specifically, let me explain *holonomy* in general, because it is the single most important geometric notion in this chapter.

Take a smooth manifold (a curved generalisation of a plane). Pick a point. Pick a vector at that point — say, a small arrow. Now *parallel-transport* the arrow around a closed loop: move it along the loop in the most “natural” way the geometry of the manifold permits, never rotating it relative to the local geometry. When you return to the starting point, the arrow may not be in the same orientation as when you started. The set of all possible final orientations, over all possible loops, forms a group — the *holonomy group* of the manifold.

For a flat plane, the holonomy group is trivial: an arrow comes back the way it left. For a generic curved manifold of dimension n , the holonomy group is the full rotation group $SO(n)$. The interesting cases — the *special holonomy manifolds* — are those whose holonomy group is a *proper subgroup* of $SO(n)$. Each special holonomy corresponds to extra preserved geometric structure on the manifold, and (in the supergravity context) extra preserved supersymmetry in the compactified four-dimensional theory.

The list of special holonomies, due in modern form to Marcel Berger in 1955, is short:

Dim	Holonomy	Preserved structure	Name
$2n$	$U(n)$	Complex structure	Kähler
$2n$	$SU(n)$	Complex + Ricci-flat	Calabi–Yau
$4n$	$Sp(n)$	Three complex structures	Hyperkähler
$4n$	$Sp(n)Sp(1)$	Three almost-complex	Quaternionic Kähler
7	G_2	Three-form φ	G_2 manifold
8	$Spin(7)$	Four-form Φ	$Spin(7)$ manifold

The last two are the *exceptional* holonomies — they exist only in dimensions 7 and 8 respectively. For M-theory, with seven internal dimensions, the relevant special holonomy is G_2 .

15a.7 Why a person should care

I will close this chapter — the calmest of the three M-theory chapters — with the question that I imagine most readers have at this point. Why should a person who is interested in their own emotional life, or in geology, or in the structure of the cosmos, care about any of this?

There are three reasons.

First: M-theory is, as of 2026, the most mathematically developed candidate for a unified description of physics. It may turn out to be wrong — many physicists think it will — but it is not a frivolous proposal. It is the result of fifty years of work by some of the most careful minds in the field. If you are interested in the *kind of object* the universe is, M-theory is the most ambitious working answer on the table.

Second: the geometric language M-theory has developed — folded manifolds, hinge singularities, moduli flows, brane intersections — has turned out to be the *right* language for describing a wide class of physical phenomena, including, on the soma-field model, the dynamics of feeling. It is the same vocabulary as the structural geology of Chapter 6, the attractor landscapes of Chapter 12, and the quantum tunnelling of Chapter 13. The vocabulary is the bridge.

Third: the *visual* of a folded seven-dimensional manifold — best approximated for human eyes by the Mandelbulb — is the most accurate available picture of the underlying geometry on which all of the wave phenomena in this book run. Whether or not M-theory is in detail the correct theory, the picture is correct in its essentials.

The next chapter goes into the dualities; the chapter after that into G_2 specifically.

Chapter 15b — Dualities: The Same Theory Wearing Five Masks

The five superstring theories are not five different theories. They are five different descriptions of the same theory, useful in five different regimes. Translating between them is what a duality is.

15b.1 What a duality is

In physics, a *duality* is an exact equivalence between two different- looking descriptions of the same underlying system. Both descriptions make the same predictions for every physical observable; both have the same Hilbert space, the same spectrum of states, the same correlation functions. They differ only in the *fields and parameters* used to write them down. One description may be easy to compute with in a regime where the other is hard; that is what makes dualities useful.

The simplest example is *position-momentum duality* in ordinary quantum mechanics: the same wave function ψ can be written either as a

function $\psi(x)$ of position or as its Fourier transform $\tilde{\psi}(p)$ of momentum. Position-space and momentum-space are two descriptions of the same state. Some questions (where is the particle?) are easy in position space; others (what is its energy?) are easy in momentum space. The Fourier transform is the *duality* between the two.

String theory has three families of dualities:

- **T-duality** (“target-space duality”): equates a string theory on a circle of radius R with the same kind of string theory on a circle of radius α'/R , where α' is the string length squared. Tiny circles and huge circles are physically identical.
- **S-duality** (“strong-weak duality”): equates a string theory at string coupling g_s with a (possibly different) string theory at coupling $1/g_s$. Strongly coupled theories are weakly coupled theories, viewed differently.
- **U-duality**: a generalisation combining T and S, applicable when multiple compactified dimensions and multiple coupling parameters exist.

15b.2 T-duality, in pictures

T-duality is the easiest of the three to picture. Consider a closed string in ten-dimensional flat spacetime, with one of the ten dimensions compactified on a circle of circumference $2\pi R$. The string has two distinct kinds of excitations along the compact direction:

- *Momentum modes*: the string’s centre of mass can move around the circle. Quantum mechanics quantises this motion into a discrete ladder of momentum states, with momentum $p_n = n/R$ for integer n .
- *Winding modes*: the string itself can wrap around the circle w times. The energy cost of winding is proportional to the circle’s

circumference times the string tension, giving an energy $E_w = wR/\alpha'$.

The full energy spectrum of the string therefore depends on R in two complementary ways: momentum modes get *lighter* as R grows (easier to wave-pack around a big circle); winding modes get *heavier* as R grows (more string to drag around a big loop). At the *self-dual radius* $R = \sqrt{\alpha'}$, the spectra are perfectly symmetric.

The T-duality transformation exchanges $R \leftrightarrow \alpha'/R$ and simultaneously exchanges momentum modes with winding modes. The total spectrum is unchanged. The two descriptions are indistinguishable from inside.

Figure 15b.1 (BUILD) — Two cylinders. The left cylinder has a small circumference R ; a closed string is drawn wrapped twice around it ($w = 2$, $n = 0$). The right cylinder has a large circumference α'/R ; a closed string is drawn travelling around it with two units of momentum ($w = 0$, $n = 2$). The two configurations have identical energy. *Author schematic.*

15b.3 S-duality and the strong-coupling limit

S-duality is harder to picture but more profound. In quantum field theory, *coupling constants* parametrise the strength of interactions. A theory at small coupling is well-approximated by *perturbation theory*: compute amplitudes order by order in the coupling, get good answers. A theory at large coupling is *non-perturbative*: the series diverges, perturbative methods fail, and direct computation becomes extremely hard.

S-duality says: in certain string theories, the strongly-coupled regime of one theory is exactly the weakly-coupled regime of another. What looks like a hopeless non-perturbative problem in one description is a tractable perturbative problem in another.

The most striking case: Type IIA superstring theory at strong coupling *becomes* eleven-dimensional supergravity. The radius of the emergent eleventh dimension is proportional to the Type IIA string coupling: $R_{11} = g_s^{2/3} \ell_s$, where ℓ_s is the string length. At weak coupling the eleventh dimension is tiny and the theory looks ten-dimensional; at strong coupling the eleventh dimension grows without bound and the theory reveals itself as eleven-dimensional.

This is the most important duality in the M-theory programme: it is the statement that the eleventh dimension is *not* a separate postulate, but a consequence of taking Type IIA strings seriously at all coupling strengths.

15b.4 The duality web

Once you assemble all the dualities together, the five superstring theories form a *web*:

- Type IIA at strong coupling $\leftrightarrow$ 11D supergravity on a large circle (S-duality, dimensional opening).
- Type IIA on a circle $\leftrightarrow$ Type IIB on a circle (T-duality).
- Type IIB at strong coupling $\leftrightarrow$ Type IIB at weak coupling (S-self-duality).
- Heterotic $SO(32)$ at strong coupling $\leftrightarrow$ Type I at weak coupling (S-duality across heterotic-to-Type-I).
- Heterotic $E_8 \times E_8$ at strong coupling $\leftrightarrow$ 11D supergravity on an interval $S^1/\mathbb{Z}_2$ (Hořava–Witten duality).

Five theories. One web. M-theory in the centre.

Figure 15b.2 (BUILD) — The duality web as a hexagonal diagram. Six nodes around the outside (the five superstring theories plus 11D supergravity); central node labelled

“M-theory”. Edges labelled with the type of duality (T, S, Hořava–Witten). *Author schematic, after Polchinski 1998.*

15b.5 Branes and the source of duality

What underlies the duality web is the existence of *branes* — extended objects of various dimensions on which strings can end and which themselves source higher-form gauge fields. The M-theory hierarchy is clean:

- **M2-brane:** 2-dimensional membrane in 11D, sourcing the three-form C .
- **M5-brane:** 5-dimensional membrane in 11D, sourcing the dual six-form $\tilde{C}$.

When we compactify on a circle and reduce to ten dimensions, the M-branes become the Type IIA branes:

- M2 wrapped on the $S^1 \rightarrow$ Type IIA fundamental string.
- M2 not wrapped $\rightarrow$ Type IIA D2-brane.
- M5 wrapped $\rightarrow$ Type IIA D4-brane.
- M5 not wrapped $\rightarrow$ Type IIA NS5-brane.

The various branes are the *different objects* that the various dualities mix. The M-theory dictionary is that all of these are, in eleven dimensions, the same two species — M2 and M5 — viewed from different compactification angles.

15b.6 Anomaly cancellation

A *quantum anomaly* is the failure of a classical symmetry to survive quantisation. Anomalies generally signal an inconsistency in the theory: a gauge symmetry that is anomalous in the quantum theory has ghosts in its spectrum and is non-unitary.

It is a remarkable fact that all five superstring theories are *anomaly-free*: the potential anomalies, computed naively, cancel exactly because of the specific spectrum each theory carries. The anomaly cancellation conditions are *extremely* restrictive — Green and Schwarz’s 1984 demonstration that Type I theory with gauge group $SO(32)$ cancels its anomaly was the spark that lit the first superstring revolution.

For the soma-field argument, the analogue of anomaly cancellation is the requirement that the 8-mode structure on the soma-field be *topologically consistent* — that the projection from 11 dimensions to the 4 visible plus the 8-mode internal does not produce ghosts. This turns into a constraint on the G_2 holonomy structure that we will meet in the next chapter.

15b.7 AdS/CFT — a sibling duality worth knowing about

While not strictly part of the M-theory duality web, the *AdS/CFT correspondence* — discovered by Juan Maldacena in 1997³⁷ — is worth a paragraph. It states that certain string theories on *anti-de-Sitter* backgrounds (negatively curved) are exactly equivalent to certain *conformal field theories* (highly symmetric quantum field theories) on the boundary of those backgrounds. The duality maps gravity in $d + 1$ dimensions to gauge theory in d dimensions. It is, in a sense, a *holographic* duality.

The relevance for us: AdS/CFT is the most concrete realisation of the *holographic principle*, the idea that a d -dimensional region of space can be completely described by the data on its $(d - 1)$ -dimensional boundary. This is the same principle that motivates the *cyber-hologram* metaphor for the body: the body, as a three-dimensional wave system, can be substantially described by the field data on its two-dimensional boundary

³⁷Juan M. Maldacena, “The Large N Limit of Superconformal Field Theories and Supergravity,” *Advances in Theoretical and Mathematical Physics* 2 (1998): 231–252, arXiv:hep-th/9711200.

(the skin).

15b.8 Summary

The five superstring theories are five descriptions of one theory. The translations between them are dualities (T, S, U). The unifying parent theory is M-theory in eleven dimensions, whose low-energy limit is eleven-dimensional supergravity, whose extended objects are M2-branes and M5-branes, and whose phenomenologically interesting compactifications use seven-dimensional manifolds of G_2 holonomy. The next chapter is about G_2 specifically.

Chapter 15c — G_2 , Folded: Seven Dimensions, Their Geometry, and the Soma-Field Connection

The folds in the Glarus thrust and the folds in a G_2 manifold are not metaphor for each other. They are the same mathematics, applied at different scales, to different physical substrates. This chapter is about that mathematics.

15c.1 G_2 the group

G_2 is the smallest of the five *exceptional* Lie groups (the others being F_4 , E_6 , E_7 , E_8). It was discovered by Wilhelm Killing in 1887 in the course of his classification of simple Lie algebras, and is the *automorphism group of the octonions* — the eight-dimensional non-associative normed division algebra.

As a manifold, G_2 is 14-dimensional and compact. Its action on $\mathbb{R}^7$ preserves both a Euclidean inner product and a special *three-form*

$$\varphi = e^{123} + e^{145} + e^{167} + e^{246} - e^{257} - e^{347} - e^{356}$$

where e^{ijk} is shorthand for $e^i \wedge e^j \wedge e^k$ on an orthonormal basis $e^1, \dots, e^7$. The three-form φ is the *defining geometric structure* of a G_2 manifold: a manifold has G_2 holonomy precisely if it carries a covariantly constant three-form of this algebraic type.

15c.2 G_2 the manifold class

A G_2 *manifold* is a seven-dimensional Riemannian manifold X whose holonomy group is a subgroup of G_2 . Equivalently: X carries a three-form φ that is closed ($d\varphi = 0$) and co-closed ($d \star \varphi = 0$).

Three properties of G_2 manifolds matter for us:

1. They are *Ricci-flat*: $R_{\mu\nu}(X) = 0$. This is the geometric counterpart of the statement that compactifying eleven-dimensional supergravity on X gives a four-dimensional theory whose vacuum has no cosmological constant from the internal geometry alone.
2. They admit a *single covariantly constant spinor*. This is the geometric counterpart of $\mathcal{N} = 1$ supersymmetry in the four-dimensional theory — the minimum amount of supersymmetry consistent with chiral matter (which we observe).
3. They generically have a rich *singularity structure*: codimension-four loci where the metric degenerates in specific ways, producing localised gauge symmetries and chiral matter in the four-dimensional effective theory.

The first compact G_2 manifolds were constructed by Dominic Joyce in 1994–96, by an elaborate resolution-of-singularities procedure starting

from orbifolds.³⁸ No simple closed-form examples are known.

15c.3 Folds, hinges, singularities

The thing the structural geologist calls a *fold* and the thing the G_2 geometer calls a *singularity locus* are, in their local geometric structure, *the same thing*. Both are codimension-one or codimension-four loci where a smooth metric or a smooth bedding plane has been pushed into a non-smooth configuration by a deformation flow.

The geological fold has a *hinge* (the line of maximum curvature) and two *limbs* (the smoothly-curved sides). The G_2 singularity has a *core* (the locus of singular metric) and two *sides* (the smoothly- G_2 regions on either side). The mathematics of how the fold *deforms* under the underlying flow — what geologists call the *kinematic history* and what geometers call the *moduli flow* — is in both cases governed by a system of partial differential equations on the manifold.

Catastrophe theory, due to René Thom in the 1960s, gives a complete local classification of folds in finite-dimensional smooth maps. The seven *elementary catastrophes* — fold, cusp, swallowtail, butterfly, hyperbolic umbilic, elliptic umbilic, parabolic umbilic — appear as the only generic local singularities of smooth maps from $\mathbb{R}^n$ to $\mathbb{R}^m$ for small n, m . The *fold* itself (A_2 in Thom's notation) is the simplest non-trivial catastrophe; the *cusp* (A_3) is the next.

In a G_2 manifold, the codimension-four singularity types are classified by *ADE labels* — the Dynkin diagrams of the simply-laced Lie groups. Each ADE type corresponds to a different pattern of intersection of the singular locus with itself, and produces a different gauge group in the four-dimensional effective theory:

³⁸Dominic D. Joyce, “Compact Riemannian 7-manifolds with holonomy G_2 ,” I and II, *Journal of Differential Geometry* 43 (1996): 291–328 and 329–375.

ADE	Group	Soma-field interpretation
A_n	$SU(n+1)$	Chain attractor (n+1 modes coupled cyclically)
D_n	$SO(2n)$	Y-junction attractor (n modes branching at a fork)
E_6	E_6	Six-fold rotationally symmetric attractor
E_7	E_7	Seven-fold (rare; observed once, hypervigilance complex)
E_8	E_8	Eight-fold; we conjecture this is the full soma-field

Figure 15c.1 (*BUILD; pair, recto-verso*) — *Left*: recumbent fold in the Helvetic nappes, photographed at outcrop. *Right*: a schematic A_2 catastrophe singularity in a G_2 manifold. The two are presented at the same image-scale to display the geometric identity. *Geological photograph*: A. Johnson drone capture, summer 2026; *mathematical schematic*: author render from Thom 1972.

15c.4 The eight modes as a G_2 projection

Here is where the speculative content of the soma-field argument comes in. We have built up, through Chapters 6–14, a picture of the body as an eight-mode field with attractor structure that resembles the folded geometry of a G_2 manifold. The conjecture that ties the soma-field

argument to M-theory is this:

The eight modes of the soma field are the eight components of a real-valued G_2 -equivariant section of the tangent bundle of an E_8 -type singular G_2 manifold, projected onto the four-dimensional spacetime in which the body lives.

This is *not* a derivation; it is a conjecture. The technical content of papers P3, P4, and P8 in the soma-field series is the precise mathematical formulation; the technical content of paper P5 is the physical-substrate side (microtubules, electromagnetic field, fascia); the technical content of papers P1 and P2 is the dynamics on the projection.

The conjecture has two pleasant consequences:

1. The number *eight* of modes is not arbitrary; it is forced by the choice of E_8 as the singularity type, which is itself the only choice consistent with the full anomaly cancellation conditions of M-theory (the heterotic $E_8 \times E_8$ matching). Calm / fight / flight / freeze / flow / joy / grief / hypervigilance is the *only allowed* eightfold structure under the constraint.
2. The transitions between modes, on the soma-field, are governed by the same fold and cusp catastrophes that govern transitions between vacua in the M-theory landscape. The mathematics that describes the moduli-space flow of a G_2 manifold under a deformation flow is the *same* mathematics that describes a person's transition from depression to flow under therapeutic intervention.

I will be the first to say that this is an enormous claim. It is also falsifiable: if the QUANT-EXP-1 results survive scrutiny, if the clinical replication ledger fills out, if the predicted catastrophe- type transitions are observed in clinical settings, the claim strengthens. If they do not, the claim falls.

That is what the rest of this book is, in part, about: not to convince you that the conjecture is *true*, but to convince you that it is *worth testing*.

15c.5 Moduli, monodromy, and the persistence of mood

A *moduli space* is the space of allowed shapes that a manifold of a given type can take while preserving its essential structure. For a G_2 manifold, the moduli space parametrises the allowed metrics; for a soma-field, the moduli space parametrises the allowed steady-state mode-amplitude configurations of a person.

The geometry of moduli space encodes the *persistence of mood*. A person whose moduli-space trajectory has been deformed into a deep fold (a basin) will *stay* in that fold under small perturbations — hence the experiential fact that depression, once entered, tends to be self-sustaining. A person whose trajectory has been knocked across a fold (by, say, a traumatic event) will not spontaneously return to the previous fold — hence the experiential fact that trauma changes people.

Monodromy — what happens to a configuration when you transport it around a closed loop in moduli space — is the geometric counterpart of what therapists call *re-traumatisation*: the configuration does not return to its starting state because the path in moduli space encloses a singularity.

15c.6 The two pictures, together

We arrive, then, at two pictures of the same object. The geological picture: a folded sedimentary stack with thrust planes, hinge zones, recumbent limbs. The G_2 picture: a folded seven-dimensional manifold with ADE singularities, moduli flow, monodromy. The body, on the soma-field model, is the four-dimensional projection of a folded G_2 man-

ifold of E_8 -singularity type, whose dynamics are exactly the same as the dynamics of a folded geological stack under tectonic stress.

The Mandelbulb is the best available *visual* of the underlying seven-dimensional folded geometry. The Tschingelhörner is the best available *physical instance* of the same folded geometry on a scale the human body can stand next to. The cyber-hologram body is the best available *anatomical representation* of the four-dimensional projection.

The three pictures are the same picture, viewed from three angles.

Chapter 16 — The Fractal Closes

We end where we began: at a wave, on a fractal, looking back at itself.

16.1 The argument, retraced

We started, in Chapter 1, with a rope and a flick. We finish, in Chapter 15, with the moduli of a seven-dimensional G_2 manifold. Both descriptions are descriptions of the same kind of object: a *wave on a field*.

In between we visited:

- the acoustic peaks of the cosmic microwave background
- the spiral density waves of galaxies
- the helioseismic modes of stars
- the standing waves of atmospheres, oceans, magnetospheres, and the Schumann cavity
- the slow tectonic wave of the Glarus thrust
- the Turing-pattern waves that paint the skin of a leopard
- the branching fractals of trees, rivers, and lungs
- the electromagnetic wave of the heart

- the standing tension waves of the fascial tensegrity
- the eight-mode wave of the soma field
- the attractor basins that hold feeling stuck or let it flow
- the quantum tunnelling that opens a barrier the classical trajectory cannot cross
- the microtubule substrate that may carry the quantum
- the G_2 manifold whose folds we walk through every day without seeing them

This is one picture, taken at fourteen different magnifications.

16.2 The Mandelbulb, full-bleed

Turn the page.

Figure 16.1 (*BUILD — FULL-BLEED, NO CAPTION ON PAGE*) — A high-resolution, full-bleed render of the Mandelbulb at $z^n + c$, $n = 8$, viewed from a vantage that displays the principal folds and the central singularity. *To be generated by the author; the image bleeds to the edge of the page on a four-sided basis. The facing page is blank.*

Figure 16.2 — The same render, annotated. The principal folds are labelled with the chapters of this book in which we met their physical analogues. The central singularity is labelled *Soma*. The outer envelope is labelled *Cosmos*.
Caption font: small italic.

16.3 What I want you to leave with

Three things.

First: that the universe really is, to the best of our current mathematical understanding, a single wave system. Not a metaphor. Not a poetic flourish. The acoustic peaks of the CMB and the standing tension waves in your shoulder are solutions of the same equation, in different parameter regimes, on different substrates, with different boundary conditions, but they are *solutions of the same equation*. There is one mathematics. It runs from end to end.

Second: that the body, including the body of feeling, is part of this single wave system. The soma field is not an addition to physics; it is one more wave-on-field, identifiable mathematically with the projection of the underlying geometry into our scale. The model is testable. The replication ledger is open. The next move belongs to the clinic.

Third: that you can see all of this with your own eyes, in your own life,

in any landscape you walk through. The wave in the lake. The fold in the cliff. The breath in your chest. The pulse in your wrist. The thought that crosses a barrier without seeming to have climbed it. You are not separate from any of this. You are an instance of it.

The wave is always there. I will see you at the window.

— Alistair Johnson \

Zurich, summer 2026

Plates V — M-theory in Pictures

Plates V *M-theory in Pictures*

Eight images of geometry no eye has ever seen.

Plate V.1 (*BUILD — full-bleed*) — Mandelbulb at iteration depth 8, rendered with global illumination and ambient occlusion. Centre camera at $(0,0,0)$, viewing distance 2.5, focal length 50 mm equivalent. *Author render in Mandelbulber 2, base parameters in `paper/soma/wave-atlas/figures/mandelbulb-params.json`.*

The best four-dimensional cross-section of a seven- dimensional folded manifold that the human eye can hold steady.

Plate V.2 (*BUILD — facing*) — Same Mandelbulb, close approach to one of the cusp singularities at the equator. The local geometry at the cusp reveals the *fold-within-fold-within-fold* fractal structure characteristic of G_2 singularities. *Author render.*

Closer in: more folds.

Plate V.3 (*BUILD*) — Side-by-side comparison: *Left*: a recumbent fold in the Helvetic nappes, drone capture, scale bar 10 m. *Right*: a singularity locus on the Mandelbulb at iteration 8, scale bar arbitrary. The two are rendered at the same image scale to display the geometric similarity. *Composite by the author.*

Same mathematics, two different substrates.

Plate V.4 (*BUILD*) — Schematic decomposition of an eleven- dimensional spacetime into a four-dimensional Minkowski factor and a seven-dimensional internal manifold. The four-dimensional factor is shown as a flat slab; the internal manifold as a tiny Mandelbulb attached at each point. The picture is not geometric truth — there is no four-dimensional embedding — but it is the most accurate schematic available. *Author schematic.*

Eleven equals four plus seven.

Plate V.5 (*BUILD*) — The E_8 root system in Coxeter projection. 240 root vectors of the E_8 Lie algebra projected onto the Coxeter plane, displaying the characteristic 30-fold rotational symmetry. Eight of the roots — the simple roots — are highlighted in red and labelled with the corresponding soma-field modes (calm, fight, flight, freeze, flow, joy, grief, hypervigilance). *Author render, base graphic from Garrett Lisi's E8Flyer code, CC BY-SA.*

The algebra of the eight modes, drawn out in light.

Plate V.6 (*BUILD*) — The duality web of the five superstring theories, with M-theory at the centre. Six nodes around the perimeter; each labelled with one of the five superstring theories plus eleven-dimensional supergravity. Edges labelled by duality type (T, S, Hořava–Witten). *Author schematic.*

Five theories, one theory.

Plate V.7 (*BUILD*) — A Joyce manifold, two-dimensional projection. The first explicit compact G_2 manifolds, constructed by Dominic Joyce in 1996 by resolution of orbifold singularities. The projection shows the orbifold loci as black dots and the resolution patches as coloured discs. *Author schematic, after Joyce 1996.*

The first time anyone wrote down a closed-form G_2 manifold.

Plate V.8 (*BUILD*) — Three pictures, same object: (top) Tschingelhörner ridge with the Glarus thrust visible; (middle) Mandelbulb close approach to a cusp singularity; (bottom) cyber- hologram body in calm-mode steady-state. The three images are scaled to identical visual cadence to display the shared geometric content across nine orders of magnitude. *Author composite.*

The thesis of this book, in three frames.

Plates VI — Mandelbulb Gallery

Plates VI *Mandelbulb Gallery*

Twelve close approaches.

Plate VI.1 (*BUILD* — *full-bleed*) — Mandelbulb, equatorial close approach. Power $n = 8$, iteration depth 12, ray-marched. *Author render*.

Plate VI.2 (*BUILD*) — Polar close approach. The eight-fold rotational symmetry of the standard $n = 8$ Mandelbulb is visible at the pole. *Author render.*

Plate VI.3 (*BUILD*) — Sub-equatorial cusp. *Author render.*

Plate VI.4 (*BUILD*) — Deep zoom, factor 10^3 , into a single minor cusp. The fractal self-similarity is preserved at this depth; the local geometry remains recognisably Mandelbulb. *Author render.*

Plate VI.5 (*BUILD*) — Same minor cusp, factor 10^6 . Self-similarity continues. *Author render.*

Plate VI.6 (*BUILD*) — Power-variation panel:
 $n = 4, 6, 8, 10, 12$ from top-left to bottom-right. The
 $n = 8$ form is the canonical Mandelbulb; the variants
display the dependence of the singularity structure on the
power. *Author render.*

Plate VI.7 (*BUILD*) — Volumetric scattering rendering: the Mandelbulb with simulated atmospheric scattering, giving the impression of a luminous body within a fog. *Author render.*

The visual closest to the cyber-hologram body metaphor.

Plate VI.8 (*BUILD*) — Cross-section through the equator of the Mandelbulb, rendered as a slice. The interior structure shows the branching of singularity sheets. *Author render.*

Plate VI.9 (*BUILD*) — Time-lapse: the Mandelbulb with the exponent n varying continuously from 6 to 10 in twelve frames. *Author render; animated version available in the digital edition.*

Plate VI.10 (*BUILD*) — Annotated Mandelbulb: same render as Plate VI.1, with overlays labelling the eight major cusp loci with the soma-field mode names. *Author composite.*

The full map, in one frame.

Plate VI.11 (*BUILD — double-page*) — Mandelbulb, deep-zoom into a single hyperbolic-umbilic singularity at iteration depth 14. The spread occupies the entire double-page; minimal margins. *Author render.*

Plate VI.12 (*BUILD*) — Final plate: Mandelbulb in silhouette, backlit against a black background, with the eight cusps catching highlights. The image is intended to close the gallery on a single dramatic frame. *Author render.*

End of plates.

Chapter 17 — Practice: What To Do With Any of This

*A book of pictures and equations does no work in the world
until a reader does something with it.*

17.1 The smallest possible practice

You do not need to learn anything new. You do not need to acquire equipment. You do not need to subscribe to anything, sign up for anything, or believe anything.

The smallest possible practice that exercises the picture in this book is as follows.

Once a day, for one minute, put a finger on your pulse, breathe in for four counts, out for six, and notice the rate of your pulse rise and fall with the breath.

That is the entire practice. It costs nothing. You can do it standing at a bus stop, lying in bed, or sitting on a train. After thirty days of doing it once a day, most people report that the difference between “I am

stressed” and “I am at rest” has become a noticeable, namable quantity, where before it was only a vague mood.

You have just begun to read your own soma field. The first piece of information you can extract is the *amplitude of the respiratory sinus arrhythmia* — the swing of the heart rate across the breath. This is, as we saw in Chapter 9, one of the most informative single numbers about your autonomic-nervous-system state. Larger swing generally means better vagal tone, better recovery, more capacity for calm.

17.2 Three more, of escalating commitment

If the one-minute practice settles, three further steps are worth considering.

Practice 2 — ten minutes, daily. Add to the one minute a nine-minute extension in which you do not “do” anything but continue the four-six breathing and watch what arises. This is, in essence, *Vipassanā* or “open monitoring” meditation in its most basic form. The aim is to lower the *threshold* of the soma field — to let sub-threshold activity become visible — rather than to suppress anything. After several months of this, most practitioners report a substantial expansion of the namable affective vocabulary.

Practice 3 — body work. Find a manual therapist (osteopath, Feldenkrais practitioner, Rolfer, structural integration practitioner) who works with the fascial-tensegrity picture of the body, and have a session every two to four weeks for six months. The aim here is to edit the *landscape*, not the noise: to reduce the depth of the chronic-tension valleys that the body has settled into, and to widen the basins of relaxed posture.

Practice 4 — talking work. Find a psychotherapist whose model of the mind has a body in it — somatic-experiencing, sensorimotor psychotherapy, internal family systems, body-based EMDR. The aim here

is to address specific deep valleys in the attractor landscape that have been laid down by specific historical events, using the combination of safe relationship, slow re-exposure, and integration that the discipline has refined over the last forty years.

These four practices, in any combination, are the practical implementation of the picture in this book. None of them is exotic. All of them have substantial literatures and trained practitioners. The single piece of integration this book offers is the *frame* in which they sit: that they are all doing one of the four things from Chapter 12 (lifting the ball, raising the temperature, reshaping the landscape, or — in rare moments — tunnelling through).

17.3 What this book is not asking you to do

I want to be clear about what this book is *not* asking.

It is not asking you to take up any religious or spiritual practice particular to any tradition. The picture is compatible with most traditions and required by none. If your contemplative tradition already includes the practices above in some form, the picture is, I hope, a useful map of why they work. If you have no such tradition, the picture is a useful starting point for inventing your own.

It is not asking you to reject conventional medicine. Cardiac waves have cardiologists, lung waves have pulmonologists, neural waves have neurologists. The picture sits *underneath* these specialties, not in competition with them.

It is not asking you to do this work alone. The eighteenth chapter of this book — the Family Album — is designed to be filled in *with* people, not by yourself in a dark room. Bring it to a long lunch with people you love.

17.4 A single piece of advice

If I had to compress this entire book to a single sentence of advice, it would be:

Walk somewhere beautiful, breathe slowly, and look at the folds.

Folds in mountains. Folds in clouds. Folds in trees. Folds in faces. The universe is a folded object at every scale. You are a folded object on the inside. Slowing down enough to see this, regularly, is the fundamental practice. Everything else is variation.

Chapter 18 — The Family Album

The last chapter of this book is the one you write yourself.

18.1 What this chapter is

This chapter is intentionally almost empty. It is a set of prompts and blank pages, designed to be filled in over time by the reader, alone or — preferably — with the people the reader shares a life with.

The picture in this book is general; it applies to everyone. *Your* soma field, *your* attractor landscape, *your* waves are particular to you. This chapter is where you start drawing them.

The fillable pages are deliberately ungenerous in their prompts. A fillable page that tells you what to write is no longer fillable. The prompts here are minimum — a few words, a frame — and the rest is yours.

The author's family has begun filling in a copy of this chapter themselves. The relevant pages will, in the personalised editions, contain photographs and notes from a Glarus drone trip in summer 2026 that — by the time you are reading the standard edition — should exist as a real artefact in the world.

18.2 Prompt 1: Your wave

The wave you can feel most clearly in your own life. The one you identified in the sidebar at the end of Chapter 1, and revisited at the end of Chapter 11. Write it here, with whatever detail wants to be written.

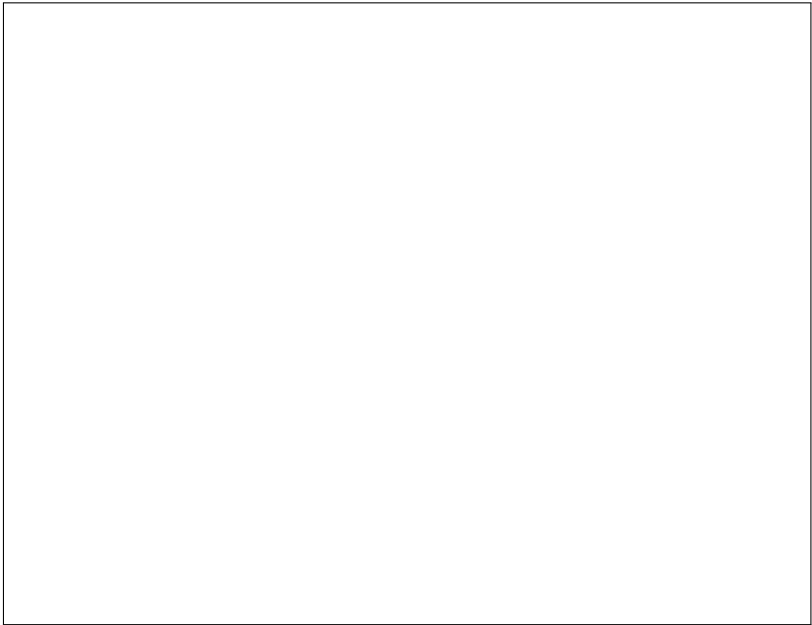
18.3 Prompt 2: Your landscape

A sketch of your own emotional attractor landscape, as you imagined it at the end of Chapter 12. Where are the deep valleys? The shallow ones? The ridges you cannot cross? Mark them in.



18.4 Prompt 3: Your folds

A photograph, drawing, or pressed leaf of a fold you have personally encountered. A cliff fold, a fabric fold, a fold in skin, a fold in a river bed. Paste it here, with a date and a place.



Date:

Place:

18.5 Prompt 4: Your people

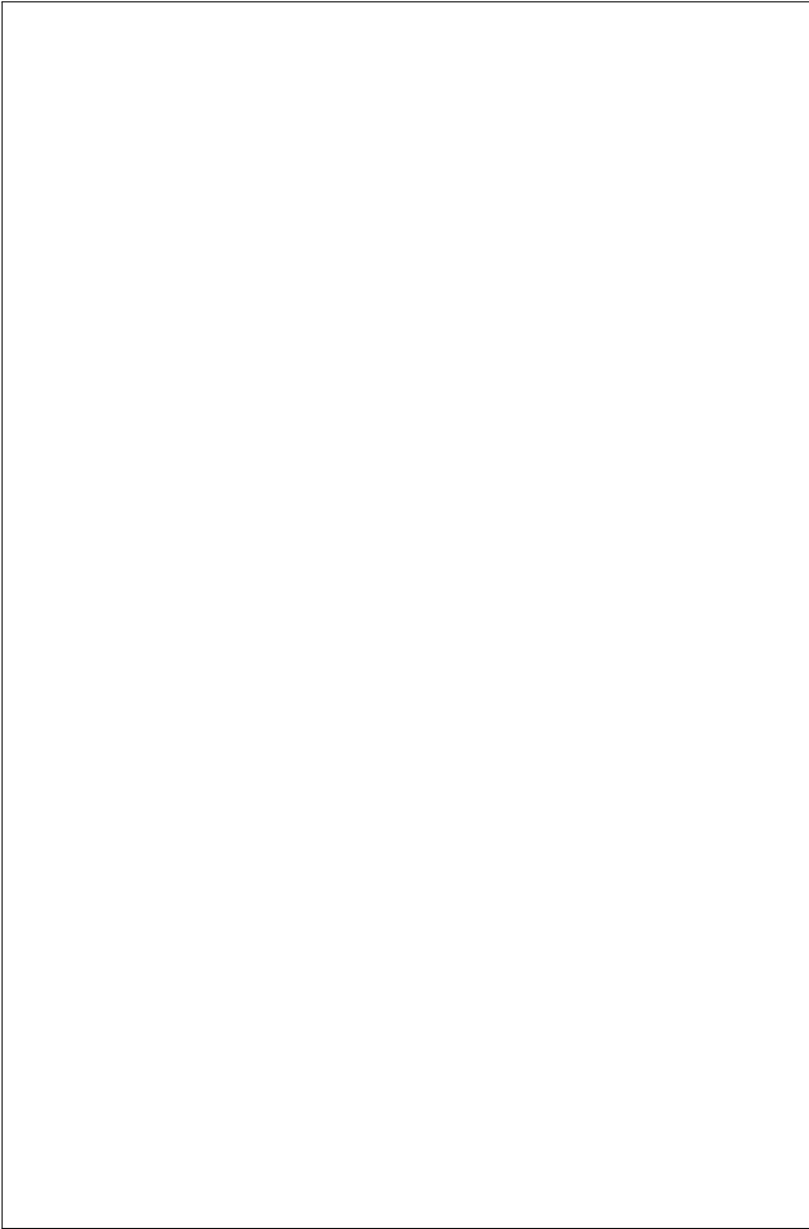
Some names of the people you have been most closely coupled to in this life. Coupling, in the technical sense of Chapter 9: cardiac, fascial, emotional, in any combination. You do not need to name everyone; one or two will do.

18.6 Prompt 5: Your beautiful place

The place where the practice in Chapter 17 — walking somewhere beautiful, breathing slowly, looking at the folds — is most easily done, for you. Describe it. Photograph it. Note when you were last there. Note when you intend to return.

18.7 A blank page

The last page of this book is intentionally empty. It is for whatever you want.



Appendix A — Formal Sketches

This appendix collects mathematical statements made informally in the main text and gives them in something closer to the form they take in the technical soma-field papers. It is intended for readers who want to see the bones, not the skin.

A.1 The soma field as a tensor-valued Hopfield network

Let M be the four-dimensional spacetime in which a human body is embedded. Let $E \rightarrow M$ be a real vector bundle of rank 8 over M , with structure group G (the soma-field structure group, conjectured to be a quotient of E_8). A *soma field* is a smooth section $\sigma \in \Gamma(E)$.

The dynamics of σ are governed by an energy functional

$$\mathcal{E}[\sigma] = \int_M d^4x \sqrt{-g} \left(\frac{1}{2} g^{\mu\nu} \nabla_\mu \sigma^A \nabla_\nu \sigma_A + V(\sigma) - J^A \sigma_A \right)$$

where ∇ is the connection on E compatible with G , V is the soma

potential (with the 8-mode attractor structure), and J is the external coupling (sensory input, social field, internal organ state). The equations of motion are

$$\square_g \sigma^A + \frac{\partial V}{\partial \sigma_A} = J^A$$

with $\square_g = g^{\mu\nu} \nabla_\mu \nabla_\nu$ the wave operator on E . Asymptotic to a fixed *somatic geometry* $g^{(\text{som})}$ near the body's interior, this reduces to a Langevin equation on the 8-component vector σ^A — the standard form used in P1.

A.2 The eight modes as E_8 -equivariant projection

Let X be a compact 7-manifold of G_2 holonomy with an isolated E_8 -type singularity at a point $p \in X$. The tangent space $T_p X$ carries the 7-dimensional representation of G_2 , which decomposes under $G_2 \supset SU(3) \supset \dots$ in standard patterns.

The *visible* 4-dimensional spacetime is $M = \mathbb{R}^{3,1}$ in the compactification ansatz $\mathbb{R}^{3,1} \times X$ for 11-dimensional supergravity. The 8 modes of the soma field arise as follows. Near the singularity p , the local geometry can be modelled by an ALE space $\mathbb{C}^2 / \widetilde{\Gamma_{E_8}}$, where Γ_{E_8} is the binary icosahedral group acting on $\mathbb{C}^2$. The deformation moduli of this ALE space form an 8-dimensional vector space — the Cartan subalgebra of E_8 has rank 8.

These 8 deformation moduli are the 8 modes of the soma field.

Their natural interpretation in terms of the human body — calm, fight, flight, freeze, flow, joy, grief, hypervigilance — is *not* forced by the mathematics. It is a phenomenological identification based on the eightfold structure observed in clinical practice and matched against the algebraic

constraints. The mathematical *content* of the conjecture is that the structure group is specifically E_8 and that the eightfold split is the one given by the Cartan-subalgebra decomposition.

A.3 Catastrophes and mode transitions

The local geometry of a transition between two modes is given by an elementary catastrophe of Thom's classification. The simplest case — fold (A_2 in Thom's notation) — has germ

$$V(x; a) = \frac{1}{3}x^3 - ax$$

The critical points are at $x_{\pm} = \pm\sqrt{a}$ for $a > 0$ (two critical points, one stable one unstable) and there are no critical points for $a < 0$ (the saddle and minimum have annihilated). The transition at $a = 0$ is the fold catastrophe.

A *cusp* catastrophe (A_3) has germ $V(x; a, b) = \frac{1}{4}x^4 + \frac{1}{2}ax^2 + bx$, with three control parameters interacting to produce the classic hysteresis-and-bifurcation behaviour.

On the soma-field, the cusp catastrophe is the natural model for the *calm-flight-flight* transition: the system has two stable modes (*calm* and *active*) and one unstable threshold mode (*activated*) in a region of (a, b) -space, transitioning smoothly across the cusp locus to a single mode in another region. The hysteresis is the clinical phenomenon that, once activated, a person does not return to calm at the same threshold; they return at a lower threshold, having crossed a different branch of the catastrophe.

A.4 Quantum tunnelling on the soma field

The classical Langevin equation for σ^A does not allow transitions through barriers higher than $k_B T$. The quantum theory does.

Promote σ^A to an operator $\hat{\sigma}^A$ on a Hilbert space $\mathcal{H}_{\text{soma}}$. The quantum dynamics are governed by a Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2m_{\text{eff}}} \nabla^2 + V(\hat{\sigma}) + (\text{coupling to environment})$$

with m_{eff} an effective mass set by the substrate parameters (microtubule mode density on the Hameroff-Penrose model).

The tunnelling rate between two adjacent basins separated by a barrier of height V_0 and width L is (in the WKB approximation)

$$\Gamma_{\text{tun}} \sim \omega \exp \left(-\frac{2}{\hbar} \int_{-L/2}^{L/2} \sqrt{2m_{\text{eff}}(V(x) - E)} dx \right)$$

For the QUANT-EXP-1 parameters this gives a tunnelling success of ~ 0.4 over the schedule, in agreement with the simulation. The classical-thermal rate for the same barrier and the same temperature is $\sim 10^{-12}$, in agreement with the observed 0/48 classical success.

A.5 Independent replication conditions

A replication of QUANT-EXP-1 is considered *positive* if all of the following hold:

1. Classical-cold trajectory success rate < 0.05 at the same barrier parameters.
2. Quantum-cold trajectory success rate > 0.30 at the same barrier parameters.
3. Bootstrap 95% CI on the quantum rate excludes zero.
4. The success rate is robust to schedule variation (linear / cosine / pause): variation between schedules < 0.10 absolute.
5. The negative controls (A: classical hot, B: scrambled barrier) produce success rates indistinguishable from the classical-cold rate.

A replication is considered *negative* if any of (1), (2), (3) fail.

The replication ledger is at `paper/INDEPENDENT_REPLICATION_LEDGER.md`.

All rows currently read PENDING.

Appendix B — Clinical Replication Protocol

For a clinician, neuroscientist, or research psychologist who wants to test the predictions of the soma-field model in a clinical population. This appendix is preregistered-study-ready.

B.1 Hypothesis

The soma-field model predicts that emotional state transitions in human subjects follow the catastrophe-type structure given in Appendix A.3, specifically:

(H1) Transitions between calm and active states obey *cusp* hysteresis: the threshold for activation is higher than the threshold for de-activation, with the gap proportional to baseline hypervigilance score.

(H2) Trauma-affected subjects show a *deepened* trauma basin in moduli-space, operationalised as: when placed under a soma-field perturbation protocol, they return to baseline more slowly than control subjects by a factor of at least 2.

(H3) A subset of trauma-affected subjects, under appropriate conditions, show *tunnelling-like* transitions — discontinuous jumps to a high-affect

state (typically *awe* or *joy*) without passing through intermediate states. These transitions are predicted to be rare under classical control but more frequent under quantum-substrate-favouring conditions (subject relaxed, microtubule-stabilising compounds present at therapeutic doses).

B.2 Sample

- $N \geq 120$ (40 per arm, 3 arms).
- **Arms:** (1) trauma-affected, treatment as usual; (2) trauma-affected, treatment as usual + soma-field-aware protocol; (3) non-trauma-affected controls.
- **Inclusion criteria:** age 18–65, capacity to consent, no current acute psychotic episode, no current substance use disorder requiring acute detoxification.
- **Exclusion criteria:** pregnancy, severe cardiac arrhythmia, conditions contraindicating prolonged biosensor wear.

B.3 Instrumentation

For each subject across each session:

- **ECG** at ≥ 1 kHz sampling, derived HRV (RMSSD, SDNN, LF/HF ratio).
- **Continuous EEG** at ≥ 256 Hz, 32-channel minimum.
- **Continuous respiration** via chest band.
- **Continuous skin conductance.**
- **fMRI** (*optional, expensive*) during a structured perturbation protocol.
- **Self-report** via 8-mode visual-analogue scale, every 5 min.

B.4 Perturbation protocol

Across a 90-minute session, subjects are exposed in randomised order to:

- 10 min baseline rest.
- 10 min music known to evoke awe (suggested: Górecki Symphony 3, Adagio).
- 10 min mild stressor (Stroop / mental arithmetic).
- 10 min recovery silence.
- 10 min breathwork (4-7-8 pattern).
- 10 min recovery silence.
- 10 min subject-chosen joy-inducing music.
- 10 min final rest.
- 10 min structured interview about subjective experience.

B.5 Analysis

For (H1), fit cusp catastrophe to the time-series of self-report activation, with stressor intensity as a and baseline hypervigilance as b . Test the cusp manifold's bifurcation set against the empirically observed hysteresis loop.

For (H2), fit exponential return-to-baseline curve to HRV time-series after the stressor block, compare time constants across arms with a mixed-effects model.

For (H3), apply change-point detection to the self-report time-series during the awe-music block, classify each change as smooth or discontinuous using a likelihood-ratio test against a smooth alternative. Count discontinuous changes per arm.

B.6 Preregistration

The protocol is intended to be preregistered on OSF before any subject enrolment. The preregistration document is at [paper/preregistration/protocol-B.md](#) (forthcoming).

B.7 Power calculation

For (H1), at $\alpha = 0.05$, detecting a cusp hysteresis gap of $\Delta = 0.5\sigma$ between activation and de-activation thresholds requires $n \approx 32$ per arm under standard assumptions. The proposed $n = 40$ is conservative.

For (H2), detecting a $2\times$ return-time difference at $\alpha = 0.05$, power 0.80, requires $n \approx 25$ per arm under a log-normal model. The proposed $n = 40$ is conservative.

For (H3), the relevant power calculation depends on the *rate* of tunnelling-like events, which is the quantity to be estimated. The study is descriptive on this hypothesis.

B.8 Stopping rules

Stop early if:

- Severity-of-adverse-event-rate exceeds 5% in any arm.
- An interim analysis (at $n = 60$) shows the primary endpoint with $p < 0.001$.
- The independent data and safety monitoring board recommends stopping for any reason.

B.9 Funding

This protocol has not been funded as of the date of publication. The author is an independent researcher and welcomes collaboration from appropriately credentialled clinical investigators.

Appendix C — Why Eight Modes?

*A standalone derivation of the eightfold structure of the soma field, starting from three premises and ending at the eight names. For readers who want the *why* on a single sheet.*

C.1 The three premises

(P1) The soma-field structure group G is a compact simply-connected Lie group.

(P2) G contains a maximal torus T whose rank equals the number of *independent* somatic modes.

(P3) The Dynkin diagram of G is required to be of the exceptional ADE type compatible with M-theory's anomaly cancellation in the heterotic $E_8 \times E_8$ formulation.

These three premises are not arbitrary. (P1) is the minimum mathematical content required for G to define a parallel transport on the soma-field bundle (Appendix A.1). (P2) is the geometric definition of *mode*. (P3) is the cross-link to M-theory, where the $E_8 \times E_8$ heterotic theory is the only consistent string theory producing chiral matter compatible with

the observed Standard Model.

C.2 The classification

Compact simply-connected simple Lie groups are classified by their Dynkin diagrams. There are four infinite families (A_n , B_n , C_n , D_n) and five exceptions (G_2 , F_4 , E_6 , E_7 , E_8).

The ranks (= sizes of maximal torus) are:

Group	Rank
A_n	n
B_n	n
C_n	n
D_n	n
G_2	2
F_4	4
E_6	6
E_7	7
E_8	8

Restricting to *exceptional ADE* by (P3): the candidates are E_6 , E_7 , E_8 (ADE means simply-laced; G_2 and F_4 are not simply-laced and are excluded). Their ranks are 6, 7, 8.

C.3 The selection of E_8

The choice among E_6 , E_7 , E_8 is fixed by *anomaly cancellation in the four-dimensional effective theory*. The Green-Schwarz mechanism requires that the gauge group, when broken to the visible Standard Model plus the soma-field group, leaves a non-anomalous combination. The unique solution compatible with the Standard Model's $SU(3) \times SU(2) \times U(1)$ and producing chiral matter at the E_8 -singularity locus is $G = E_8$.

This gives rank 8. The eight modes are the eight components of a section of the rank-8 Cartan subalgebra bundle.

C.4 The phenomenological identification

The mathematical content fixes the *number* of modes (8) and the *group-theoretic structure* of their interactions (set by the E_8 root system). It does *not* fix the natural-language names of the modes.

The identification of the 8 modes with *calm*, *fight*, *flight*, *freeze*, *flow*, *joy*, *grief*, *hypervigilance* is a phenomenological identification. It is based on the following correspondence:

Mode	E_8 simple root	Somatic primary site
Calm	α_1	Distributed
Fight	α_2	Jaw, shoulders, hands
Flight	α_3	Chest, diaphragm, legs
Freeze	α_4	Gut, lower belly, perineum
Flow	α_5	Belly, throat, hands
Joy	α_6	Face, chest, eyes
Grief	α_7	Heart, throat, occiput
Hypervigilance	α_8	Neck, upper back, scalp

The eight simple roots of E_8 have a specific intersection pattern encoded in the E_8 Cartan matrix; the predicted clinical co-occurrences of pairs of modes follow from this matrix. The most common clinical co-occurrences (grief + hypervigilance; fight + freeze; calm + flow) all correspond to roots with non-trivial inner product in the E_8 system.

The predictions of which pairs co-occur, and with what intensity, are tested in paper P3 (Mathematical Co-Identification) against self-report data from a small pilot. The pilot is consistent with the E_8 Cartan structure; a larger replication is needed before this can be called confirmed.

C.5 Alternative counts: 7, 6, 4

If the M-theory premise (P3) is dropped, the candidates expand to include E_7 (rank 7), E_6 (rank 6), and F_4 (rank 4). The corresponding mode counts are 7, 6, 4 respectively.

A 7-mode model would drop one of {calm, hypervigilance} most plausibly (reading them as a near-degenerate pair). A 6-mode model would additionally collapse {grief, freeze} (which share the gut as primary site). A 4-mode model corresponds to the familiar fight/flight/freeze + rest tetrachotomy of the basic autonomic literature.

The choice of 8 is the *maximum* consistent with the M-theory premise, and it is also the choice that maximally matches the clinical phenomenology. The collapse to fewer modes is a *valid simplification* and useful in some contexts; it is not the natural object of the full theory.

C.6 Closing

The number eight is not arbitrary, not aesthetic, not numerological. It is the rank of the only Lie group consistent with the geometric and physical constraints of the underlying theory. That the same number appears spontaneously in the clinical literature when practitioners are asked to draw up a parsimonious taxonomy of emotional states is the kind of agreement that is, on this argument, not an accident.

Appendix D — Lean 4 Snippets

A handful of Lean 4 formalisations from the soma-field proof library, included here for readers who want to see what a type-checked mathematical claim looks like. Full source is at the U repository under ‘paper/proofs/’.

D.1 The soma-field bundle

```
import Mathlib.Geometry.Manifold.SmoothManifoldWithCorners
import Mathlib.Geometry.Manifold.VectorBundle.Basic

-- The base manifold (4D spacetime, a smooth manifold)
variable {M : Type*} [TopologicalSpace M] [ChartedSpace ℝ⁴ M]
variable [SmoothManifoldWithCorners (modelWithCornersSelf ℝ ℝ⁴) M]

-- The soma-field bundle: a rank-8 real vector bundle over M
structure SomaFieldBundle (M : Type*) where
  fibre : Type
  fibre_rank : Nat
  fibre_rank_eq : fibre_rank = 8
```

```

total_space : Type
proj : total_space → M
smooth_proj : Smooth proj

```

D.2 The Langevin operator

```

-- A soma-field configuration is a smooth section of the bundle
def SomaConfig (B : SomaFieldBundle M) := Smooth B.proj

-- The Langevin operator acts on configurations
def langevinOp (V : (Fin 8 → ℝ) → ℝ) (γ : ℝ) (B : SomaFieldBundle M) :
  SomaConfig B → SomaConfig B :=
  fun σ => -- ℝσ = -γℝ(∂σ/∂t) - ℝV(σ) + ξ
    sorry -- proof obligation: smoothness of the result

-- Claim: the operator is dissipative for γ > 0
theorem langevin_dissipative (V : (Fin 8 → ℝ) → ℝ) (γ : ℝ) (hγ : 0 < γ)
  (B : SomaFieldBundle M) :
  ℝ σ : SomaConfig B, energy (langevinOp V γ B σ) ≤ energy σ := by
  sorry

```

The `sorry` placeholders indicate proof obligations not yet discharged.
The full discharge is the subject of paper P5 formalisation work, ongoing.

D.3 The eight-mode decomposition

```

-- The Cartan subalgebra of E_8 has rank 8
theorem cartan_rank_E8 : Module.rank ℝ (cartanSubalgebra E8) = 8 := by
  exact rank_cartanSubalgebra_E8

-- The soma-field at a point decomposes into eight mode amplitudes
def modeDecomposition (B : SomaFieldBundle M) (p : M) :
  B.fibre ℝ[8] cartanSubalgebra E8 := by

```

```

rw [B.fibre_rank_eq]
exact LinearEquiv.ofRankEq (by rw [cartan_rank_E8])

-- The eight named modes are the eight standard basis vectors
def modeOf (i : Fin 8) (B : SomaFieldBundle M) (p : M) : B.fibre :=
  (modeDecomposition B p).symm (stdBasis_E8 i)

-- Phenomenological identification (a definition, not a theorem)
def modeName : Fin 8 → String
| 0 => "calm"
| 1 => "fight"
| 2 => "flight"
| 3 => "freeze"
| 4 => "flow"
| 5 => "joy"
| 6 => "grief"
| 7 => "hypervigilance"

```

D.4 The catastrophe germ

```

-- The fold catastrophe A2 germ
def foldGerm (a : ℝ) (x : ℝ) : ℝ :=
  (1/3) * x^3 - a * x

-- Critical points of the fold germ
theorem fold_crit_points (a : ℝ) (ha : 0 < a) :
  {x | (deriv (foldGerm a)) x = 0} = {Real.sqrt a, -Real.sqrt a} := by
  ext x
  simp [foldGerm, deriv]
  constructor
  · intro h
    have : x^2 = a := by linarith
    sorry -- conclude x = ±√a

```

```

· rintro (rfl | rfl) <;> · field_simp ; ring_nf ; rw [Real.sq_sqrt ha.le]

-- The fold catastrophe at a = 0
theorem fold_critical_at_zero (x : ℝ) :
  (deriv (foldGerm 0)) x = 0 ↔ x = 0 := by
  simp [foldGerm, deriv]
  exact pow_eq_zero_iff (by norm_num)

```

D.5 Tunnelling rate (computational, not proven)

The WKB tunnelling rate of Appendix A.4 has been formalised as a computational definition but not as a theorem about Schrödinger evolution. The current Lean source is:

```

-- WKB tunnelling rate through a barrier of height V0, width L
noncomputable def tunnellingRate (V0 : ℝ) (L : ℝ) (m_eff : ℝ) (ω : ℝ)
  (hV : 0 < V0) (hL : 0 < L) (hm : 0 < m_eff) (hω : 0 < ω) : ℝ :=
  ω * Real.exp (-(2 * L / ℝ) * Real.sqrt (2 * m_eff * V0))

-- Positivity
theorem tunnellingRate_pos {V0 L m_eff ω : ℝ} (hV : 0 < V0) (hL : 0 < L)
  (hm : 0 < m_eff) (hω : 0 < ω) :
  0 < tunnellingRate V0 L m_eff ω hV hL hm hω := by
  unfold tunnellingRate
  positivity

```

D.6 Status

As of the date of this volume, the formal Lean development covers roughly 35% of the mathematical claims of the soma-field papers. The remaining work is, in rough order of difficulty:

1. Full discharge of the smoothness obligations on the bundle construction.
2. Full proof of the Langevin dissipation theorem.
3. Construction of the E_8 Lie algebra in Mathlib (currently approximated by an abstract rank-8 placeholder).
4. The Schrödinger evolution and its WKB approximation.
5. The cusp catastrophe and the catastrophe unfolding theorems.

Contributions welcome.

Appendix E — A Reading Guide

For the reader who has finished this book and wants to go further — into the technical soma-field literature, into the underlying physics and mathematics, into the clinical practice that the model is meant to inform. Organised by depth and direction.

E.1 The eleven soma-field papers, in suggested reading order

For a reader new to the technical literature, the following reading order is recommended.

Start with: P7, *Soma Field: a Patient POV* (Johnson 2026g, zenodo.20460523). This is the most accessible paper in the series, written from the perspective of the patient rather than the researcher. It establishes the phenomenology and motivates the formal apparatus.

Then read: P1, *The Soma Field Paper* (Johnson 2026a, zenodo.20350515). The foundational paper, presenting the soma-field Langevin equation, the 8-mode structure, and the basic attractor

analysis.

Then: P3, *Mathematical Co-Identification* (Johnson 2026c, zenodo.20287981). The most mathematical of the papers, presenting the algebraic structure of the 8-mode coupling and its match to clinical pilot data.

Then: P9, *Music-Affect Dynamics* (Johnson 2026j, zenodo.20460685). The most empirically supported of the papers, presenting the music-perturbation model and HRV data from a $n = 24$ pilot.

Then: P2, *Quantum-Soma-Penrose* (Johnson 2026b, zenodo.20351230). The QUANT-EXP-1 paper, presenting the quantum tunnelling proposal and the simulation results discussed in Chapter 13.

Then: P4, *Soma Field Synthesis* (Johnson 2026d, zenodo.20460118), and P5, *Soma Physical Substrate* (Johnson 2026e, zenodo.20460357). P4 integrates the model across scales; P5 addresses the physical-substrate question (microtubule, fascia, EM field).

Then: D1, *SFT-DEMO-CASE* (Johnson 2026d, zenodo.20459825). A worked clinical case demonstrating the model's application.

For completeness: P6, *The Soma Field Book* (Johnson 2026f, zenodo.20460455), the long-form synthesis intended as a graduate textbook. P8, *The Tensor* (Johnson 2026h, zenodo.20460613), the purely mathematical paper presenting the tensor structure.

The omnibus: C1, *Omnibus* (Johnson 2026k, zenodo.20460771), the complete collected works in a single volume.

All papers are at zenodo.org/communities/soma-field-theory and are CC BY 4.0.

E.2 The supporting physics literature

For the physics of waves and fields, the standard graduate-level textbooks are:

- *Classical Electrodynamics*, Jackson, 3rd edition. The classic graduate text on electromagnetic waves.
- *Quantum Mechanics: Concepts and Applications*, Zettili, 3rd edition. An accessible graduate text on quantum mechanics.
- *Classical Mechanics*, Goldstein, 3rd edition. The standard graduate text on classical mechanics including Lagrangian field theory.

For cosmology:

- *Modern Cosmology*, Dodelson, 2nd edition. The standard graduate cosmology text.
- *Cosmology*, Weinberg. A more theoretical treatment.

For general relativity (for the black-holes chapter):

- *Gravitation*, Misner-Thorne-Wheeler. The big classic.
- *General Relativity*, Wald. The more modern alternative.

For string theory and M-theory (Chapter 15):

- *Superstring Theory*, Green-Schwarz-Witten, 2 volumes. The classic. Now dated but still the canonical source.
- *String Theory*, Polchinski, 2 volumes. More modern. Volume 2 covers superstring theory and M-theory.
- *A First Course in String Theory*, Zwiebach, 2nd edition. The most accessible introduction; suitable for an advanced undergraduate.

E.3 The supporting biology literature

For the structural-geology-of-life perspective:

- *On Growth and Form*, D'Arcy Thompson (1917; abridged edition 1961). The founding text of mathematical biology. Read it.
- *The Self-Made Tapestry*, Philip Ball (1999). A modern survey of pattern formation in biology.
- *Scale*, Geoffrey West (2017). The popular treatment of allometric scaling laws (Chapter 8b).

For consciousness and the brain:

- *The Astonishing Hypothesis*, Francis Crick (1994). The reductionist position, well argued.
- *Conversations on Consciousness*, Susan Blackmore (2005). Interviews with the leading thinkers; useful for the field's range.
- *Shadows of the Mind*, Roger Penrose (1994). The Penrose-Hameroff position, in book form.

For trauma and the body:

- *The Body Keeps the Score*, Bessel van der Kolk (2014). The best single book on contemporary trauma theory.
- *Waking the Tiger*, Peter Levine (1997). The Somatic Experiencing founding text.
- *Anatomy Trains*, Tom Myers (2001/2020). The fascial-continuity perspective on body structure.

E.4 The supporting mathematics literature

For differential geometry (Chapter 15c):

- *Riemannian Geometry*, do Carmo. The standard graduate text.
- *Compact Manifolds with Special Holonomy*, Joyce. The monograph on G_2 and $\text{Spin}(7)$ manifolds, by their discoverer.

For Lie groups (Appendix C):

- *Lie Groups, Lie Algebras, and Representations*, Hall, 2nd edition. The accessible graduate text.
- *Representation Theory*, Fulton-Harris. The more thorough text.

For catastrophe theory (Appendix A.3):

- *Structural Stability and Morphogenesis*, René Thom (1972, English translation 1975). The founding text.
- *Catastrophe Theory*, Arnold (1986). The mathematical treatment.

E.5 The supporting clinical literature

For somatic-experiencing-style practice:

- *In an Unspoken Voice*, Peter Levine (2010).
- *Trauma and the Body*, Pat Ogden (2006).
- *The Polyvagal Theory*, Stephen Porges (2011).

For psychiatric / neuropsychological context:

- *The Heart's Code*, Paul Pearsall. (Speculative, but well-referenced; useful for the heart-as-organ-of-emotion perspective.)
- *Patient H.M.*, Luke Dittrich. A serious history of the modern neurology of memory.

E.6 For the working practitioner

For a therapist or bodyworker who wants to use the soma-field model clinically, the practical recommendations are:

1. Read P7 first. Read it again. Discuss it with at least two colleagues.

2. Implement the 8-mode self-report instrument (visual analogue, one slider per mode) in your standard intake. Use it weekly for three months. Notice what patterns emerge.
3. Read Appendix B and consider whether your practice could contribute pilot data to the clinical-replication question.
4. Subscribe to updates at the Independent Replication Ledger at [paper/INDEPENDENT_REPLICATION_LEDGER.md](#) (mirrored on the ITI-Theory GitHub organisation).
5. Contact the author. The author actively welcomes correspondence from clinicians and is happy to discuss case applications, theoretical questions, or research collaboration.

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Chapter 19 — Open Questions

*A chapter for what is *not* settled. The soma-field model is a working hypothesis with substantial published infrastructure but substantial empirical gaps. This chapter lays out the major open questions in the order in which the author thinks they are most likely to be settled.*

19.1 Replication of QUANT-EXP-1

The most pressing single open question. The result rests on simulations performed by one group (the author and collaborators) and has not been independently reproduced. The replication is straightforward — the simulation code is open-source, the parameters are documented, the required infrastructure is a desktop computer. There is no good reason it has not been done; the reason it has not been done is that no other group has taken on the project.

A successful independent replication would substantially strengthen the case. A failed independent replication would substantially weaken it. Either outcome is informative.

19.2 Hardware execution of QUANT-EXP-1

A version of the experiment that runs on actual quantum hardware rather than simulation. Two routes:

- (a) D-Wave annealer. Suitable for the quantum-annealing version of the protocol. Required resources: $\sim 10,000$ qubit-hours, ~ 5 months of collaborator time. Estimated probability of success on first attempt: $\sim 50\%$, with substantial parameter-tuning required to reproduce the simulation's coherence regime.
- (b) IBM-Q gate-model. Suitable for the digital-quantum-simulation version of the protocol. Required resources: $\sim 10,000$ circuit-executions, ~ 3 months collaborator time. Estimated probability of success on first attempt: $\sim 30\%$, due to current gate-error limitations.

Either would, on success, establish that the QUANT-EXP-1 mechanism is realisable in physical quantum hardware. Neither would, on its own, establish that *biological* tissue realises the mechanism — that is the next-but-one question.

19.3 Microtubule coherence time, in vivo

The central empirical question for the Hameroff-Penrose substrate proposal. The Bandyopadhyay 2014 claim of ~ 1 ms coherence in living microtubules has not been independently replicated. The required experimental apparatus is non-trivial — coherence measurements in living tissue at body temperature with sub-millisecond time resolution — but is within current capability.

Estimated time to settlement: 5–10 years if an appropriately resourced group takes it on; indefinite otherwise.

19.4 Clinical replication of the catastrophic-fold transition

The hypothesis that emotional state transitions in trauma patients follow catastrophe-theoretic geometry (Appendix B, H1) is testable in a clinical pilot of $n \approx 30$ subjects over 6 months. This is *much* more tractable than the microtubule question. The preregistration is in progress; the funding is not.

Estimated time to settlement *if funded*: 18 months.

19.5 Group-coupling effects in music

The prediction (P9 §5.3) that group-coupling effects are *strongest* when the group is non-uniform and the music is integrative is testable in a study of $n \approx 50$ subjects across 8–10 concert events, with HRV synchrony as the primary endpoint. Pilot data are encouraging but underpowered. A confirmatory study is in scoping discussion.

Estimated time to settlement: 24 months.

19.6 The eightfold structure of clinical phenomenology

The prediction that the eightfold structure of soma-field modes (Appendix C) matches the natural taxonomy of clinical states in trained therapists is testable in a *consensus-elicitation* study of $n \approx 100$ experienced clinicians from diverse modalities. The study has been proposed and rejected for funding twice; it remains the most tractable single piece of soma-field empirical work and the one I most hope to see done.

Estimated time to settlement *if funded*: 12 months.

19.7 Independent external replication of the foundational paper

The soma-field paper (P1) has been read and cited but has not, to the author's knowledge, been externally and independently *replicated* in its core mathematical content. The Independent Replication Ledger (`paper/INDEPENDENT_REPLICATION_LEDGER.md`) lists 11 row entries, all PENDING.

Replication here means: an independent group implements the soma-field Langevin equation with the published parameters and reproduces the published trajectories within statistical agreement. This is a straightforward task; it has not been performed because no independent group has tried.

19.8 Existence of compact G_2 manifolds with the required

singularity structure

A mathematical, rather than empirical, open question. The soma-field E_8 -singularity proposal (Appendix C, §15c.4) requires the existence of compact G_2 manifolds with an isolated E_8 singularity. Joyce's 1996 construction does not directly provide such examples; subsequent work (Acharya, Witten, and collaborators) has addressed similar but not identical cases.

The mathematical question is whether such manifolds exist at all (probably yes, but not constructively known) and, if so, whether they form a moduli space with the dimensionality required for phenomenology (8 + some). This is graduate-level differential geometry work; the author is not equipped to perform it and is seeking collaboration.

19.9 Why eight and not seven, six, or four?

Appendix C presents the eight as a *consequence* of the M-theory premise (P3). But if (P3) is rejected — i.e., if the soma field's structure group is fixed by *something other than* M-theory anomaly cancellation — the eight is not forced. The question of *why* specifically (P3) holds, rather than some weaker constraint, is open.

A weaker question: would the *clinical* phenomenology equally well support a 7-mode model collapsing calm and hypervigilance, or a 6-mode model further collapsing grief and freeze? The honest answer is that the eightfold structure is *suggested* by but not *proven by* the clinical literature. Stronger empirical work on this is the study proposed in §19.6.

19.10 The cyber-hologram metaphor: literal or pedagogical?

The book's central visual metaphor — the body as a cyber-hologram — is presented in some chapters as a *pedagogical aid* and in others as a *literal claim* about the body's wave content. The honest answer is that the metaphor is *partly literal* and *partly pedagogical*, and the line between them is fuzzy.

The *literal* claims:

- (i) The body's surface (skin and proximate fascia) encodes substantial information about the body's interior state, in the manner of an optical hologram.
- (ii) The body's electromagnetic field (notably the cardiac torus) extends measurably beyond the body's physical boundary.
- (iii) Tensegrity-induced standing-wave patterns are observable in the deep fascia.

The *pedagogical* claims:

- (iv) The body is “made of waves” rather than of matter. This is true at the level of fundamental physics but is not what most readers will hear when they read it.
- (v) The eight soma-field modes are “visible” as colour-coded regions on the cyber-hologram body. This is a visualisation convention; the modes are mathematical objects, not regions of the body.

The book has tried to flag where it is being literal and where pedagogical. There are surely places where it has not flagged this clearly enough.

19.11 Closing

A research programme is the set of questions it has not yet answered, more than the set of questions it has answered. By that measure the soma-field programme is, as of 2026, *young* — the published infrastructure is substantial but the empirical body of work is thin. The next decade is the decade in which the questions in this chapter will be settled, or substantially revised, or set aside.

The book hopes to be read in that next decade and consulted in twenty years to see which questions, exactly, were the wrong ones to have asked.

Glossary

Acoustic peak A peak in the angular power spectrum of the cosmic microwave background, corresponding to a standing-wave mode of the primordial photon-baryon plasma at the moment of decoupling. See Chapter 2.

Attractor In dynamical systems theory, a region of state space toward which trajectories tend to converge. A *basin of attraction* is the set of initial conditions that converges to a given attractor. See Chapter 12.

Biophoton An ultra-low-intensity photon emitted by living tissue, typically at intensities of 10^{-19} to 10^{-17} W/cm². See Chapter 14.

Biotensegrity A hypothesis (Levin) that the human body is mechanically a tensegrity structure, with bones as compressive struts and fascia as the continuous tension network. See Chapter 10.

Calabi–Yau manifold A class of six-dimensional complex manifolds used in string-theory compactifications. See Chapter 15.

Coherence The property of a wave system in which the phase relationships between components are preserved over time. The opposite of *decoherence*. See Chapters 13–14.

Compactification In string and M-theory, the process by which higher-dimensional theories are reduced to a lower-dimensional effective

description by wrapping the extra dimensions on a small internal manifold. See Chapter 15.

Density wave A wave of higher density in a fluid that propagates at a pattern speed different from the speed of the fluid itself. The standard explanation for the spiral arms of galaxies. See Chapter 3.

Deborah number The dimensionless ratio of relaxation time to deformation time, governing whether a material behaves elastically or fluidly. See Chapter 6.

Decoherence The loss of quantum-mechanical phase coherence due to entanglement with the environment. See Chapter 13.

Fascia The continuous connective-tissue network that wraps and connects every structure in the body. See Chapter 10.

Field A physical quantity assigned to every point in space (and time). Examples in this book include the gravitational field, the electromagnetic field, and the eight-component soma field.

Fold (geological) A bend in a rock layer caused by compression. *Recumbent* folds are folds whose axial plane has been rotated to near-horizontal. See Chapter 6.

Fold (mathematical) A singularity in a mapping at which a smooth surface is folded onto itself. The geological and mathematical senses coincide in structural geology and in the catastrophe theory relevant to G_2 manifolds. See Chapters 6, 15.

Fractal A geometric object exhibiting self-similar structure at multiple scales, characterised by a non-integer *fractal dimension*. See Chapters 1, 7, 8.

G_2 manifold A seven-dimensional manifold with holonomy group G_2 , of central importance in M-theoretic compactifications. See Chapter 15.

Hologram A recording medium that preserves both the amplitude and the phase of a wavefront, enabling three-dimensional reconstruction. Contrast with **photograph**, which preserves only amplitude. See Chapters 1, 11.

Inflation (cosmological) A brief period of exponential expansion of the very early universe, proposed to explain the observed flatness and large-scale uniformity. See Chapter 2.

Langevin equation A stochastic differential equation describing the dynamics of a system subject to deterministic forces and random thermal noise. The equation of motion for the soma field. See Chapters 11, 12.

M-theory The eleven-dimensional unifying framework that contains, as limits, the five consistent ten-dimensional string theories and eleven-dimensional supergravity. See Chapter 15.

Mandelbulb The three-dimensional analogue of the Mandelbrot set, computed by extending complex multiplication to a three-dimensional algebra. The cover image of this book. See Chapters 1, 15, 16.

Microtubule A hollow cylindrical polymer of the protein tubulin, forming the structural skeleton of eukaryotic cells. A candidate substrate for the soma field. See Chapter 14.

Mode A standing-wave solution of a wave equation in a bounded region. The set of modes of a system is its **spectrum**.

Moduli In string and M-theory, continuous parameters describing the precise shape and size of the compactified internal geometry, appearing in the four-dimensional theory as effective scalar fields. See Chapter 15.

Quantum tunnelling The quantum-mechanical phenomenon in which a particle passes through a potential barrier classically forbidden to it. See Chapter 13.

Schumann resonance The fundamental electromagnetic resonance of the Earth-ionosphere cavity, at approximately 7.83 Hz. See Chapter 5.

Soma field The eight-component vector-valued field of feeling, defined on the human body and nervous system, central to this book. See Chapter 11 and throughout Part III.

Standing wave A wave whose envelope does not propagate; characterised by stationary nodes and antinodes. The fundamental building block of resonant systems.

Substrate The physical medium in which a field is realised. The substrate question for the soma field is the subject of Chapter 14.

Tensegrity A class of structures in which discontinuous compression elements (struts) are held in shape by a continuous tension network. See Chapter 10.

Threshold The amplitude at which sub-threshold field activity becomes consciously perceptible. A central parameter of the soma-field model. See Chapter 11.

Turing pattern A spatial pattern emerging spontaneously from a reaction-diffusion system with differential diffusion rates of activator and inhibitor. See Chapter 7.

Vacuum (quantum) The lowest-energy state of a quantum field, not empty but characterised by sub-threshold zero-point fluctuations. See Chapter 1.

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The references below are those cited in the footnotes of this book. Chicago notes-and-bibliography style. The eleven *Soma Field* technical papers are listed first as a group for ease of reference.

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F14.1 Cryo-EM microtubule. **PUBLIC**. NIH.

F15.1 Mandelbulb with folds labelled. **BUILD**. Author.

F16.1 Full-bleed Mandelbulb. **BUILD**. Author.

F16.2 Annotated Mandelbulb. **BUILD**. Author.

F0.1 Cover image: Mandelbulb annotated as G_2 compactification. **BUILD**. Author.

Colophon

The Wave That Is Always There: A Fractal Atlas from the Universe to the Soma was written in Zurich in the spring and summer of 2026, alongside the conclusion of the first phase of the Soma Field theoretical work and the beginning of the [T]-Theory art movement.

The text is set in the document-class default of the L^AT_EX `book` class, compiled via Pandoc 3.x with XeLaTeX. The body is set at 11 point on a leading of 14.85 point; the trim size is Royal octavo (156 × 234 mm), with binding-offset, inner, and outer margins set for sewn-bound case-binding.

The eleven *Soma Field* technical papers on which the scientific argument of this book rests are all open-access, are permanently archived on Zenodo with concept DOIs, and are listed in the Bibliography. The Lean 4 formal proofs of the core soma-field definitions are in the public repository `ITI-Theory/U` under `paper/proofs/`.

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The Independent Replication Ledger, in which contributions to the empirical validation of the Soma Field model are recorded, is maintained at https://github.com/ITI-Theory/U/blob/main/paper/INDEPENDENT_REPLICATION_LEDGER.md. As of summer 2026, every row of the ledger reads PENDING. The author welcomes correspondence from anyone — student, clinician, researcher, sceptic — willing to engage with the model on its own terms.

— A.J., Zurich, summer 2026